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# CHAPTER ATA 29 HYDRAULIC POWER

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY



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## Hydraulic Power - General

### Introduction

The Dash 8 Q400 has four hydraulic systems, three main systems and an auxiliary system. The No. 1 and No. 2 independent main systems provide power to operate the:

- Flight Controls
- Landing Gear
- Nose Wheel Steering
- Brake Systems.

The No. 3 main hydraulic system supplies pressure to the left and right elevators if the No. 1 and / or No. 2 hydraulic system(s) fail. The auxiliary hand operated hydraulic system provides power to the emergency main landing gear extension system.

### General Description

Refer to Figure 29-1

Main hydraulic power is provided by three independent hydraulic systems, designated No. 1 (left), No. 2 (right) and No. 3 (aft) . The No. 1 and No. 2 hydraulic systems are normally pressurized by a single Engine-Driven Pump (EDP) for each system. System pressure is maintained at  $3000 \pm 50$  psi ( $20684 \pm 345$  kPa).

The No. 3 hydraulic system is powered by an accumulator that is pressurized by a DC-Motor-Driven-Pump (DCMP). A pressure switch controls the DCMP operation to keep the accumulator pressure to within 2600 to 3250 psi (17926 to 22407 kPa). An electrically driven Standby Power Unit (SPU) operates as a backup to the No. 1 hydraulic system. The SPU operates during the takeoff and landing phases, or if the No. 1 EDP fails.

A Power Transfer Unit (PTU) operates as a backup to the No. 2 hydraulic system in the landing and take-off phases, or if No. 2 EDP or No. 2 engine fails. The PTU is powered by the No. 1 hydraulic system. If a dual engine failure

occurs, where both EDPs and the SPU are unavailable, the DCMP in No. 3 hydraulic system provides sufficient hydraulic power to the elevators for pitch control.

The No. 1 system powers the systems that follow:

- Flaps
- Inboard roll spoilers
- Main wheel brakes / Anti-skid
- Rudder (Lower Power Control Unit (PCU))
- Elevators (Outboard PCUs).

The No. 2 system powers the systems that follow:

- Landing gear
- Nose wheel steering
- Outboard roll spoilers
- Emergency / Parking brakes
- Rudder (Upper PCU)
- Elevators.

The No. 3 system supplies backup hydraulic power to the elevators.

- Left Elevator (Inboard PCU)
- Right Elevator (Inboard PCU)

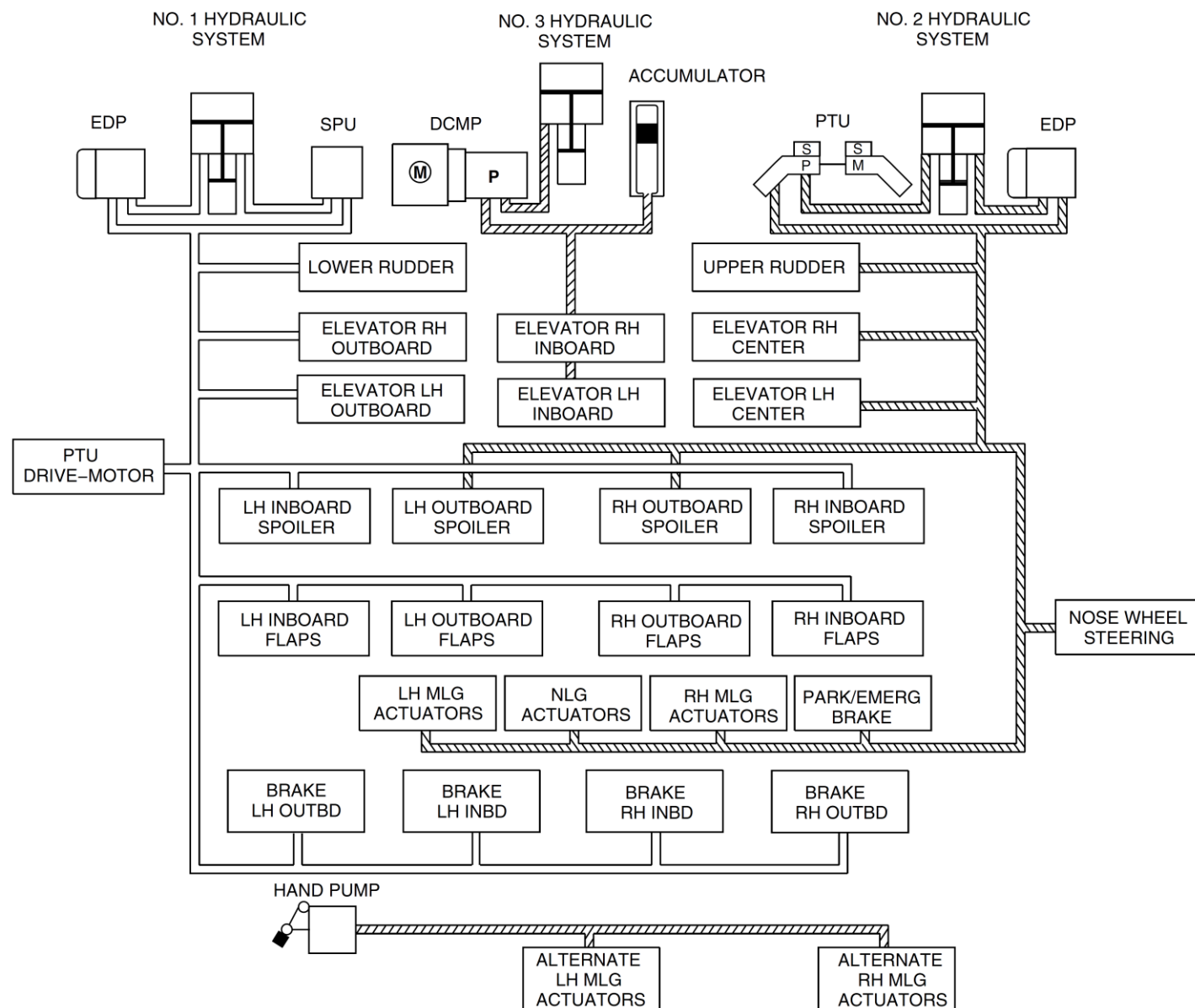
The auxiliary hydraulic system is provided as an alternate landing gear extension system. It supplies hydraulic power to extend the main landing gears when main hydraulic power is not available. A hand pump is used to manually pump up the hydraulic pressure in the auxiliary system, if the landing gear LEFT and/or RIGHT green locked down advisory lights do not turn on.

All systems use a synthetic phosphate ester-based fluid.

The hydraulic system contains the systems that follow:

- Main Hydraulic Systems
- Auxiliary Hydraulic Systems
- Indicating





**Figure 29-1 Hydraulic Systems Block Diagram**

## Main Hydraulic Systems 1 And 2

### Introduction

The No. 1 and No. 2 hydraulic systems supply hydraulic power to:

- Flight controls
- Landing gear
- Brakes
- Nose wheel steering.

### Detailed Description

The hydraulic system reservoirs store hydraulic fluid and supply the necessary fluid volume to the hydraulic systems. The No. 1 hydraulic system reservoir is installed in the left engine nacelle while the No. 2 reservoir is installed in the right engine nacelle. The reservoir uses system output pressure  $3000 \pm 50$  psi ( $20684 \pm 345$  kPa) to pressurize itself and provide a 55 psi (379 kPa) suction pressure to the EDPs.

Hydraulic fluid over temperature in each reservoir, is indicated by No. 1 or No. 2 HYD FLUID HOT caution light coming on.

The volume of the No. 1 system reservoir is 8 US qt (7.57 L). The volume of the No. 2 system reservoir is 12 US qt (11.36 L).

The No.1 and No.2 Engine Driven Pumps (EDP) supply the primary power for the hydraulic systems. Secondary No.1 hydraulic system pressure is supplied by the Standby Power Unit (SPU). The SPU operates during takeoff and landing mode and during an emergency EDP1 failure mode. The Power Transfer Unit (PTU) supplies secondary pressure to the No.2 hydraulic system during periods of high system flow demand and takeoff and landing modes. The PTU motor is powered by No.1 hydraulic system. The SPU and PTU can be turned on, by operation of the STBY HYD PRESS and PTU switch lights on the HYDRAULIC CONTROL panel, or set to the "normal" mode.

The No. 1 and No. 2 hydraulic systems contain the components that follow:

- Engine-Driven Pumps
- Firewall Shut-off Valves
- Isolation Valves
- No.1 Hydraulic System Reservoir
- No.2 Hydraulic System Reservoir
- Ecology Bottles
- Filter Manifolds
- Pressure Filter Elements
- Return Filter Elements
- Case-Drain Filter Elements
- Standby Power Unit
- Heat Exchangers
- Heat Exchanger Bypass Valves
- No.1 Hydraulic System Balanced Relief Valve
- Power Transfer Unit
- Hydraulic PTU Select Valve
- 225 °F (107 °C) Temperature Switch
- 275 °F (135 °C) Temperature Switch

### System Operation

[Refer to Figures 29-2 & 29-3](#)

Hydraulic fluid, is supplied by the pressurized reservoir through the Firewall Shutoff Valve to the EDP. Fluid under pressure is now directed to its subsystems at a nominal pressure of  $3000 \pm 50$  psi ( $20684 \pm 345$  kPa). before returning to the reservoir. Individual hydraulic pressure and quantity indicators are provided to monitor No. 1 and No. 2 hydraulic systems. Park brake and standby hydraulic pressure are also monitored. All indicators are presented on the co-pilot's Multi-Function-Display (MFD). If an EDP fails, the #1 or #2 ENG HYD PUMP caution light comes on.

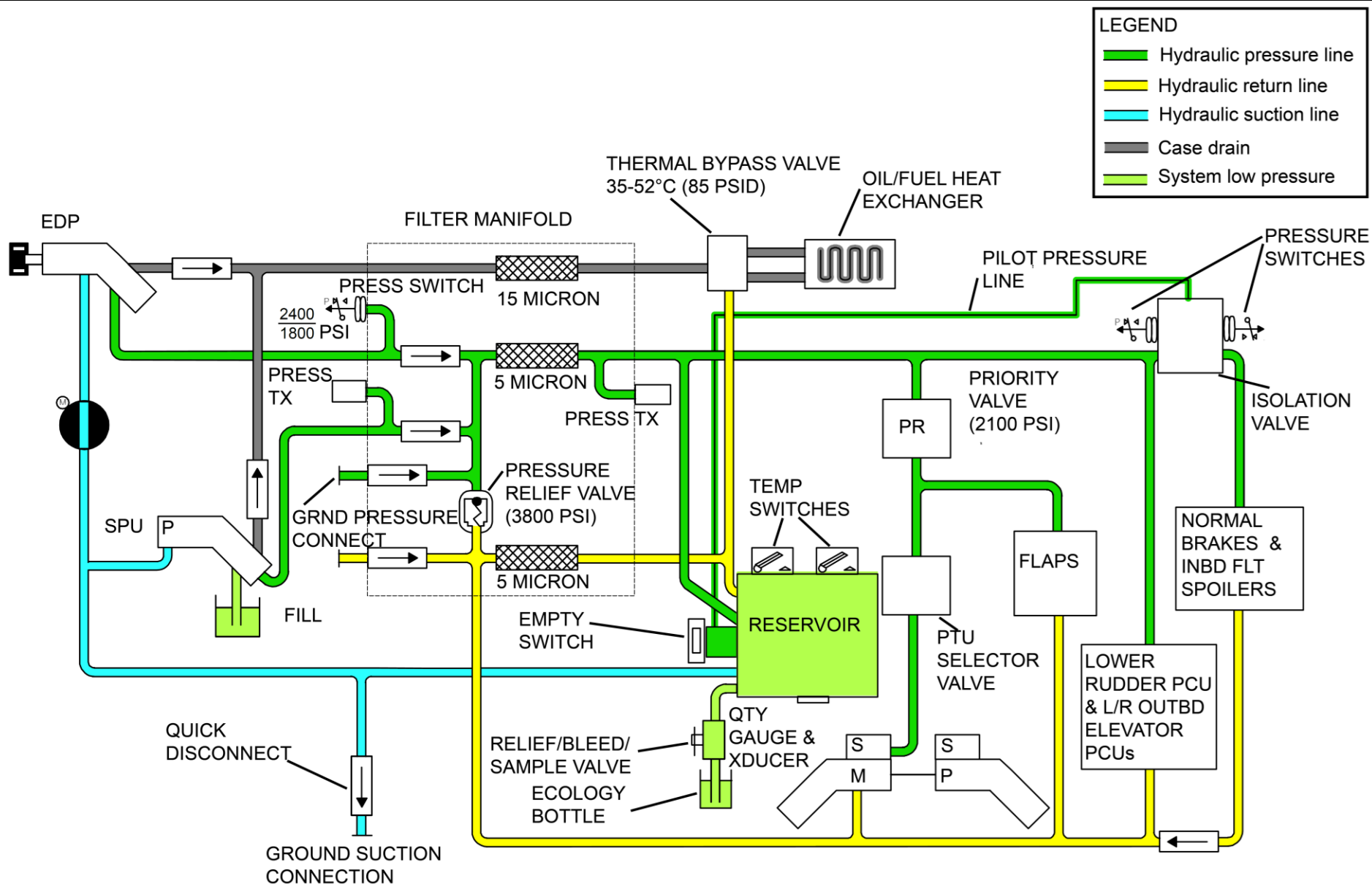
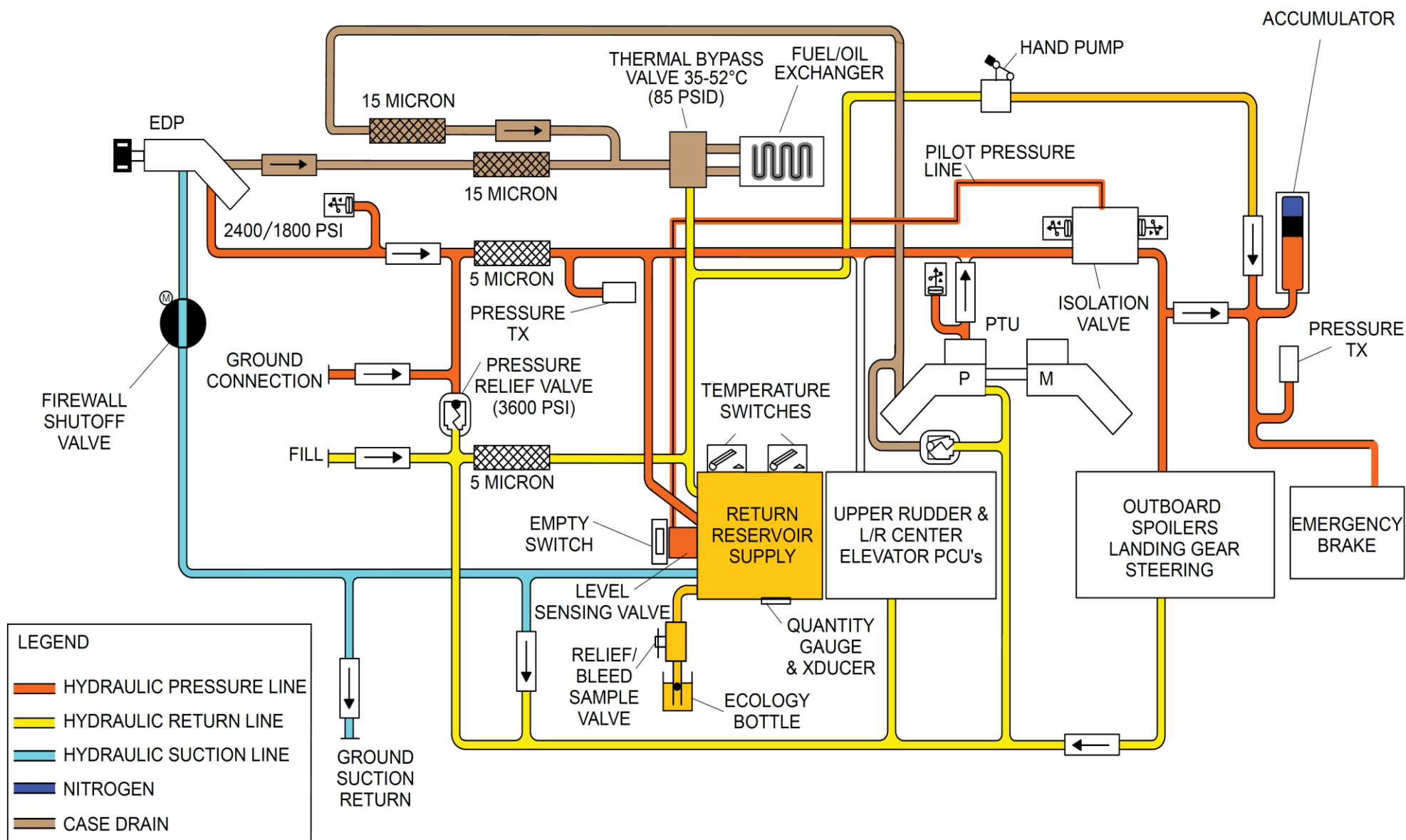


Figure 29-2 Hydraulic Synoptic No. 1



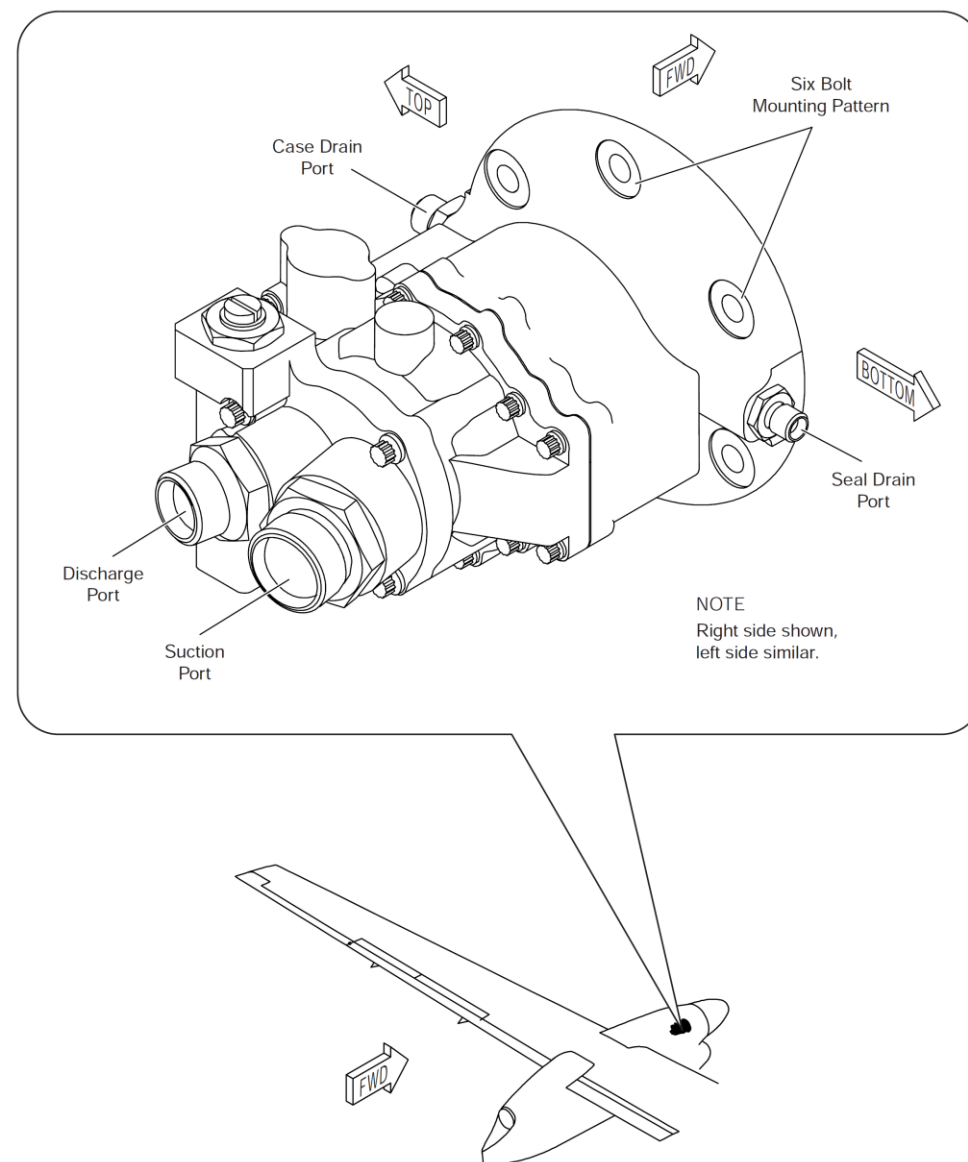
## Hydraulics System – Component Description

### No. 1 and No. 2 Hydraulic System Engine–Driven Pumps

Refer Figure 29-4

The Engine–Driven Pumps (EDPs) are installed on the accessory gearbox on both the No. 1 and No. 2 engines. Each pump supplies full, primary system pressure of  $3000 \pm 50$  psi ( $20684 \pm 345$  kPa). The pumps can change the volume of the fluid flow to keep the necessary system pressure. The suction, pressure and case drain ports of the pumps are connected by hoses to the hydraulic systems.

On aircraft with Modsum 4–113690, the EDPs suction, pressure and case drain ports are connected with the self-sealing fitting hose assembly. The self-sealing fitting hose assembly will decrease the loss of hydraulic fluid during the EDP change and bleeding. Also it reduces the bleeding time.



**Figure 29-4 Engine Driven Pump**

## Firewall Shutoff Valves

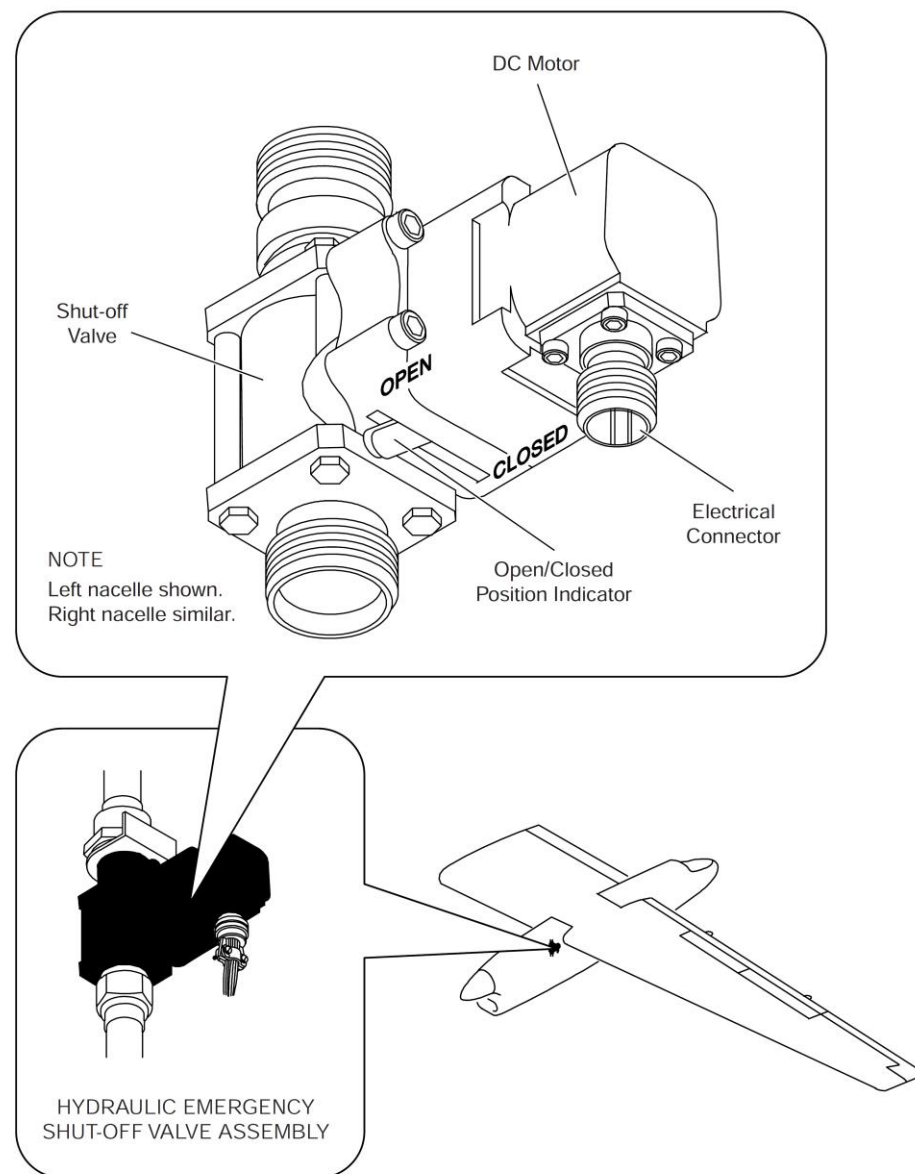
[Refer Figure 29-5](#)

The No. 1 and No. 2 hydraulic systems each have a firewall shut-off valve that, when closed, stops the flow of hydraulic fluid to the EDPs. Two HYD advisory lights for each firewall shut off valve are provided on the Fire Protection Panel, one green and one white. The green advisory light turns on when the firewall valve is OPEN and the white when the valve is closed. If an engine is shut down due to an engine fire, hydraulic fluid is shut off to the EDP by pulling the ENGINE 1 or ENGINE 2, PULL FUEL/HYD OFF handle out to its stop. The green advisory light will go out and the white advisory light will turn on.

[Refer to Figures 29-6, 29-7 & 29-8](#)

The Firewall Shutoff Valve is powered by the battery bus and will close when:

- ENGINE 1 or ENGINE 2 PULL FUEL/HYD OFF handle is pulled
- The hydraulic fluid reservoir quantity switch gives an empty indication.
- A hydraulic fluid high temperature condition causes the 275 °F (135 °C) temperature switch to close.



**Figure 29-5 Firewall Shutoff Valve**

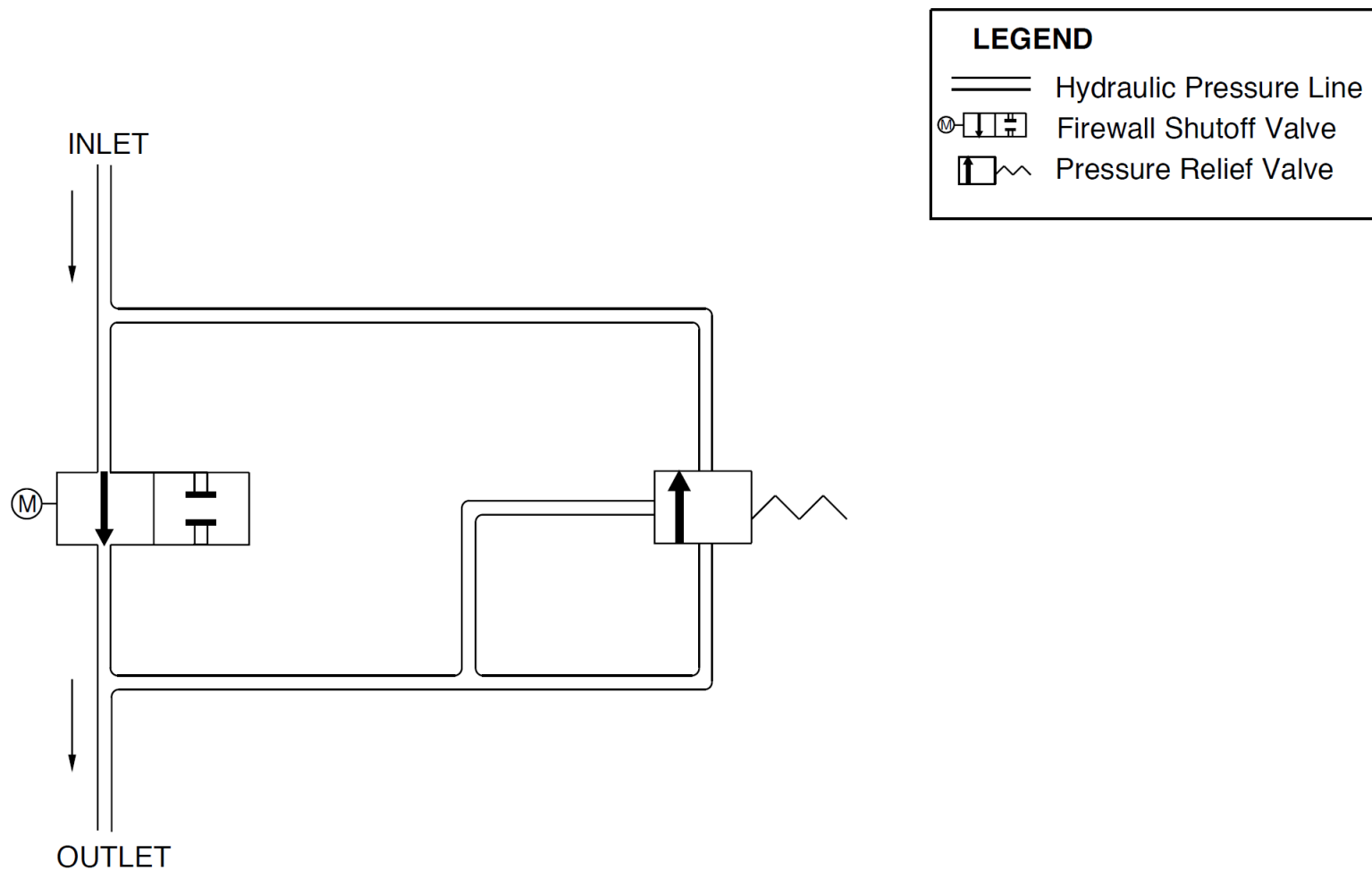


Figure 29-6 Firewall Shutoff Valve Hydraulic Schematic

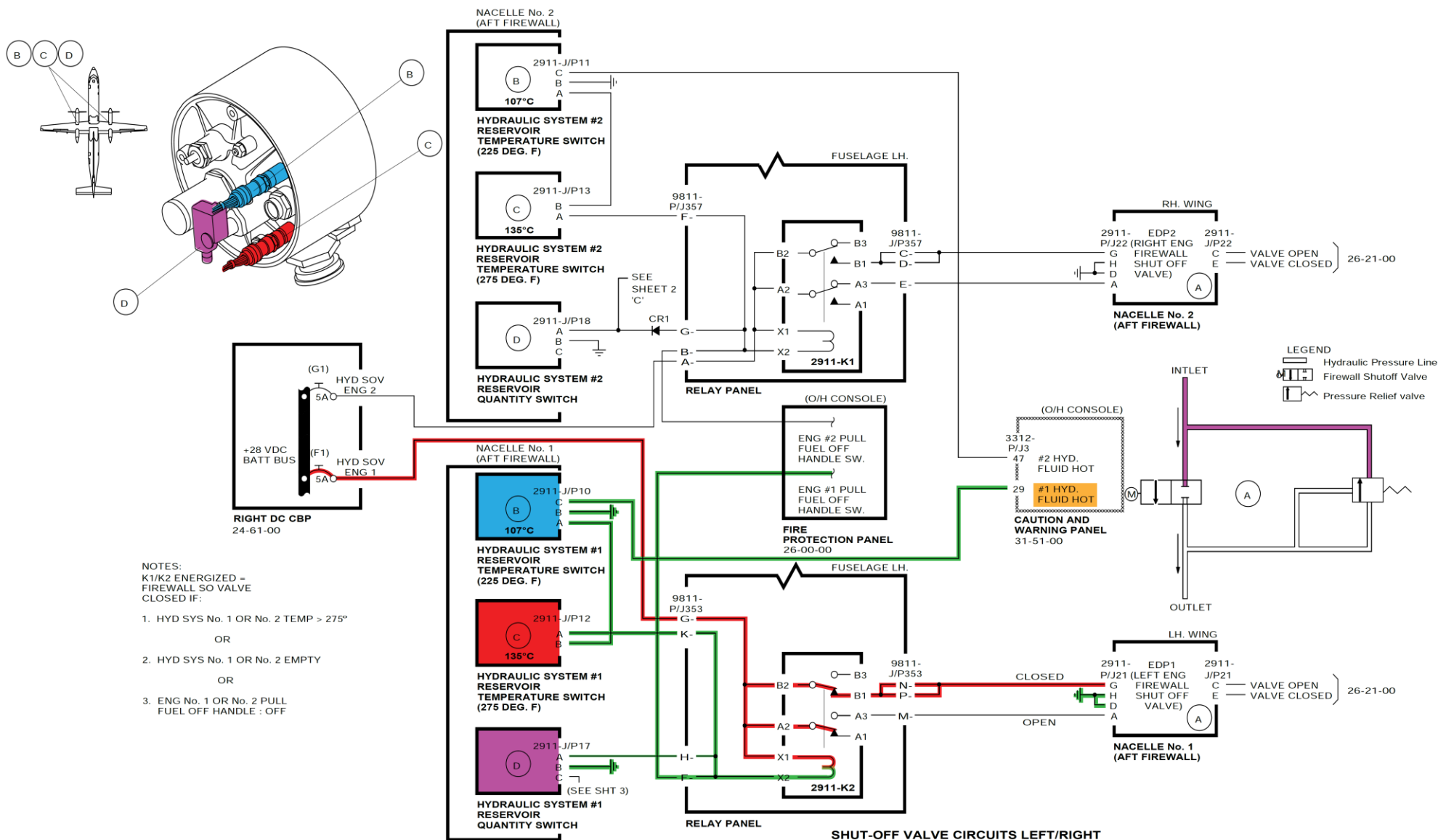


Figure 29-7 Firewall Shutoff Valve Electrical Schematic



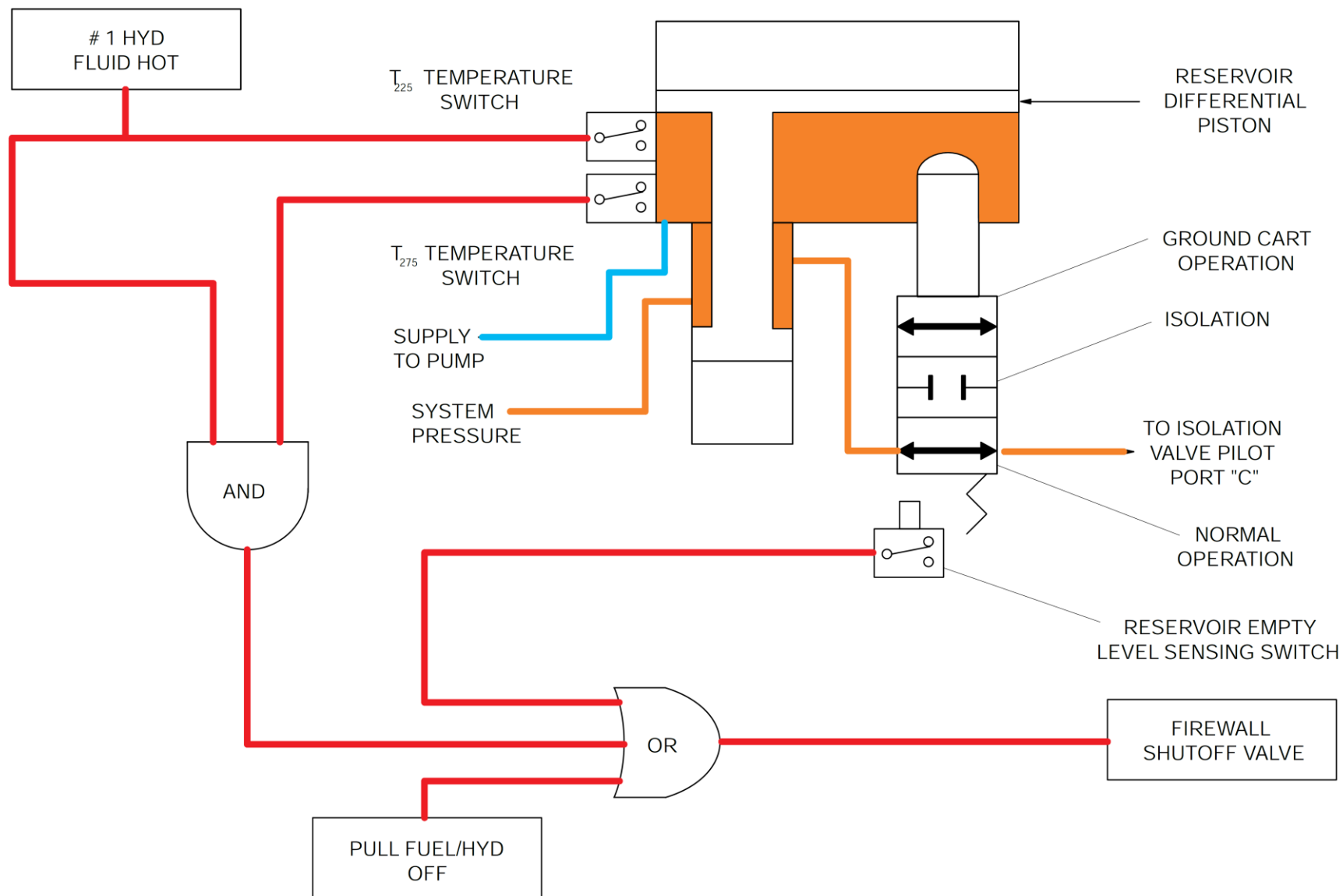


Figure 29-8 Firewall Shutoff Valve & Isolation Valve Close Logic

## No. 1 and No. 2 Hydraulic System Isolation Valves

**Refer Figure 29-9**

The No. 1 and No. 2 hydraulic systems each include an isolation valve. The isolation valve stops the flow of hydraulic fluid to part of the hydraulic system if the reservoir contents drop below 80 in<sup>3</sup> (1311cm<sup>3</sup>).

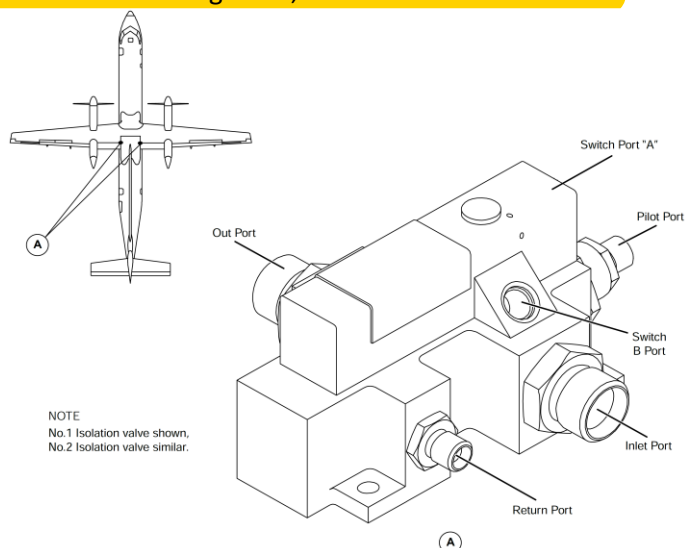
No. 1 hydraulic system isolation only allows fluid flow to the components that follow:

- Flaps
- PTU
- Rudder
- Elevators.

No. 2 hydraulic system isolation only allows fluid flow to the components that follow:

- Rudder
- Elevators

The isolation valve is operated by spring force and pilot pressure controlled by a reservoir level sensing valve, installed in the reservoirs.



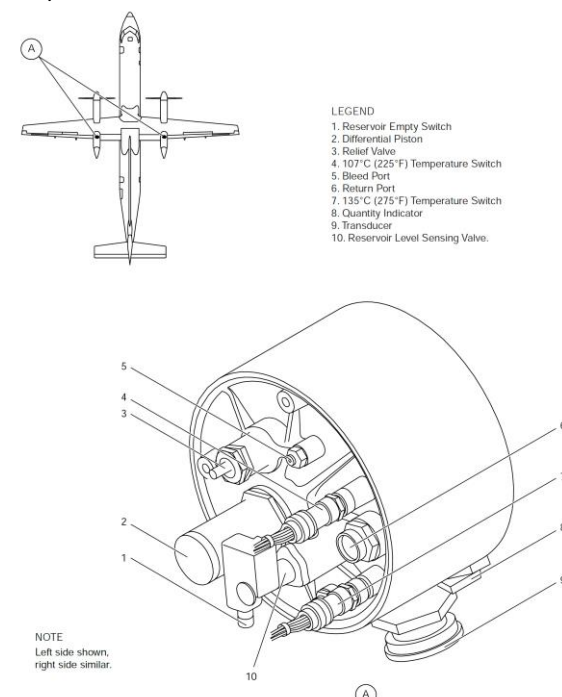
**Figure 29-9 Isolation Valve (No.1 & No.2 System)**

## No.1 and No.2 Hydraulic System Reservoirs

**Refer Figure 29-10**

The hydraulic system reservoirs store hydraulic fluid and supply the necessary fluid volume to the hydraulic systems. The No. 1 hydraulic system reservoir is installed in the No. 1/left side engine nacelle.

The No. 2 hydraulic system reservoir is installed in the No. 2/right side engine nacelle. The reservoir is a bootstrap type that provides a 55 psi (379 kPa) suction pressure to the EDPs. A pressure bleed/relief valve is installed in the reservoir to supply pressure relief of the hydraulic system at 80 psi (552 kPa). The bleed/relief valve may also be manually opened to bleed air from the reservoir or to take a hydraulic fluid sample. The volume of the No. 1 system reservoir is 8 US qt (7.57 L). The volume of the No. 2 system reservoir is 12 US qt (11.4L).

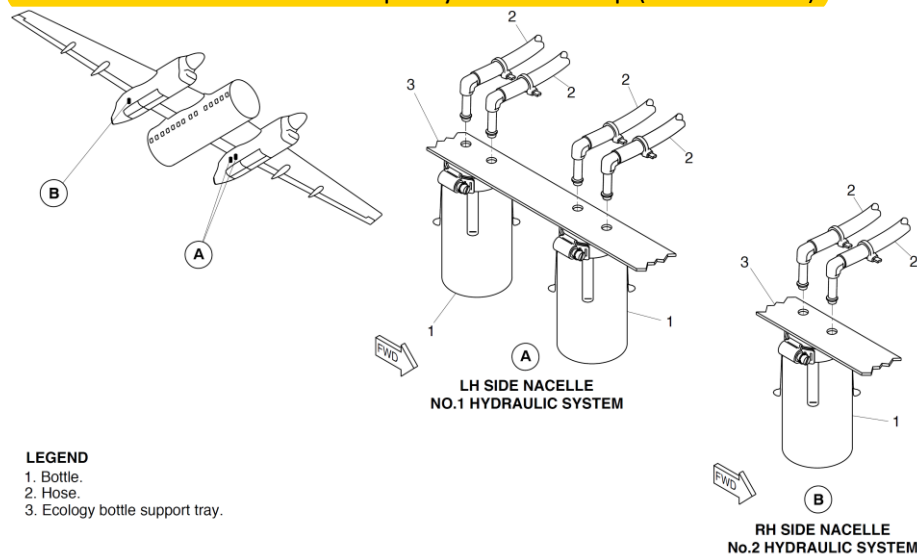


**Figure 29-10 Hydraulic System Reservoir**

## No. 1 and No. 2 Hydraulic System Ecology Bottles

Refer Figure 29-11

The No. 1 hydraulic system ecology bottle and the SPU ecology bottle are installed in the No.1/left side engine nacelle. The No. 1 ecology bottle collects leakage from the No. 1 reservoir bleed/relief valve and the balanced relief valve. The SPU ecology bottle collects leakage from the SPU shaft seal. The No. 2 hydraulic system ecology bottle is installed in the No. 2/right side engine nacelle and collects drainage from the No. 2 reservoir bleed/relief valve. Each bottle has a fluid capacity of 0.793 US qt (750 millilitres).



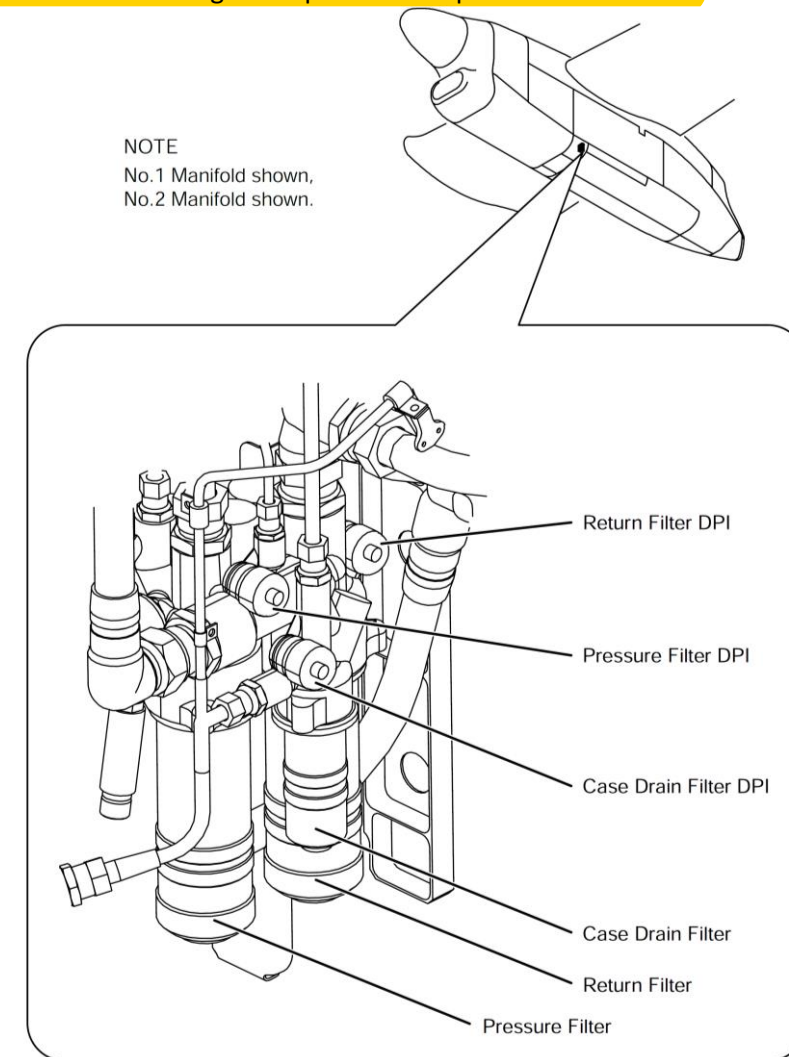
**Figure 29-11 Ecology Bottles (No. 1 & No. 2 Systems)**

## No.1 and No.2 Hydraulic System Filter Manifolds

Refer Figure 29-12

The No. 1 and No. 2 hydraulic systems each include a filter manifold. The primary function of the filter manifold is to remove contaminants from the pressure, return, case drain, filling and ground power fluid circuits. The filter manifold is also the supply point for flow from the pumps and the collection point for return flow from the system to the reservoir. The No. 1 hydraulic system filter manifold is installed in the No. 1 engine nacelle. The No. 2

hydraulic system filter manifold is installed in the No. 2 engine nacelle. A Differential Pressure Indicator(DPI) is installed for each of the filter bowls on the filter manifold. A red indicator extends to give a visual warning of clogged filter elements or large fluid pressure drops across the filters.

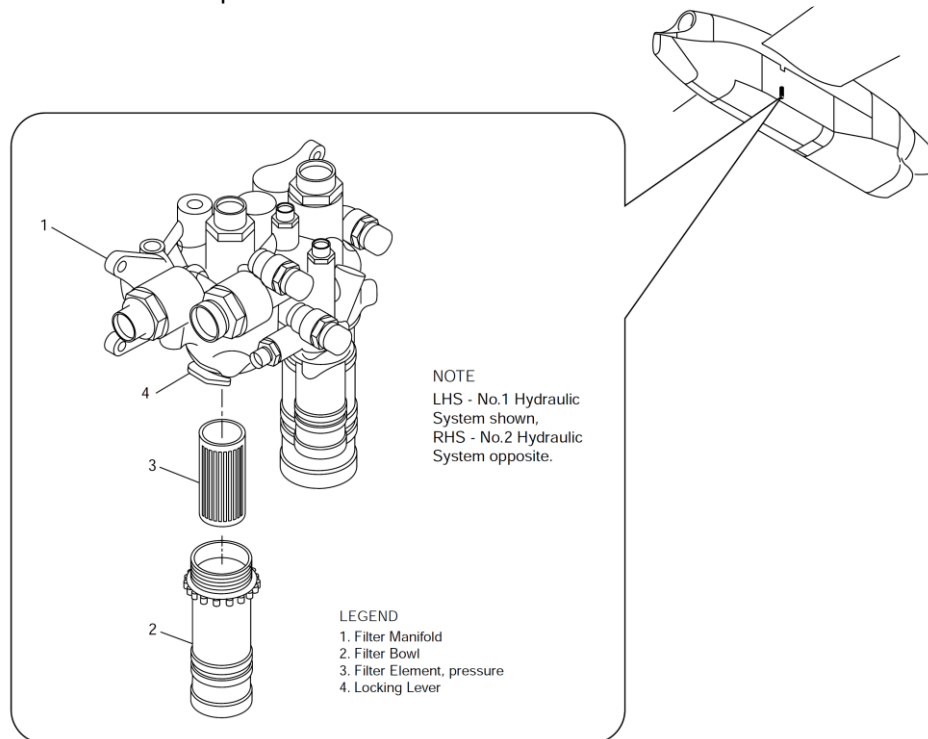


**Figure 29-12 Filter Manifold**

## No. 1 and No. 2 Hydraulic System Pressure Filter Elements

[Refer Figure 29-13](#)

One pressure filter is installed in each filter manifold. The filter element removes 100% of all contamination particles larger than 5microns. The filter elements are disposable.

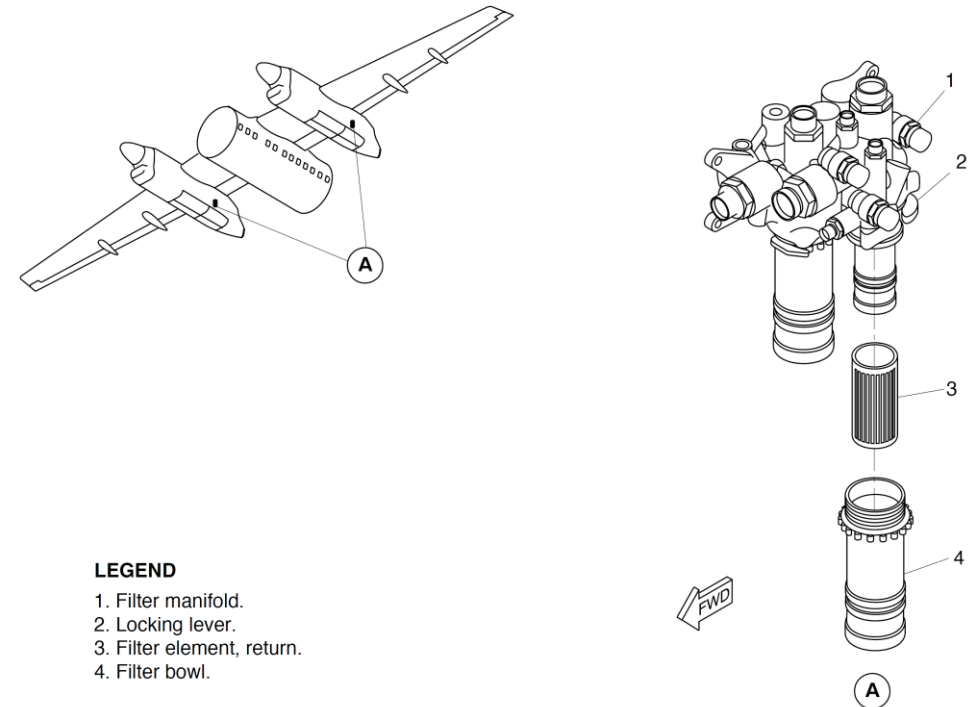


**Figure 29-13 Pressure Filter Element**

## No. 1 and No. 2 Hydraulic System Return Filter Elements

[Refer Figure 29-14](#)

One return filter is installed in each filter manifold. The filter element removes 100% of all contamination particles larger than 5 microns. The filter elements are disposable.



**Figure 29-14 Return Filter Element**

[Refer Figures 29-15 & 29-16](#)

## No. 1 Hydraulic System Case-Drain Filter Elements

One case-drain filter is installed in filter manifold. The filter elements remove 100% of all contamination particles larger than 15 microns. The filter elements are disposable.

## No. 2 Hydraulic System Case-Drain Filter Elements

One case-drain filter is installed in filter manifold. The filter elements remove 100% of all contamination particles larger than 15 microns. The filter elements are disposable.

On aircraft with Modsum 4-113545 or Post SB84-29-21, two case-drain

filters are installed, one in filter manifold and the other one in the forward equipment area of the right engine nacelle. The case-drain filter fitted in the forward equipment area of the right engine nacelle will decrease the back pressure and increase its cooling flow. This will decrease the EDP operating temperature and the number of EDP failures because of overheating. The filter elements remove 100% of all contamination particles larger than 15 microns. The filter elements are disposable.

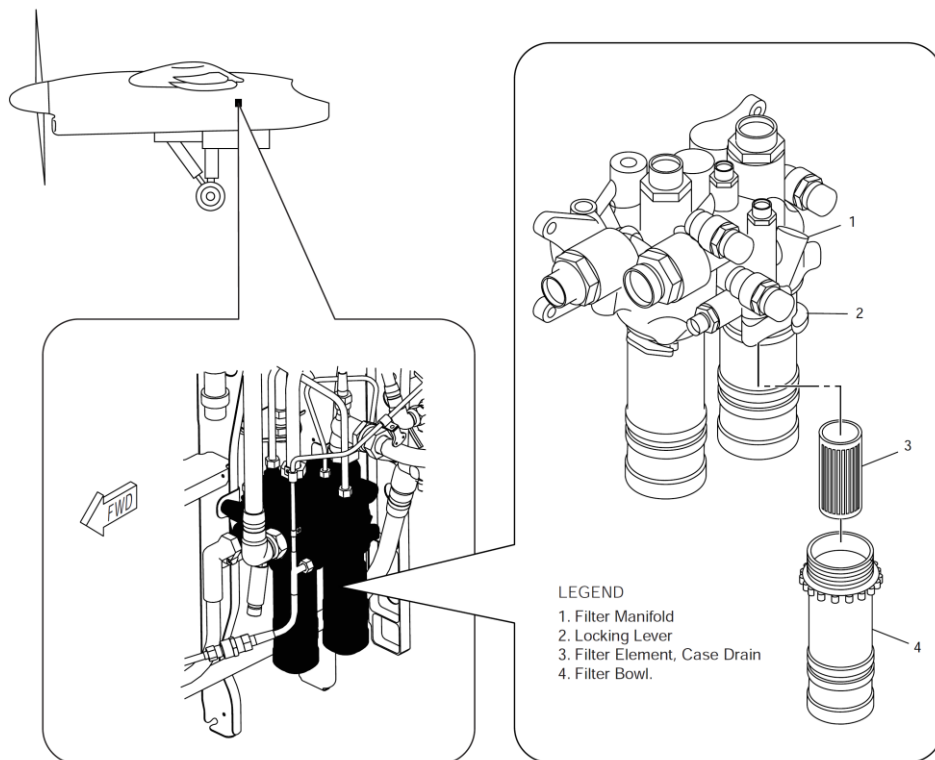


Figure 29-15 No.1 & No.2 (Pre-Modsum) Case Drain Filter

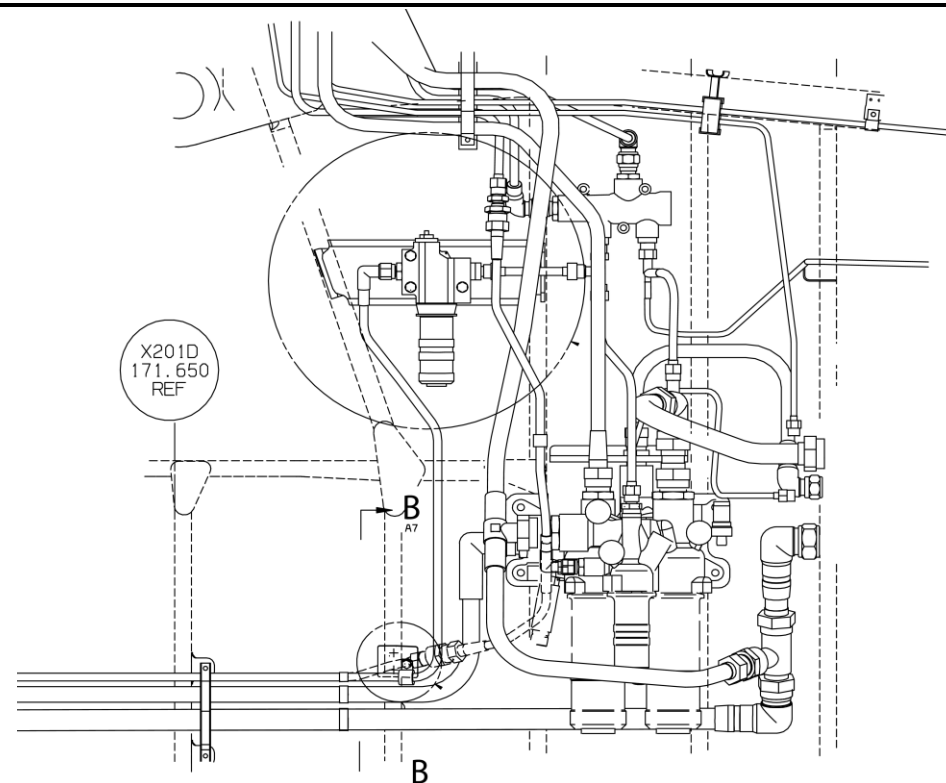


Figure 29-16 No.2 Case Drain Filter (Post-Modsum)

## Hydraulic System Heat Exchangers

[Refer Figure 29-17](#)

The No. 1 and No. 2 hydraulic systems each include an oil-to-fuel heat exchanger, located in the fuel tank that is used to cool the hydraulic fluid of each system. Hydraulic fluid from the case drain line, flows through the heat exchanger before returning to the hydraulic reservoir. A heat exchanger bypass valve, controls the flow of hydraulic fluid to and from the heat exchanger. The Engine Driven Pumps (EDPs), the Standby Power Unit (SPU), and the Power Transfer Unit (PTU) together or independently can pressurize the case drain line. The Ground Hydraulic Power Unit (GHPU) does not pressurize the case drain line.

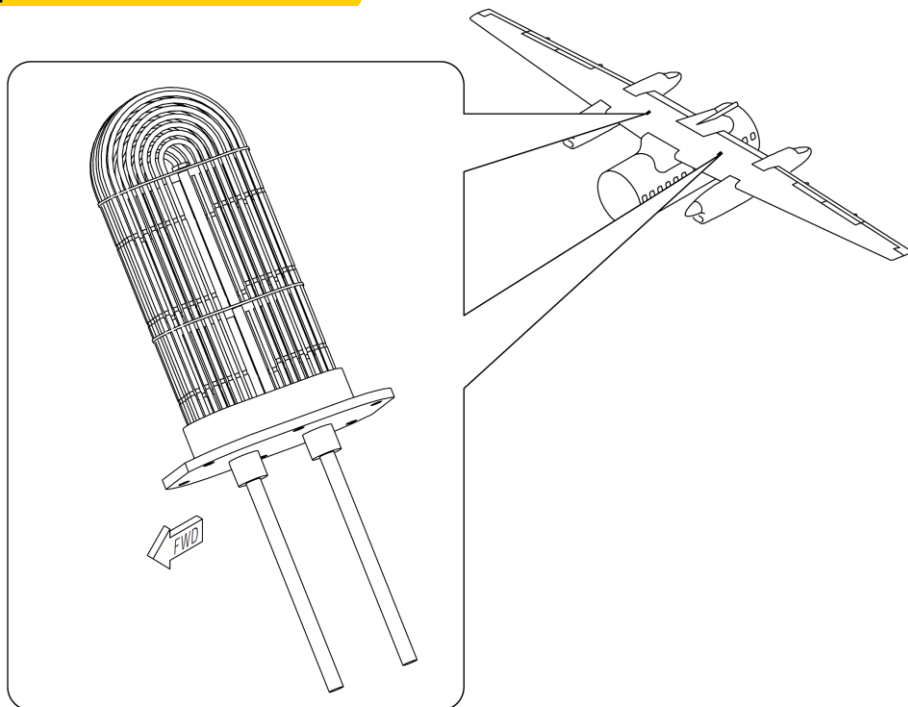


Figure 29-17 Heat Exchanger

## No. 1 and No. 2 Hydraulic System Heat Exchanger Bypass Valves

[Refer Figure 29-18](#)

The heat exchanger bypass valve is installed between the heat exchanger inlet and outlet line on each of the No. 1 and No. 2 hydraulic systems. The valve controls the flow of hydraulic fluid from the case-drain to the heat exchanger as a function of fluid temperature. The valve also functions as a relief valve to let fluid bypass the heat exchanger if there is too much of a pressure decrease. The valve starts to open at 95 °F (35.0 °C) and is fully open at 125 °F (52 °C).

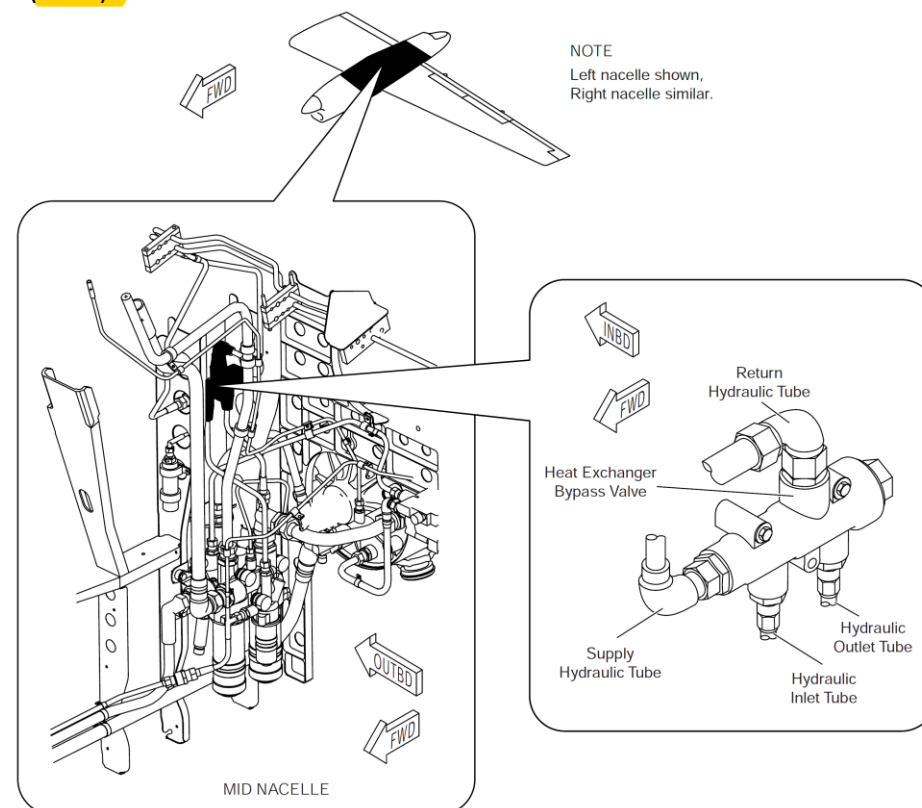


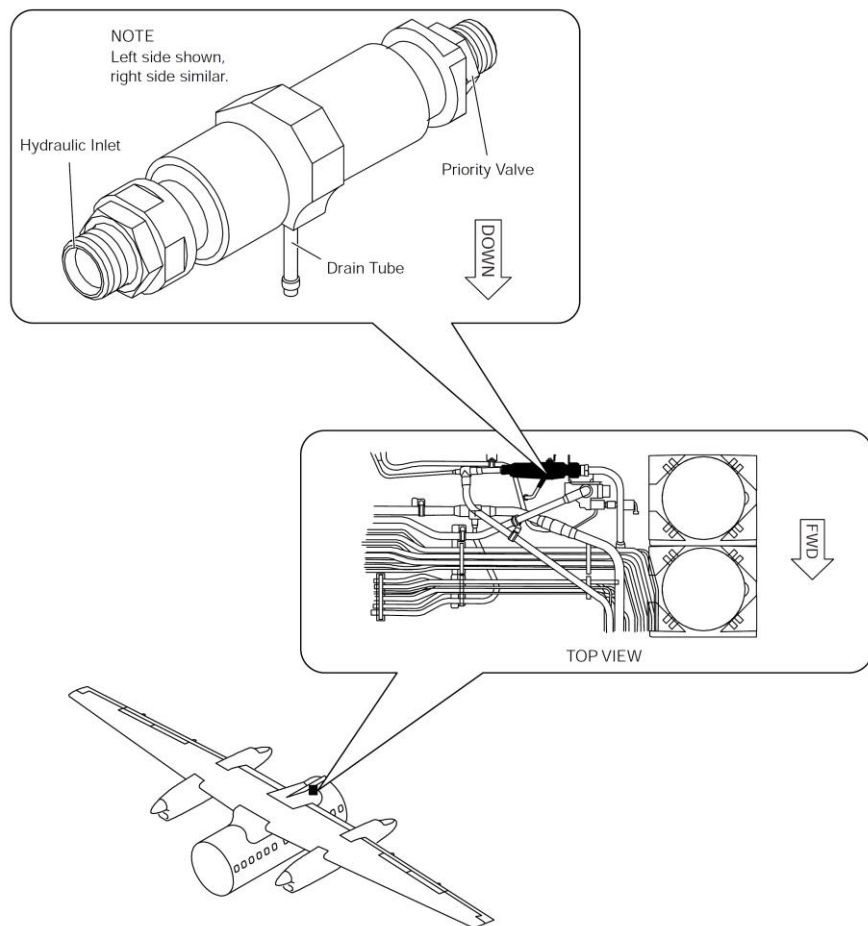
Figure 29-18 Heat Exchanger Bypass Valve



## No. 1 Hydraulic System Balanced Relief Valve (Priority Valve)

**Refer Figure 29-19**

A priority valve is included in the No. 1 hydraulic system and is normally open. If the hydraulic pressure in No. 1 system decreases below 2100 psi (14478 kPa) because of system demand, the priority valve closes. This shuts off hydraulic power to the flaps and PTU. Hydraulic pressure is maintained to the elevators, rudder, inboard spoilers and brakes.



**Figure 29-19 No. 1 Hydraulic System Balanced Relief Valve (Priority Valve)**

## Standby Power Unit

**Refer Figure 29-20**

No. 1 hydraulic System, uses a variable frequency AC motor-driven pump (SPU) which functions as a backup source for providing pressurized hydraulic fluid in response to system demand. The SPU supplies backup power to the No. 1 hydraulic system and is selected on for takeoff and landing. The SPU is electrically powered by the right 115 Vac bus. Backup electrical power to the SPU is supplied by the left 115 Vac bus. The SPU is installed in the No. 1 engine nacelle.

### NORMAL OPERATION

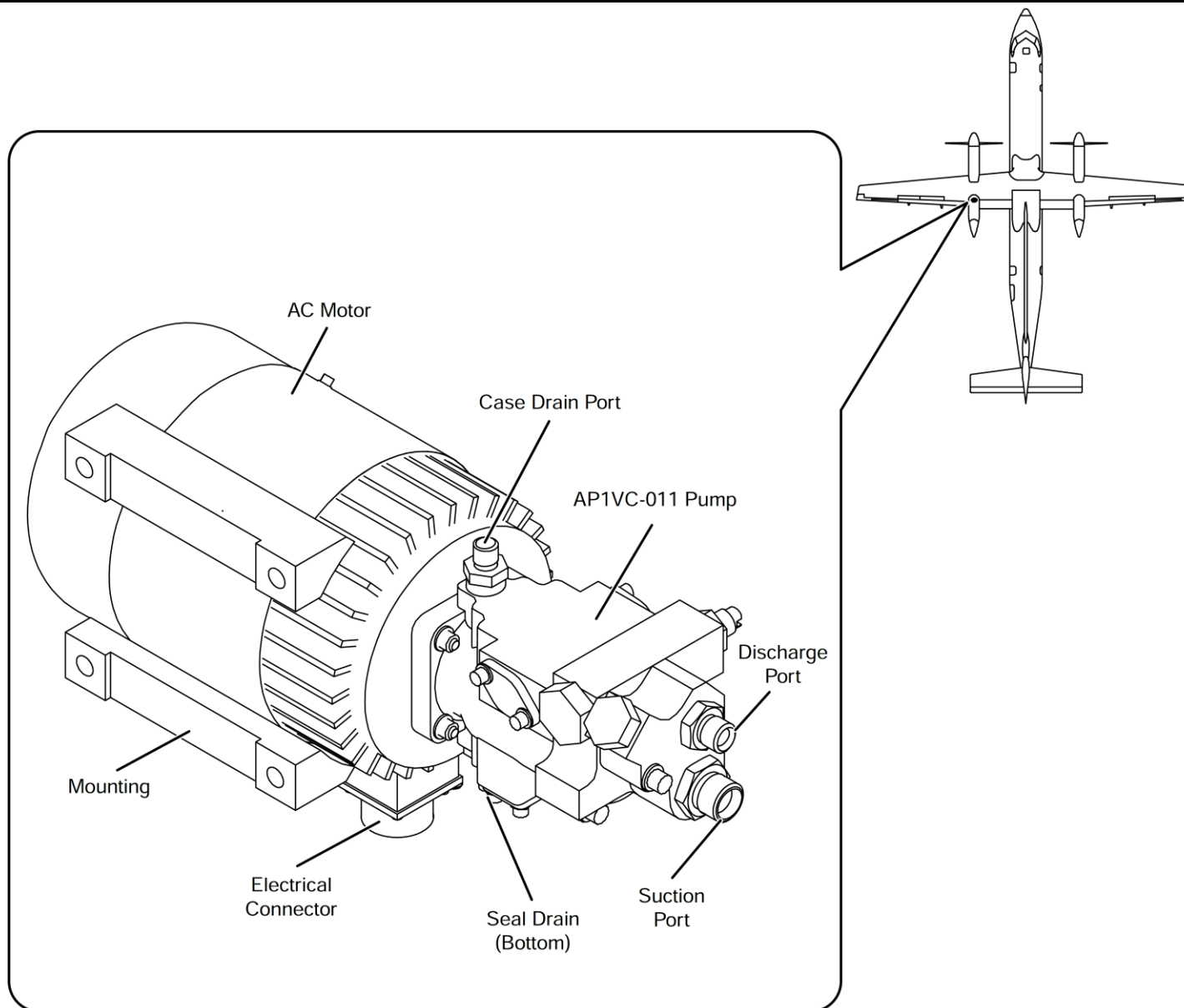
The SPU is normally selected on for takeoff and landing. When the STBY HYD PRESS switch light on the HYDRAULIC CONTROL panel is pushed, the SPU is energized on. A green ON legend in the switchlight turns on. When the SPU is activated on automatically, the green ON legend in the switchlight will not turn on. After takeoff, the STBY HYD PRESS switch is selected off to deactivate the SPU. Nominal system pressure is 3000 psi (20684 kPa).

**If not selected on, the SPU will automatically turn on if:**

- No. 1 engine fails during flight
- Flaps are selected to positions greater than 0 degrees when park brakes selected off and hydraulic reservoir No.1 not empty.

When the STBY HYD PRESS switch is manually selected on confirm green legend in switchlight shows ON.

If the SPU pump windings should overheat, the #1 STBY HYD PUMP HOT caution light will turn on.



**Figure 29-20 Standby Power Unit (SPU)**



## SPU Manual Operation – Electrical Schematic

[Refer Figure 29-21](#)

The SPU system operates manually if selected on with the switchlight and if 115 VAC power is available.

Power from the right 28 VDC Main Bus (CB Q4) to:

- Pin K on Hydraulic Control Panel

- Pin 41 on right AC contactor box to relay K-7.

Pushing the STBY HYD PRESS switchlight:

- Provides a ground through the ACU to cause the switchlight ON indication to illuminate
- Provides a ground to energize relay K-7 that will close the three contacts supplying 115 VAC to energize the SPU

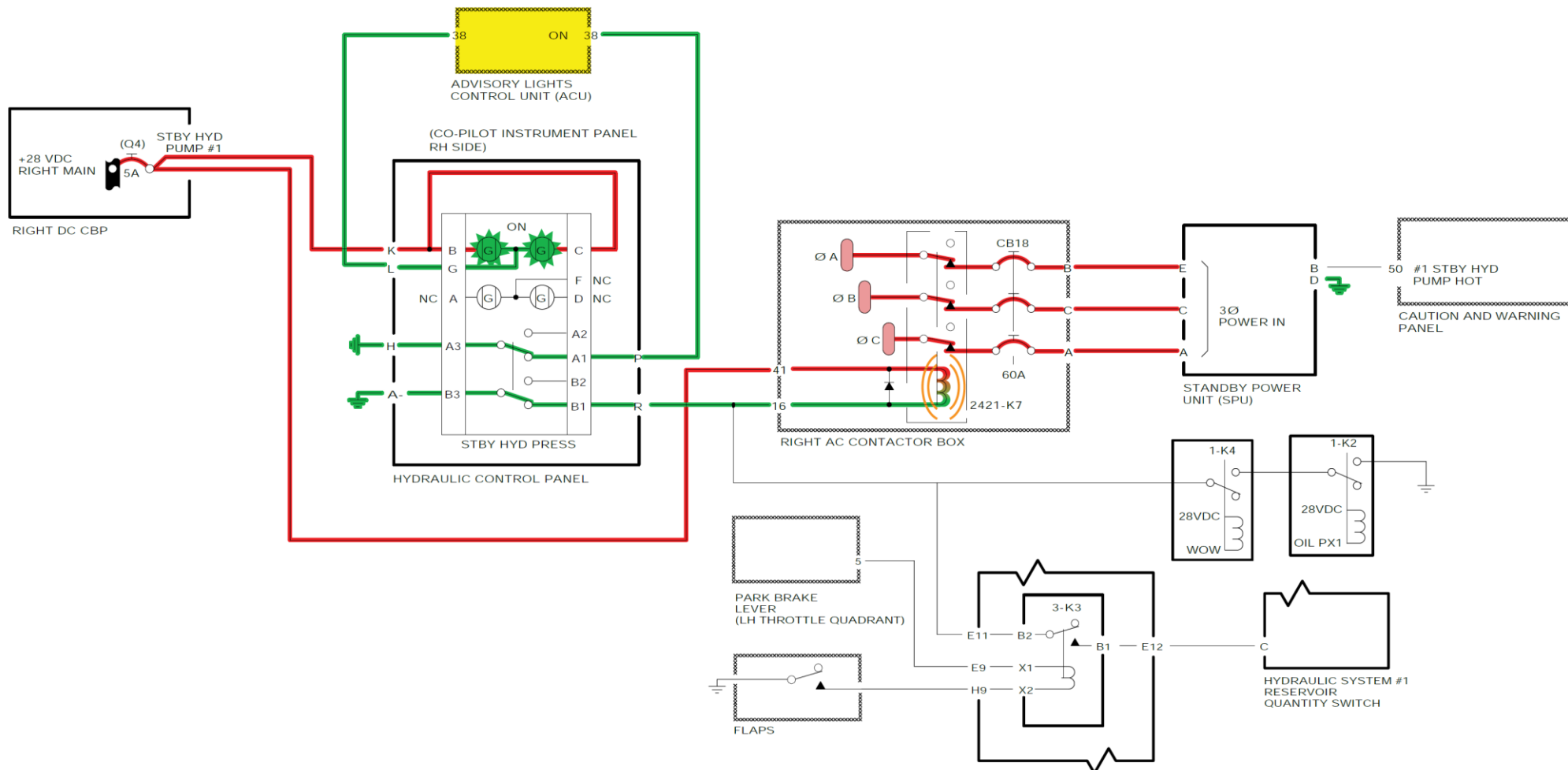


Figure 29-21 SPU Manual Operation Electrical Schematic

## SPU Automatic Operation (FLAP >0° & PRK BRK -OFF) – Electrical

### Schematic

[Refer Figure 29-22](#)

The SPU operates automatically if flaps are greater than 0° and the parking brake is selected OFF and 115VAC power is available and No.1 hydraulic reservoir is not empty.

- Park brake OFF supplies power from pin 5 to pin E9 on relay 3-K3.
- Flap > 0° provides ground to energize relay 3-K3.

Relay 3-K3 energized:

- Contacts B2-B1 provides ground for relay K7 supplying 115 VAC to energize the SPU (Reservoir not empty).

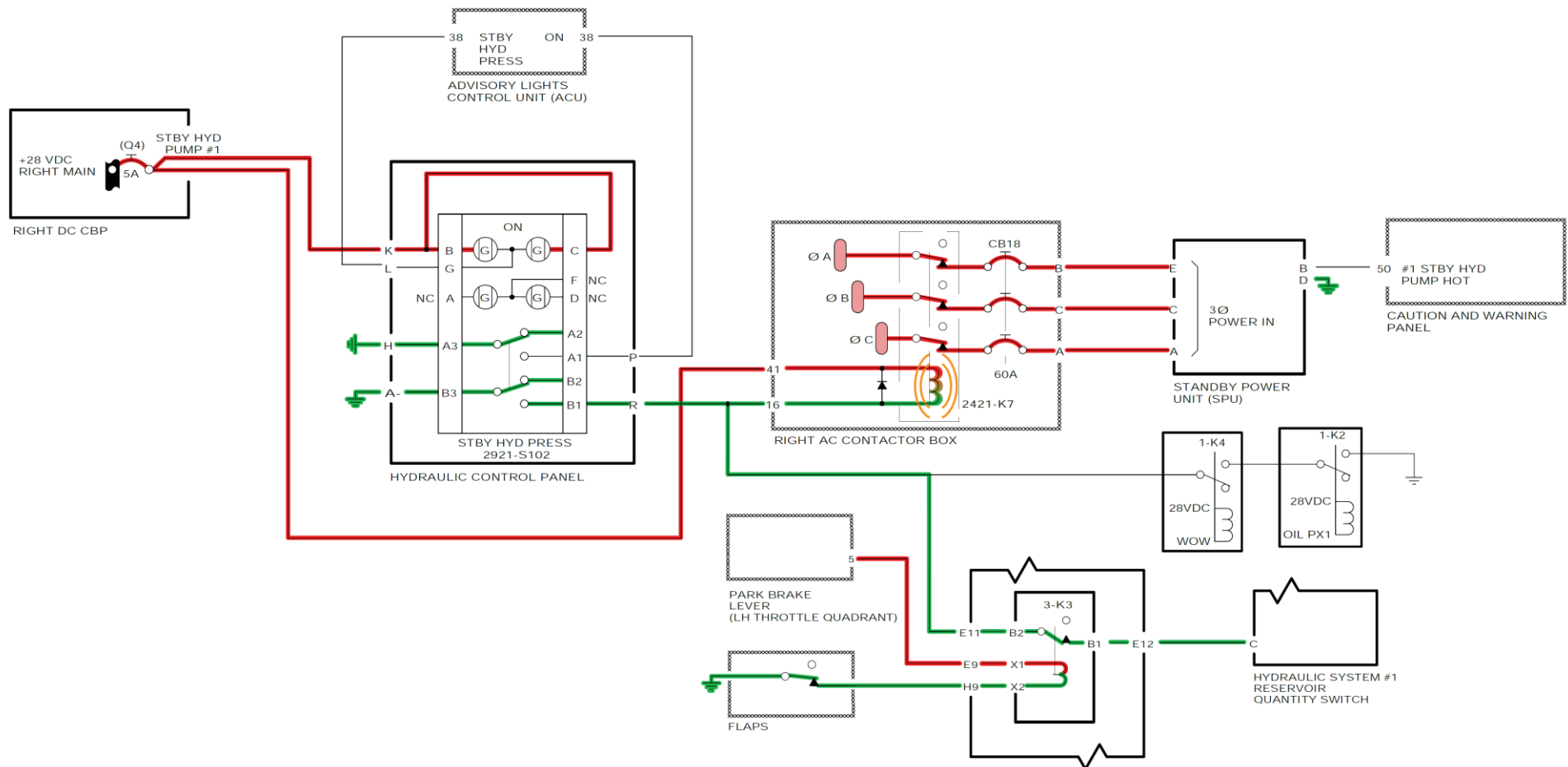


Figure 29-22 SPU Automatic Operation (FLAP >0° & PRK BRK -OFF) - Electrical Schematic

## SPU Automatic Operation (Engine Failure in Air) – Electrical Schematic

Refer Figure 29-23

The SPU comes on automatically if:

- The engine No.1 oil pressure is low and
- The aircraft is WOFFW.

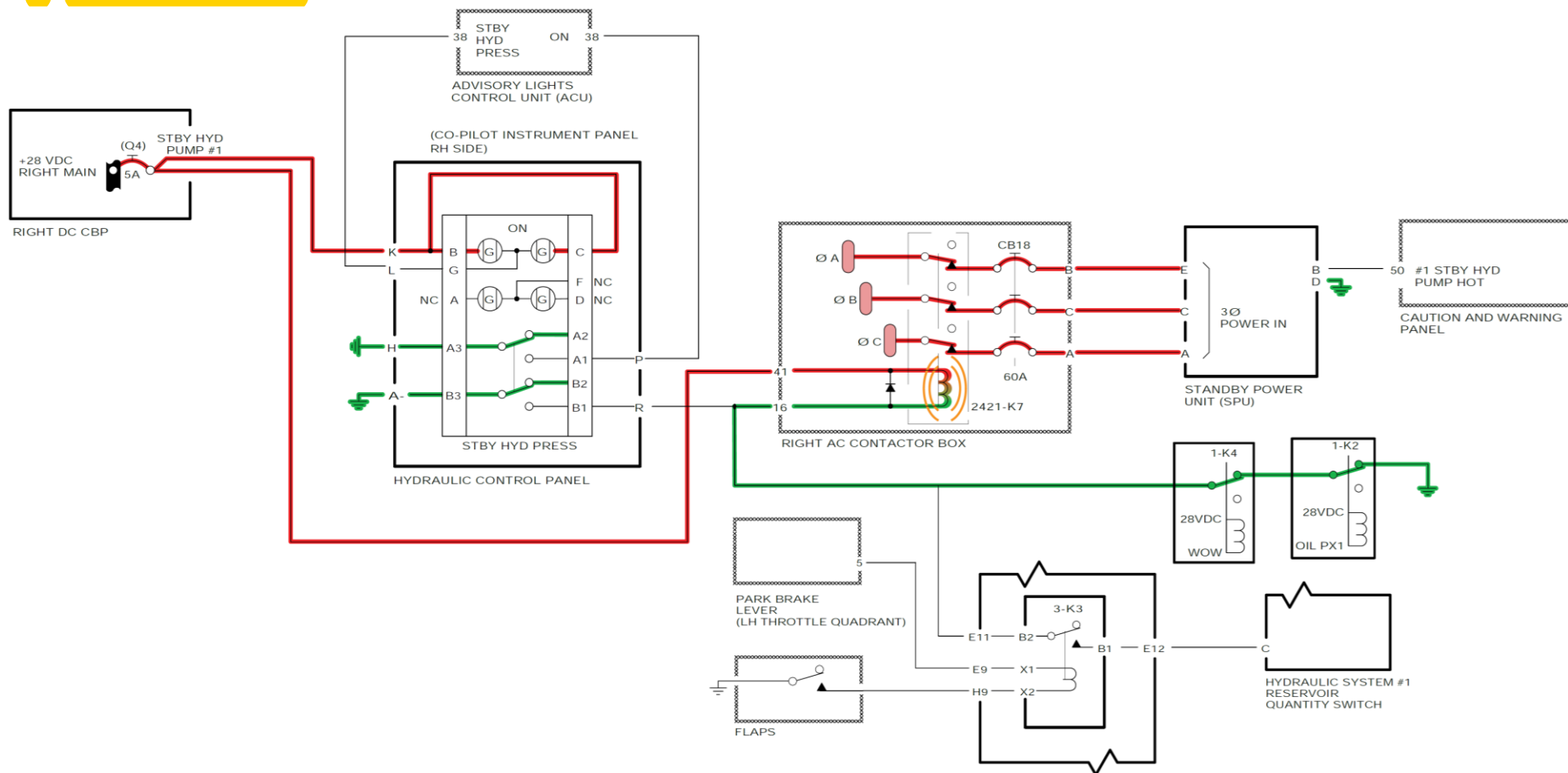


Figure 29-23 SPU Automatic Operation (Engine Failure in Air) – Electrical Schematic

## Power Transfer Unit (PTU)

[Refer Figures 29-24 & 29-25](#)

A Power Transfer Unit (PTU) operates as a backup hydraulic pressure to the No. 2 hydraulic system during takeoff, landing and periods of high system flow demand. The PTU uses hydraulic pressure from the No. 1 system to power a hydraulic motor. The motor then operates a hydraulic pump to pressurize the No. 2 system. Hydraulic fluid is not shared or transferred between No. 1 and No. 2 hydraulic systems during PTU operation. Hydraulic fluid must be available in the No. 2 system for PTU operation.

The PTU is installed in the fuselage-to-wing fairing. The PTU transfers hydraulic power from the No. 1 hydraulic system to the No. 2 hydraulic system without sharing fluid between the systems. Hydraulic pressure from the No. 1 system powers the PTU motor that operates the pump-side of the PTU to power the No. 2 hydraulic system.

The PTU is a fixed displacement, one-directional device used to supply backup power to the No. 2 hydraulic system. The PTU has two separate constant displacement hydraulic units connected by a drive shaft. The motor side is slightly larger in displacement than the pump side. The output of the pump is 16.6 gpm (63 L/min) at a discharge pressure of 1900 psi (13100 kPa) minimum with an inlet pressure of  $50 \pm 15$  psi ( $345 \pm 103$  kPa). The motor inlet pressure is 2700 psi (18616 kPa) and return pressure is  $150 \pm 25$  psi ( $1034 \pm 172$  kPa).

Pressurized fluid is applied to the pressure port on the motor side of the unit. When the cylinder turns, the angled bearing surface on the cam moves the piston and the shoe subassemblies back and forth. A spline drive shaft transmits the torque to the pump side of the PTU.

The pump has an in-line cylinder barrel, which is attached to and operated by the input shaft. The cylinder barrel contains nine fluid displacement piston and the shoe subassemblies. The piston and the shoe assemblies are held to the bearing surface by compression during the outlet stroke and a retainer plate during the intake stroke.

During the intake stroke of the piston, fluid goes into the cylinder bores

through the holes in the port cap, from the inlet port side of the pump. At the end of the intake stroke, the cylinder bore becomes full of fluid. When the cylinder barrel continues to turn, the piston moves towards the port cap and releases the fluid from the cylinder bore. The fluid goes through the port cap and the outlet port to the hydraulic system.

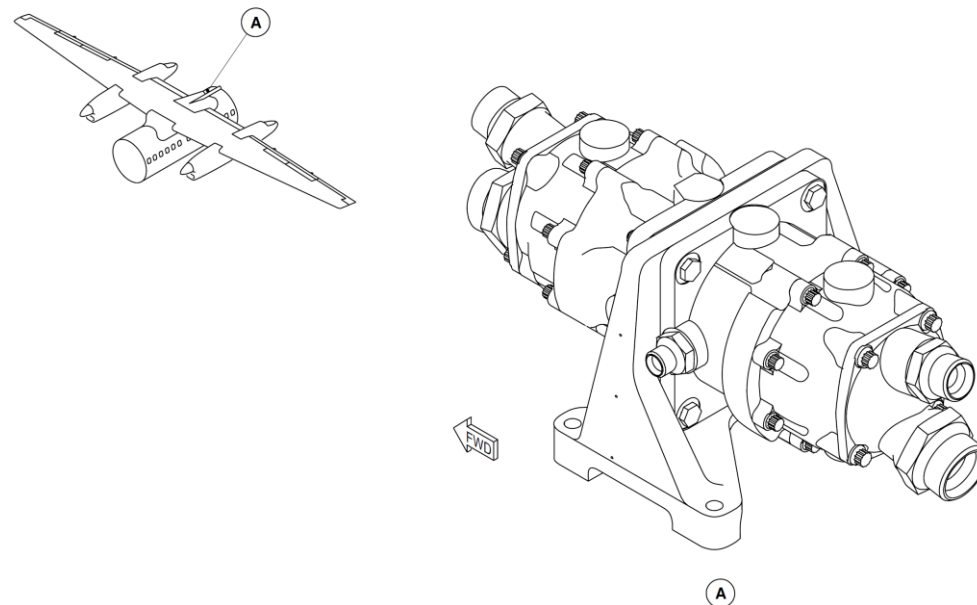


Figure 29-24 Power Transfer Unit

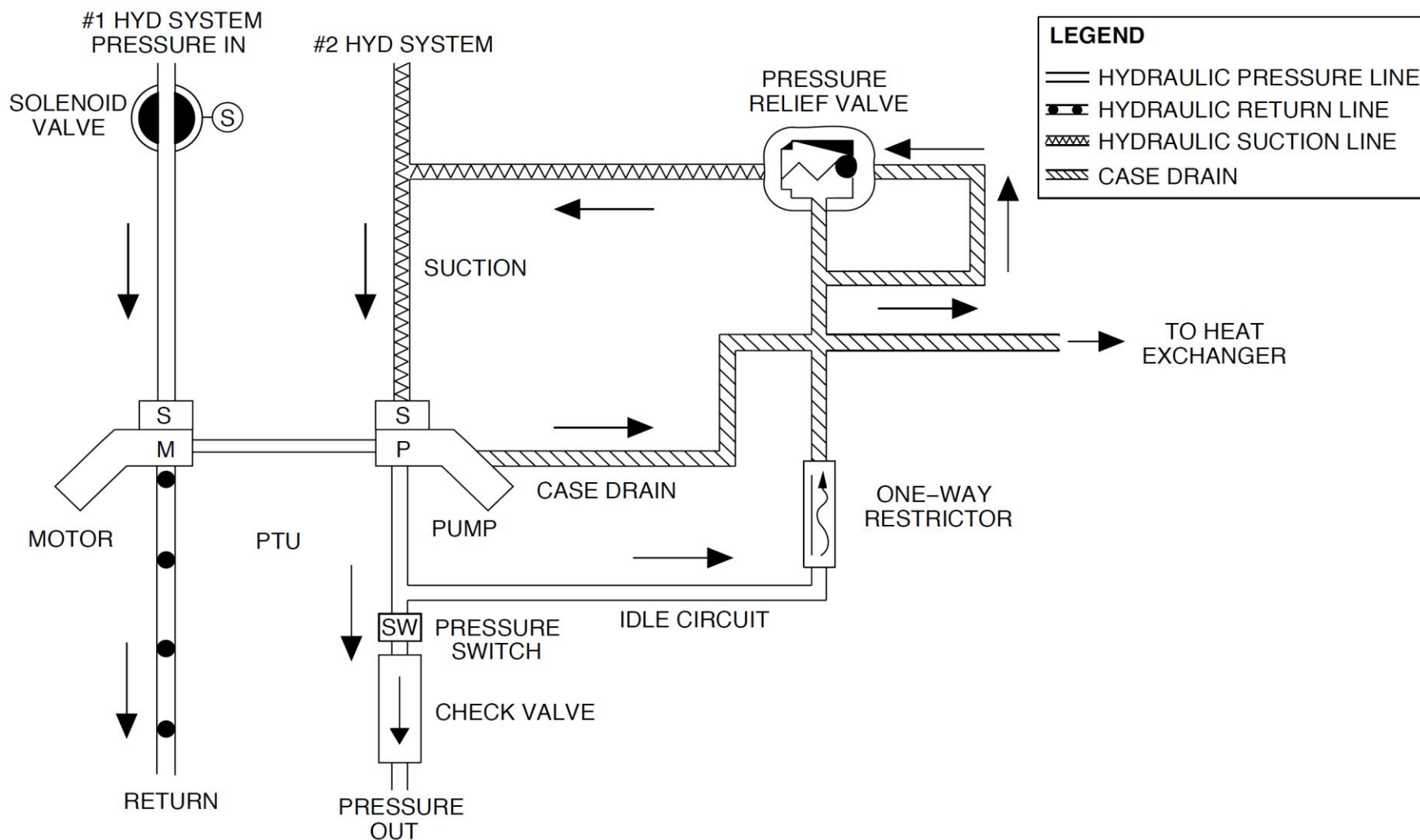


Figure 29-25 PTU Hydraulic Synoptic

## Power Transfer Unit Operation

On aircraft without ModSum 4Q126354 and SB84–29–22 incorporated:

The PTU may be selected on manually, or is actuated automatically during take-off and approach. Manual selection of the PTU is by pushing the PTU CNTRL switchlight on the HYDRAULIC CONTROL panel. A green ON legend in the switchlight turns on. Manual actuation of the PTU occurs when:

- The PTU CNTRL switchlight on the HYDRAULIC CONTROL panel is “ON”, and No. 2 Hydraulic System Reservoir is not empty on ground, or
- The PTU CNTRL switchlight on the HYDRAULIC CONTROL panel is “ON”, and No. 2 Hydraulic System Reservoir is not empty and No. 1 EDP pressure is not less than 2400 psi (16547.4 kPa).

Automatic actuation of the PTU occurs when:

- No. 2 Hydraulic System Reservoir is not empty, and
- The park brake is selected off, and
- Flaps are selected greater than 0 degrees, and
- No. 1 EDP Pressure Switch indicating high pressure (greater than 2400 psi (16547.4 kPa)), and
- PTU Pressure Switch does not indicate low pressure (less than 1500 psi (10342.1 kPa)) for longer than 30 seconds.

On aircraft with ModSum 4Q126354 and SB84–29–22 incorporated:

The PTU may be selected on manually, or is actuated automatically during single engine taxi operation, take off and approach. Manual selection of the PTU is by pushing the PTU CNTRL switchlight on the HYDRAULIC CONTROL panel. A white ON legend and green hashed bar in the PTU CNTRL switchlight turns on.

Manual actuation of the PTU occurs when:

- The PTU CNTRL switchlight on the HYDRAULIC CONTROL panel is “ON”, and No. 2 Hydraulic System Reservoir is not empty on ground, or

- The PTU CNTRL switchlight on the HYDRAULIC CONTROL panel is “ON”, and No. 2 Hydraulic System Reservoir is not empty and No. 1 EDP pressure is not less than 2400 psi (16547.4 kPa).

Automatic actuation of the PTU occurs when:

- Flaps are selected greater than 0 degrees and No. 1 EDP pressure is not low and No. 2 Hydraulic System Reservoir is not empty and park brake is selected off, or
- The aircraft is in air and No. 1 EDP pressure is high (greater than 2400 psi (16547.4 kPa)) and No. 2 Hydraulic System Reservoir is not empty and park brake is selected off and No. 2 main oil pressure is low, or
- The aircraft is in air and No. 1 EDP pressure is high (greater than 2400 psi (16547.4 kPa)) and No. 2 Hydraulic System Reservoir is not empty and park brake is selected off and No. 2 EDP pressure is low.

If the PTU does not produce pressure within 5 seconds, the PTU will be turned off for the automatic and manual selection.

---

## PTU Manual Operation – Electrical Schematic

[Refer Figure 29-26](#)

Power is supplied to the PTU SELECT SWITCH from the R ESS Bus.

This power is also supplied to the PTU SELECTOR VALVE.

- When the PTU SELECT SWITCH is selected ON, a ground is supplied to the ON light of the PTU SELECT SWITCH through the ADVISORY LIGHTS CONTROL UNIT.
- A ground is supplied through the PTU PRESSURE SWITCH, through the ADVISORY LIGHTS CONTROL UNIT to the SELECT SWITCH to bring on the green bar.
- A ground is supplied through the EDP #1 PRESSURE STATUS de-energized relay K12 to SYS #2 QTY STATUS de-energized RELAY K11, through the SELECT SWITCH, through the PTU FAIL RELAY K5 to the PTU SELECTOR VALVE.
- The PTU comes ON.





## PTU Automatic Operation (FLAP >0 & PRK BRK -OFF) – Electrical Schematic

[Refer Figure 29-27](#)

With the PARK BRAKE selected OFF ESS power from the ESS BUS is supplied through the de-energized PTU FAIL RELAY K5 to the PTU ENABLE RELAY K13, the ground for K13 is through the de-energized EDP #1 PRESSURE STATUS RELAY K12 and the flap lever in the 0 degree position.

- The PTU SELECTOR VALVE receives its ground through the de-energized PTU FAIL RELAY K5, the energized PTU ENABLE RELAY K13, and the de-energized SYS #2 QTY STATUS RELAY K11 and EDP #1 PRESSURE STATUS RELAY K12.
- The PTU comes ON.
- The green bar has power from the R ESS Bus and finds a ground through ADVISORY LIGHTS CONTROL UNIT and the PTU PRESSURE SWITCH in high.



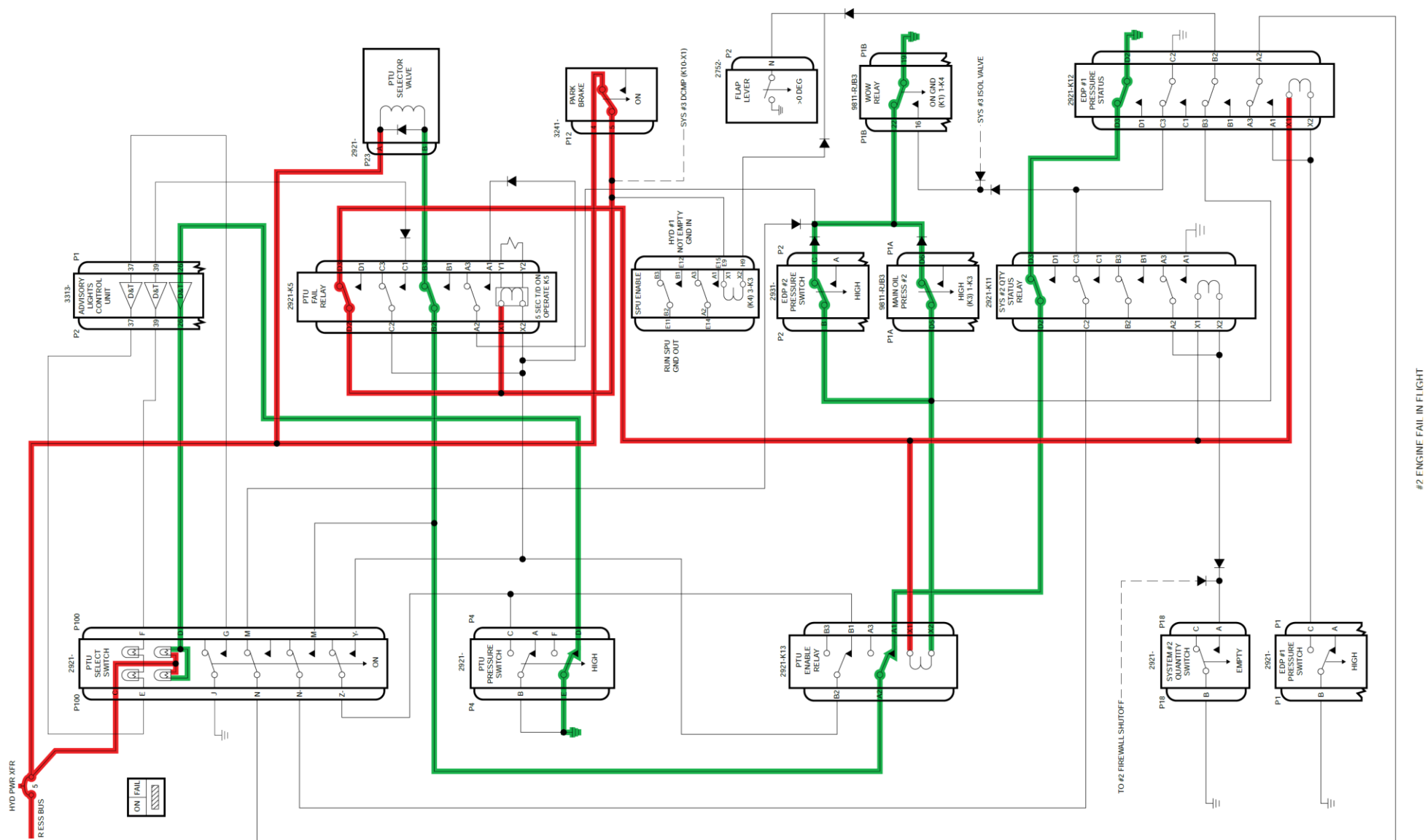
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## PTU Automatic Operation (No. 2 EDP fail or No. 2 Engine Fail in Air) – Electrical Schematic

[Refer Figure 29-28](#)

The power supply remains the same as previously described, but the ground is different.

- The ground for the PTU ENABLE RELAY K13 is through the MAIN OIL PRESS #2 SWITCH in low, or the EDP #2 PRESSURE SWITCH in low and then through the WOW RELAY to ground.



**Figure 29-28 PTU Automatic Operation (No. 2 EDP fail or No. 2 Engine Fail in Air) – Electrical Schematic**

## PTU FAIL Indication – Electrical Schematic

[Refer Figure 29-29](#) & [29-30](#)

- Power to the PTU FAIL RELAY K5 is through the PARK BRAKE switch when the park brake is selected OFF.
- In normal operation of the PTU there is no ground for the K5 relay because of the position of the PTU PRESSURE SWITCH in the high pressure mode.
- If the pressure from the PTU fails, the PTU PRESSURE SWITCH will provide a ground to the K5 RELAY through the PTU ENABLE RELAY K13.
- After 5 seconds time delay, K5 relay will energize removing the power from the PTU SELECTOR VALVE to shutdown the PTU, and provide a ground for the FAIL indication on the PTU SELECT SWITCH.

In flight a ground through the WOW 1-K4 will keep relay K5 energized following a loss of PTU pressure, isolating the PTU SELECTOR VALVE

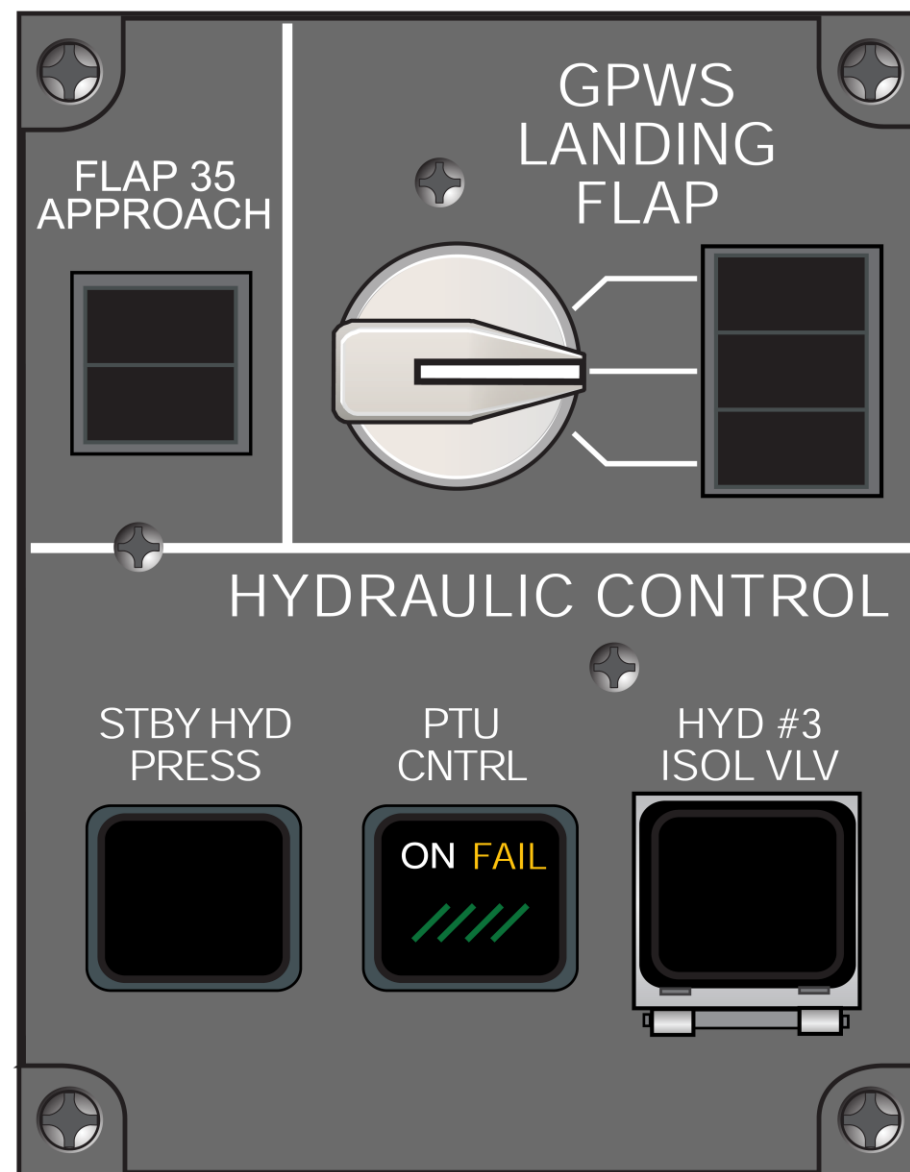


Figure 29-29 Hydraulic control panel - PTU FAIL Indication

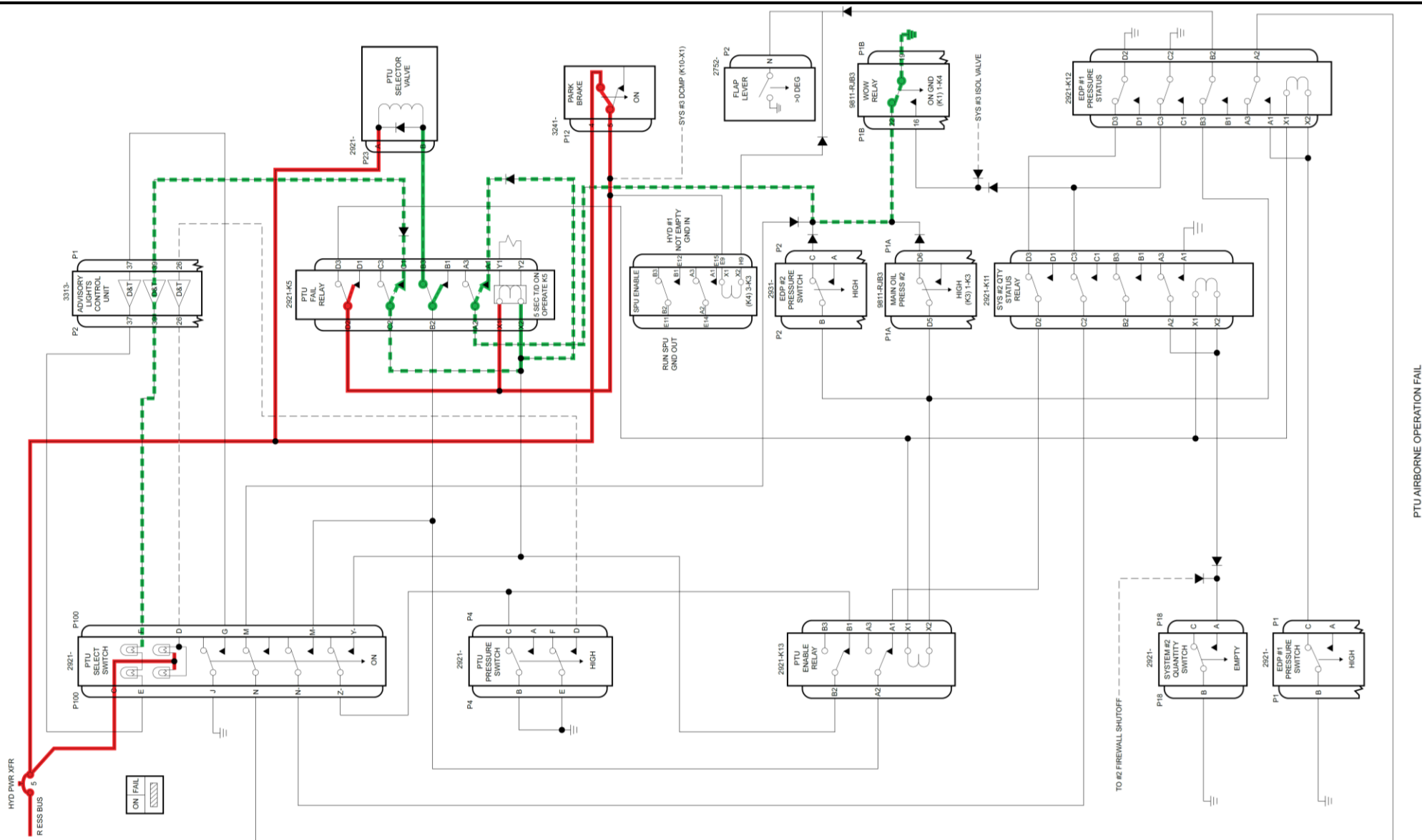


Figure 29-30 PTU FAIL Indication – Electrical Schematic



### Standby Hydraulic Pressure Switch light (green)

[Refer Figure 29-31](#)

PUSH ON segment (green)

- Turns Standby Power Unit (SPU) on manually
- SPU functions as backup source, providing hydraulic pressure to No. 1 hydraulic system
- PUSH ON segment (blank)
- Turns SPU off manually
- ON segment (blank)

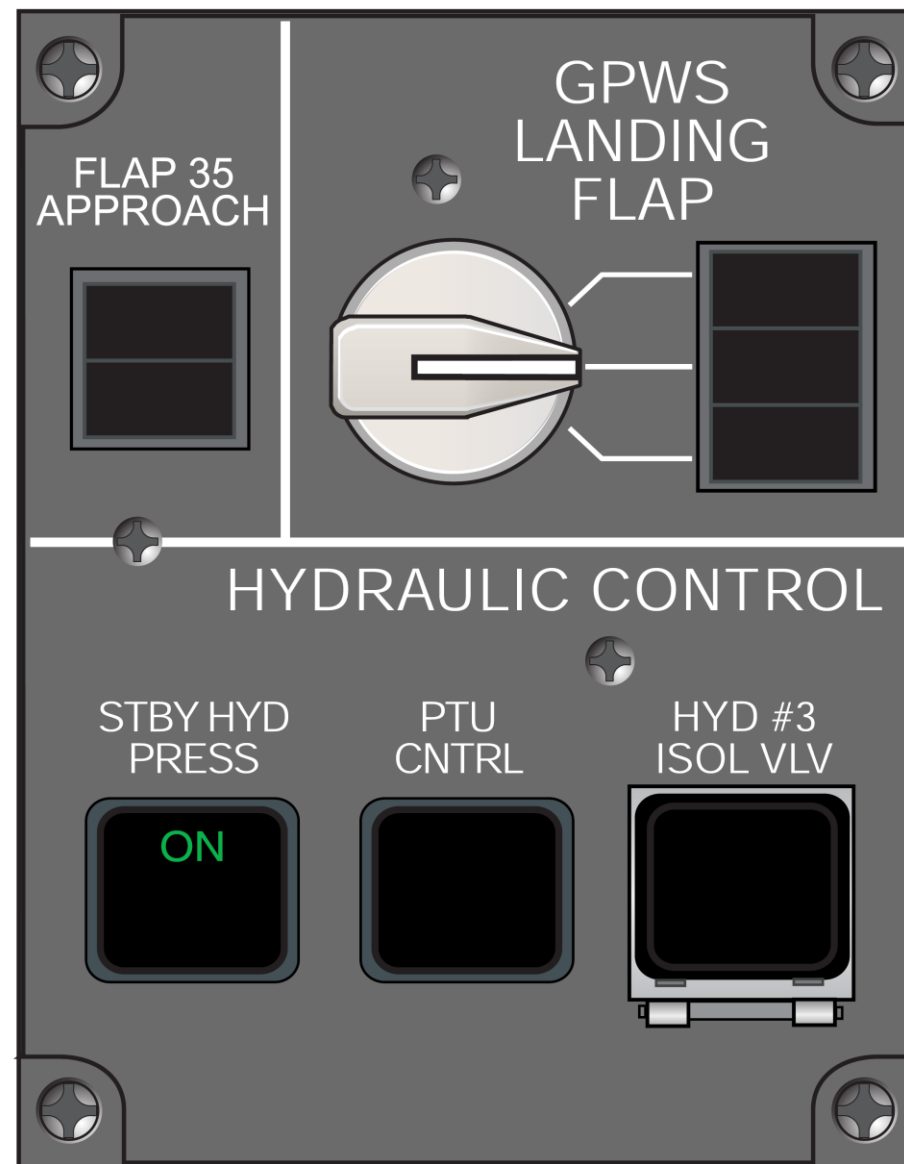


Figure 29-31 SPU Switchlight

## Power Transfer Unit Switch light

[Refer Figure 29-32 & 29-33](#)

On aircraft without ModSum 4Q126354 and SB84-29-22 incorporated:

- PUSH ON segment (green)
- Turns PTU on
- Indicates PTU is operating
- PTU provides pressure to No. 2 system
- When not selected on, the PTU will turn on automatically if No. 2 system fails, when flap is selected to greater than 0 degrees.

On aircraft with ModSum 4Q126354 or SB84-29-22 incorporated:

- PUSH ON segment (white)
- Turns PTU on
- Indicates PTU is operating
- PTU provides pressure to No. 2 system
- Hashed bar (green)
- Indicates PTU pressure switch senses a pressure of approximately 2700 psi (18615.8 kPa) in the No. 2 system.
- FAIL segment (amber)
- Indicates PTU pressure switch senses a pressure less than 1500 psi (10342.1 kPa) in the No. 2 system, and the PTU stops operating
- When not selected on, the PTU will turn on automatically if No. 2 system fails, when flap is selected to greater than 0 degrees.

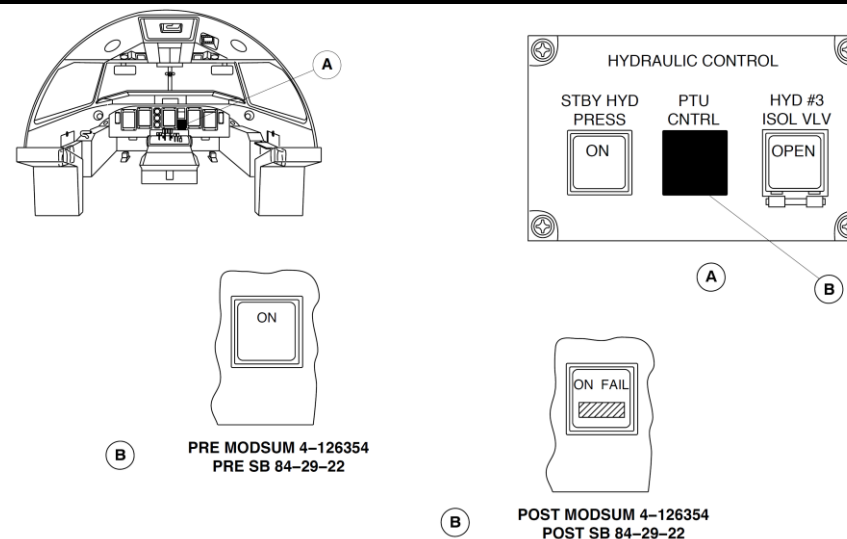


Figure 29-32 Hydraulic Panel Switches

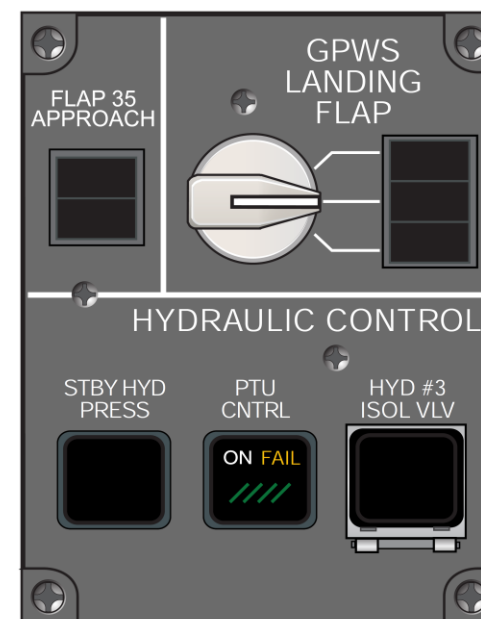
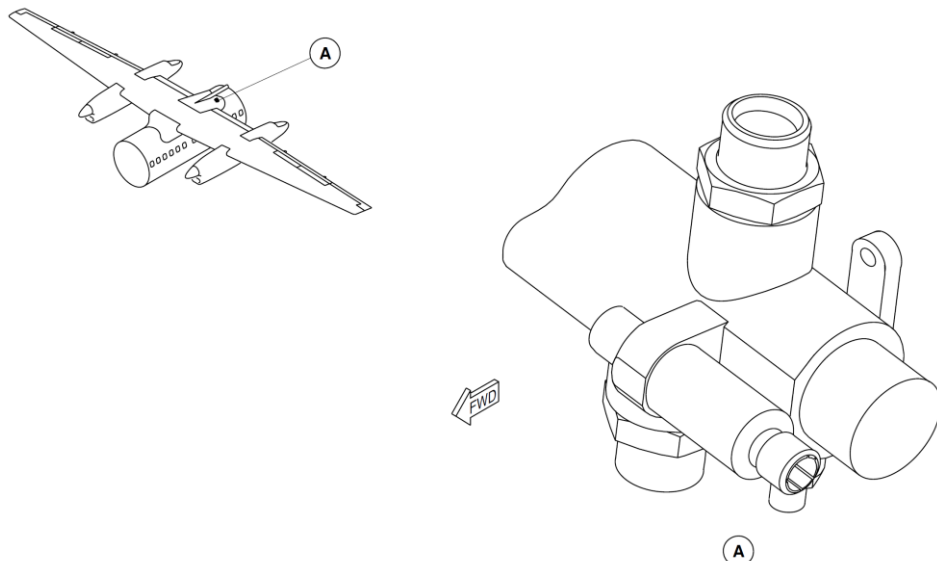


Figure 29-33 PTU Switchlight (Post Modsum 4Q126354)

## Hydraulic PTU Select Valve

[Refer Figure 29-34](#)

The No. 1 hydraulic system uses a solenoid-operated PTU select valve to control the operation of the PTU motor. The PTU select valve is a 3-way, 2-position, pilot-operated spool valve. The valve is normally closed and pressure-operated to the open position. The PTU select valve is installed in the fuselage-to-wing fairing.

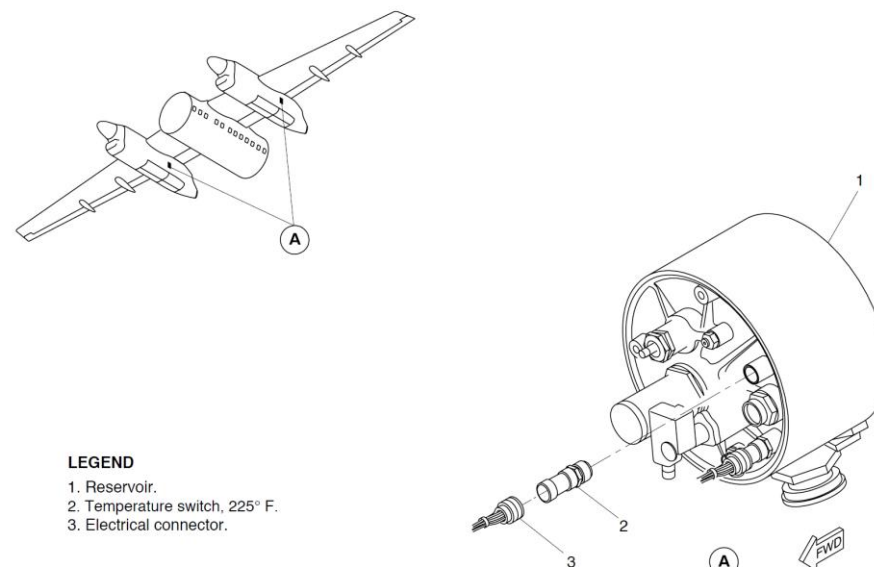


**Figure 29-34 Hydraulic PTU Select Valve**

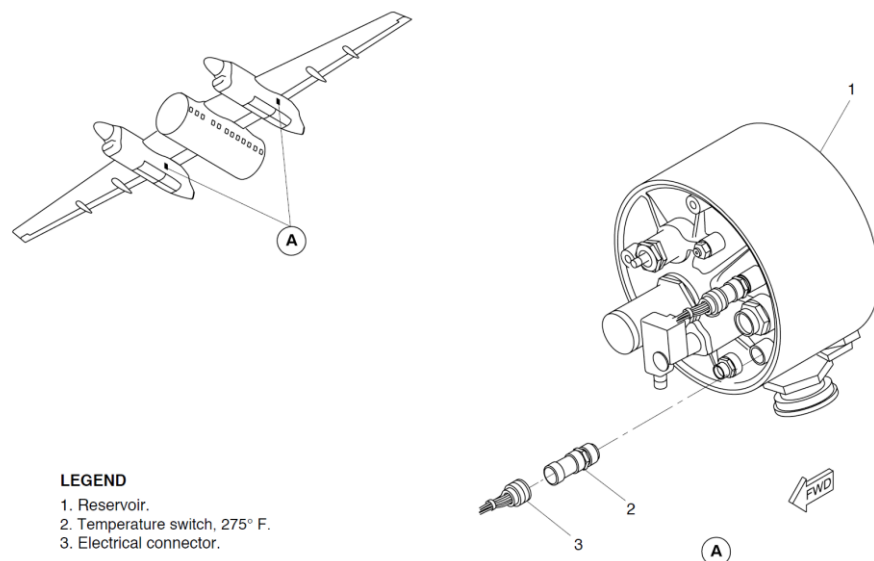
## No. 1 and No. 2 Hydraulic System Temperature Switches

[Refer Figures 29-35 & 29-36](#)

The No. 1 and No. 2 hydraulic systems each contain a 225 °F (107 °C) and a 275 °F (135 °C) temperature switch, installed on the hydraulic system reservoir. The switches monitor the temperature of the hydraulic fluid inside the reservoir. If fluid temperature reaches 225 °F (107 °C) the switch sends a signal to the No. 1 or No. 2 HYDRAFLUID HOT caution light. If fluid temperature reaches 275 °F (135 °C), the temperature switch sends a signal which closes the firewall shut-off valve.



**Figure 29-35 Temperature Switch – 225°F**



**Figure 29-36 Temperature Switch – 275°F**

## HYDRAULIC SYSTEM 1 RESERVOIR

### FILLING INSTRUCTIONS

USE ONLY APPROVED PHOSPHATE ESTER FLUID  
FILLING SOURCE MUST BE FILTERED AND LIMITED TO  
0.5 GPM AND 70 PSI

FOR COMPLETE INSTRUCTIONS SEE MAINTENANCE MANUAL CHAPTER 12

CAUTION: DO NOT OPERATE SYSTEM IF RESERVOIR HAS LESS THAN 2.3 QUARTS  
WITH SYSTEM UNPRESSURIZED

1. ATTACH THE FILLING SOURCE TO THE RESERVOIR FILL RECONNECTABLE FITTING.
2. NOTE AMBIENT TEMPERATURE AND VOLUME WITH NO SYSTEM PRESSURE.
3. FLUID SHOULD BE AT OR NEAR AMBIENT TEMPERATURE (AT LEAST ONE HOUR AFTER LAST OPERATION).
4. CHECK THAT MAIN BRAKES ARE NOT APPLIED AND THAT INBOARD FLIGHT SPOILERS ARE CLOSED.
5. WITH SYSTEM DEPRESSURIZED, ADD FLUID TO RESERVOIR TO FULL, 0 PSI LINE.
6. APPLY EXTERNAL ELECTRIC POWER.
7. START SPU. VERIFY THAT COCKPIT INDICATION SHOWS 2850 TO 3050 PSI.
8. CHECK RESERVOIR QUANTITY ON INDICATOR ON RESERVOIR.
9. IF CHANGE IN VOLUME (Step 8 minus Step 2) EXCEEDS 0.8 QT, BLEED SYSTEM PER MAINTENANCE MANUAL.
10. FLUID LEVEL MUST BE WITHIN FULL, 3000 PSI AND REFILL, 3000 PSI LIMITS FOR FLUID TEMPERATURE AS SHOWN ON GRAPH LOCATED NEAR RESERVOIR.
11. FILLING IS REQUIRED IF THE QUANTITY IS ON OR BELOW THE REFILL, 3000 PSI LINE.
12. IF FILLING IS REQUIRED, PUMP FLUID IN UNTIL THE QUANTITY IS JUST ABOVE THE FULL, 3000 PSI LINE.
13. OPERATE THE BLEED VALVE ON THE RESERVOIR UNTIL THE QUANTITY IS JUST ON THE FULL, 3000 PSI LINE.
14. REPEAT STEPS 12 AND 13 IF FLUID FROM RESERVOIR APPEARS TO HAVE AIR ENTRAINED AS SHOWN BY FROTHY, BUBBLY APPEARANCE.
15. TURN OFF SPU.
16. WHEN COMPLETE, DISCONNECT FILLING SOURCE AND REPLACE ALL PROTECTIVE CAPS.

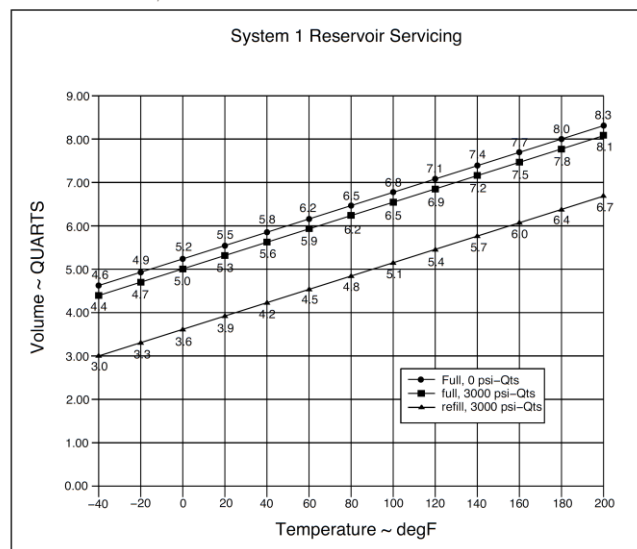


Figure 29-37 No. 1 System Reservoir Servicing Instructions

## HYDRAULIC SYSTEM 2 RESERVOIR

### FILLING INSTRUCTIONS

USE ONLY APPROVED PHOSPHATE ESTER FLUID  
FILLING SOURCE MUST BE FILTERED AND LIMITED TO  
0.5 GPM AND 70 PSI

FOR COMPLETE INSTRUCTIONS SEE MAINTENANCE MANUAL CHAPTER 12

CAUTION: DO NOT OPERATE SYSTEM IF RESERVOIR HAS LESS THAN 2.8 QUARTS  
WITH SYSTEM UNPRESSURIZED

1. ATTACH THE FILLING SOURCE TO THE RESERVOIR FILL RECONNECTABLE FITTING.
02. NOTE AMBIENT TEMPERATURE AND VOLUME WITH NO SYSTEM PRESSURE.
03. FLUID SHOULD BE AT OR NEAR AMBIENT TEMPERATURE (AT LEAST ONE HOUR AFTER LAST OPERATION).
04. VERIFY LANDING GEAR ARE CHOCKED.
05. OPERATE PARKING BRAKE 10 TIMES TO DEplete BRAKE ACCUMULATOR.
06. PRECHARGE BRAKE ACCUMULATOR PER PLACARD.
07. WITH LANDING GEAR EXTENDED CHECK THAT PARKING BRAKES ARE NOT APPLIED AND THAT OUTBOARD FLIGHT SPOILERS ARE CLOSED.
08. WITH SYSTEM DEPRESSURIZED, ADD FLUID TO RESERVOIR TO FULL, 0 PSI LINE.
09. APPLY EXTERNAL ELECTRIC POWER.
010. START SPU AND PTU. VERIFY THAT COCKPIT INDICATION SHOWS 2750 TO 3050 PSI FOR SYS2.
11. CHECK RESERVOIR QUANTITY ON INDICATOR ON RESERVOIR.
12. IF VOLUME CHANGE (Step 11 minus Step 2) EXCEEDS 1.3 QTS, BLEED SYSTEM PER MAINTENANCE MANUAL.
13. FLUID LEVEL MUST BE WITHIN FULL, 3000 PSI AND REFILL, 3000 PSI LIMITS FOR FLUID TEMPERATURE AS SHOWN ON GRAPH LOCATED NEAR RESERVOIR.
014. FILLING IS REQUIRED IF THE QUANTITY IS ON OR BELOW THEREFILL, 3000 PSI LINE.
015. IF FILLING IS REQUIRED, PUMP FLUID IN UNTIL THE QUANTITY IS JUST ABOVE THE FULL, 3000 PSI LINE.
016. OPERATE THE BLEED VALVE ON THE RESERVOIR UNTIL THE QUANTITY IS JUST ON THE FULL, 3000 PSI LINE.
014. REPEAT STEPS 15 AND 16 IF FLUID FROM RESERVOIR APPEARS TO HAVE AIR ENTRAINED AS SHOWN BY FROTHY, BUBBLY APPEARANCE.
015. TURN OFF SPU AND PTU.
016. WHEN COMPLETE, DISCONNECT FILLING SOURCE AND REPLACE ALL PROTECTIVE CAPS.

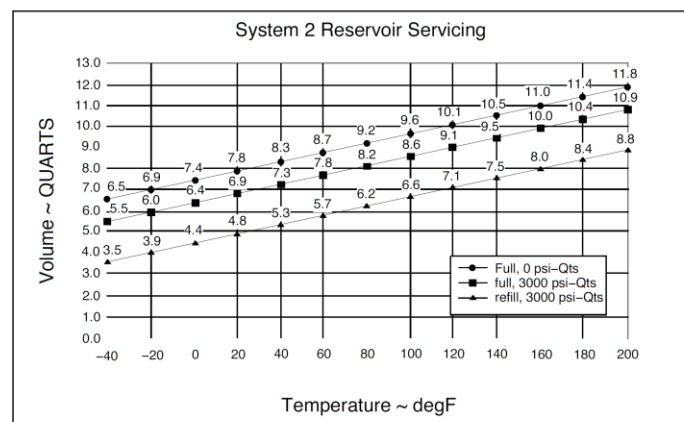


Figure 29-38 No.2 System Reservoir Servicing Instructions

## Main Hydraulic System 3

### Introduction

The No. 3 hydraulic system supplies emergency hydraulic power to the elevators.

### General Description

The No. 3 hydraulic system is installed in the aft fuselage. The system operates automatically during an emergency condition when the No. 1 and/or No. 2 hydraulic systems are unavailable, or when both engines indicate low oil pressure. The No. 3 hydraulic system can also be engaged manually when the HYD #3 ISOL VLV switch, located on the HYDRAULIC CONTROL Panel in the flight compartment, is set to OPEN. The No. 3 hydraulic system uses an accumulator which is pressurized by a DC Motor Driven Pump (DCMP).

The No. 3 hydraulic system contains the components that follow:

- No. 3 Hydraulic System Accumulator
- No. 3 Hydraulic Bootstrap Accumulator
- No. 3 Hydraulic System DCMP
- No. 3 Hydraulic System Accumulator Isolation Valve
- No. 3 Hydraulic System Reservoir
- No. 3 Hydraulic System Pressure Filter
- No. 3 Hydraulic System Ecology Bottle
- No. 3 Hydraulic System Pressure Transducer
- No. 3 Hydraulic System Pressure Switches

### System Operation

[Refer Figure 29-39](#) & [29-40](#)

The No. 3 hydraulic system is an independent system. It supplies hydraulic power only to the left and right elevator inboard PCUs if No.1 and No. 2 hydraulic systems fail. The system operates automatically during an emergency condition when the No. 1 and/or No. 2 hydraulic systems fail, or if a dual engine failure occurs. The No. 3 hydraulic system can also be engaged manually by pushing the HYD #3 ISOL VLV switchlight on the HYDRAULIC

CONTROL panel. An amber OPEN legend on the switchlight will turn on. An accumulator and an isolation valve are also installed in the No. 3 hydraulic system.

A 28 volt DC Motor Driven Pump (DCMP) operates automatically to pressurize the accumulator and keep the accumulator pressurized from 2600 to 3250 psi (17926 to 22408 kPa). When the DCMP is not operating, the accumulator holds a reserve of pressure. The volume of the No. 3 system reservoir is 2.6 qt (2.46 L). The DCMP operates intermittently and is controlled by two pressure switches installed on the accumulator isolation valve. One pressure switch signals the DCMP to operate if system pressure drops to 2600 psi (17926 kPa) and any one of the following conditions apply:

- No. 3 hydraulic system isolation valve is de-energized, or
- Aircraft is in-Air mode with the parking brake not in "PARK", or
- Either No. 1 or No. 2 engine oil pressure switches indicate high pressure with the parking brake not in "PARK".

The pressure switch commands the DCMP to turn off when system pressure reaches 3250 psi (22408 kPa). The other switch turns on the #3 HYD PUMP caution light if system pressure falls to 900 psi (6205 kPa), or the DCMP has been operating for longer than 60 seconds on the ground. Electrical power is supplied to the DCMP by the standby battery.

On aircraft with SB84-29-12 or Modsum 4-126281 incorporated:

The automatic operation of the DCMP is inhibited when the parking brake is set to 'PARK'.

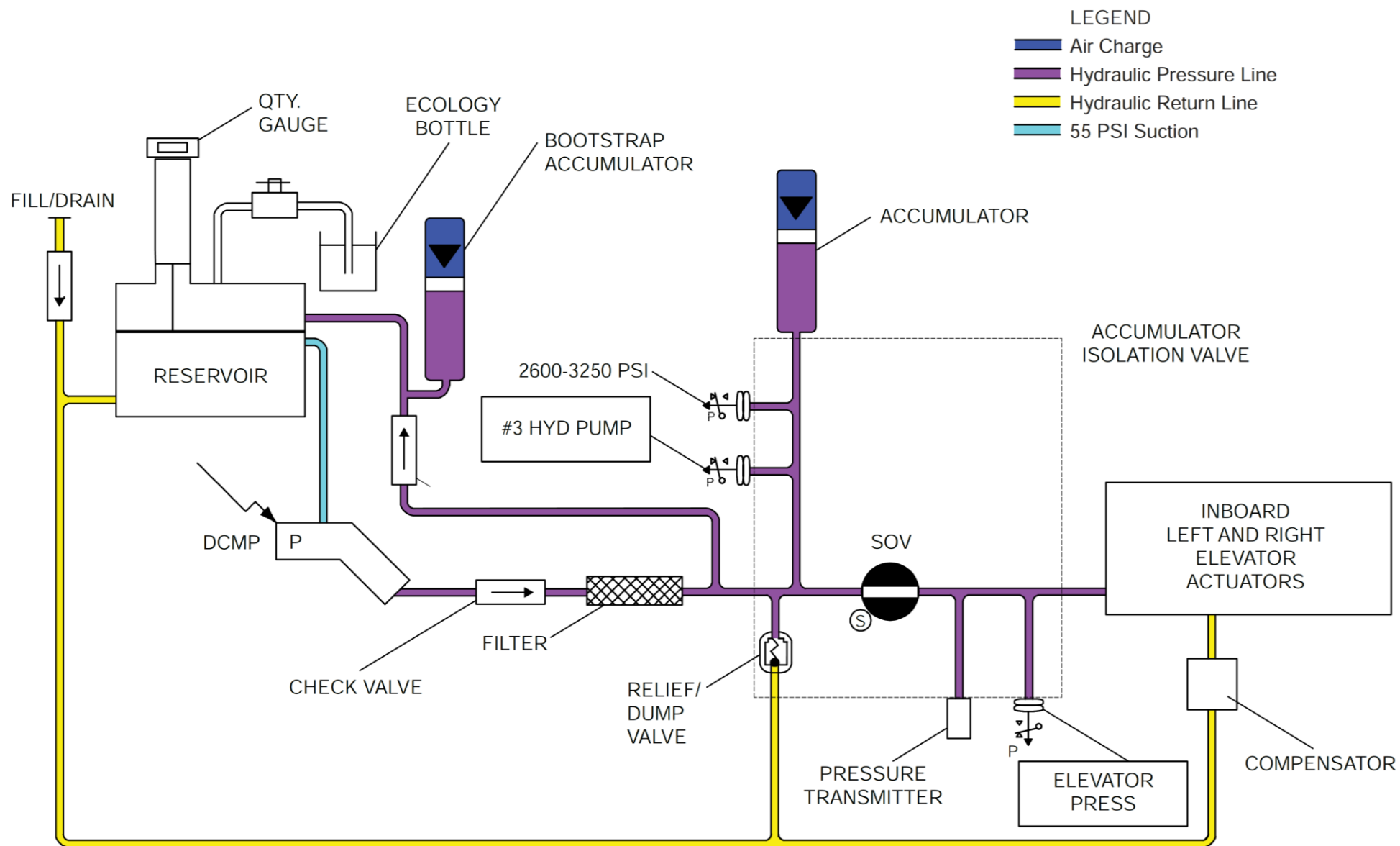


Figure 29-39 No.3 Hydraulic System Synoptic (Pre Modsum 4-190397)

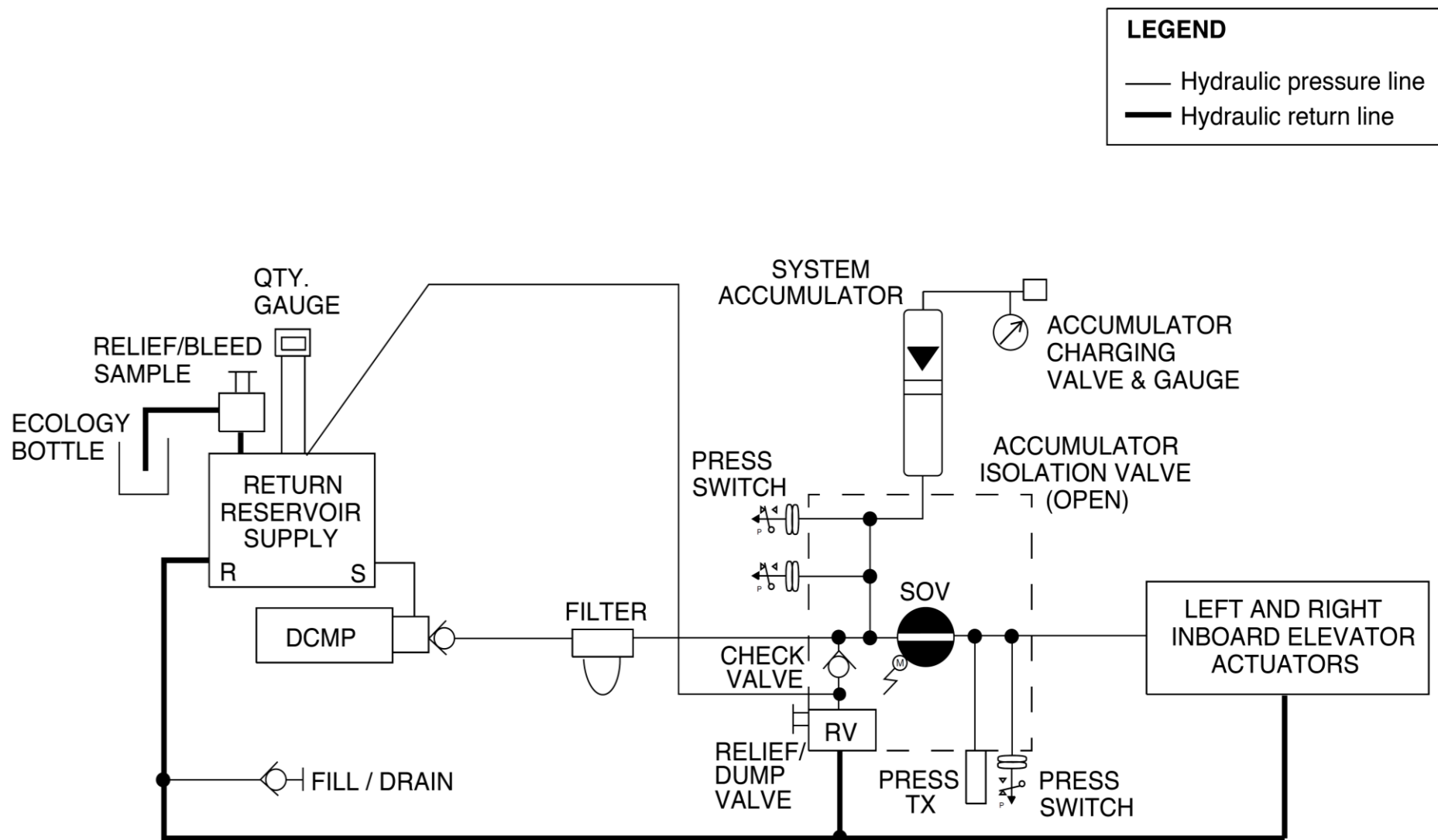


Figure 29-40 No.3 Hydraulic System Synoptic (Post Modsum 4-190397)

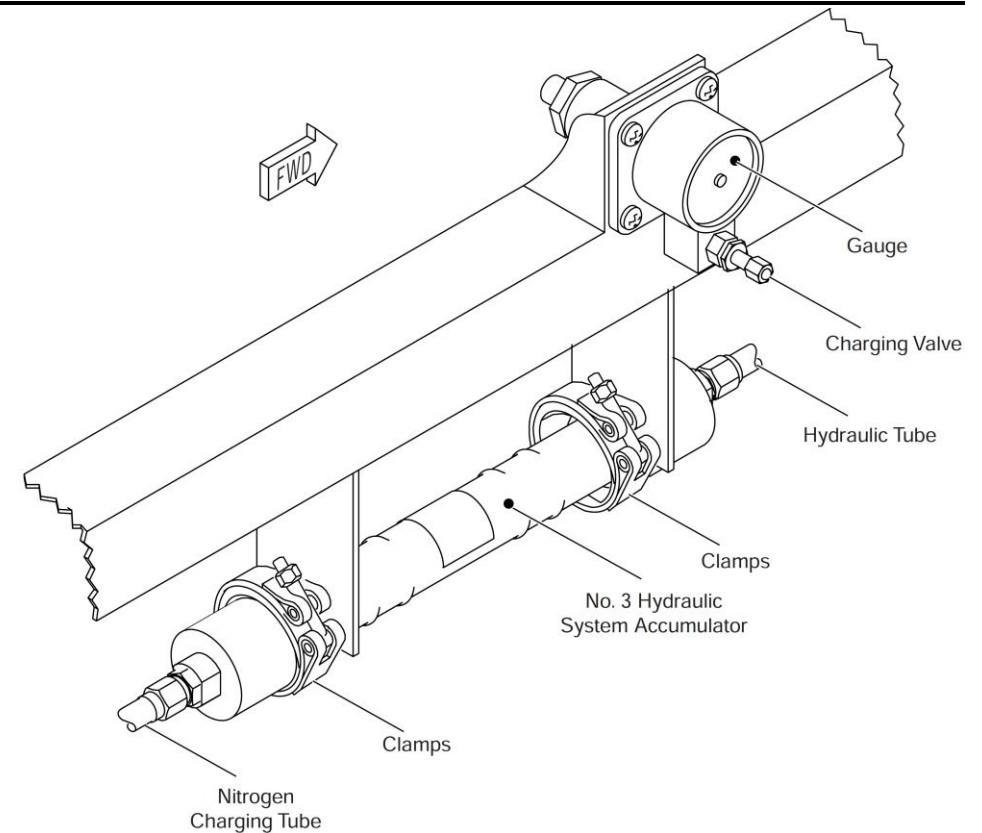


### No.3 Hydraulics System – Component Description

#### No. 3 Hydraulic System Accumulator

[Refer Figure 29-41](#)

The No. 3 hydraulic system accumulator keeps the system pressurized during normal flight operation. The accumulator is charged with dry nitrogen gas. The capacity of No. 3 hydraulic system accumulator is 50 in<sup>3</sup> (819 cm<sup>3</sup>). There is an accumulator charging-gauge manifold to charge the accumulator. The accumulator charging-gauge manifold has a pressure gauge and a charging valve. It has an attachment for an external dry nitrogen source to supply the gas precharge to the accumulator. A service placard, attached to the charging manifold, provides accumulator charging instructions. Access to the charging-gauge manifold is through a door on the bottom of the rear fuselage.



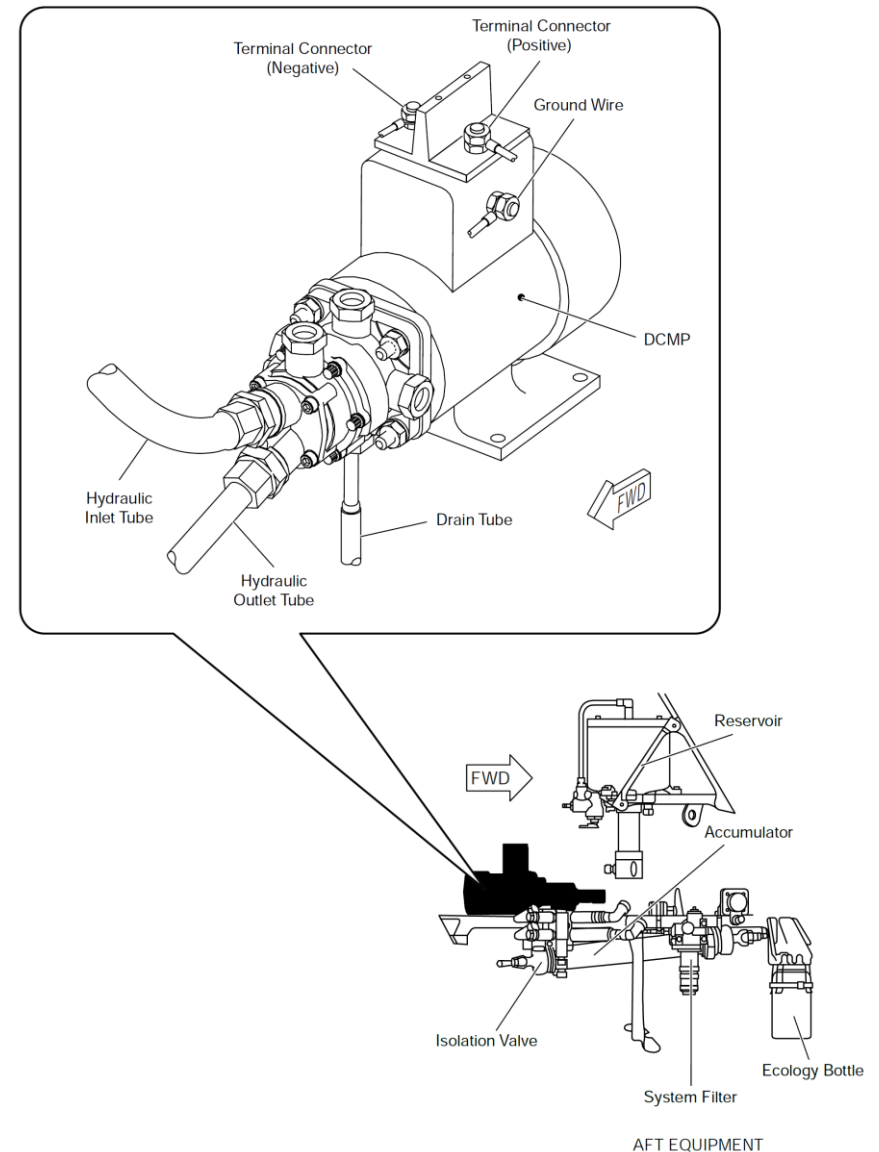
**Figure 29-41 No.3 Hydraulic System Accumulator**

## No. 3 Hydraulic System DCMP

Refer Figure 29-42

The DC Motor Pump (DCMP) is used in the No. 3 hydraulic system, with the accumulator, to pressurize the No. 3 system. Electrical power is supplied to the DCMP by the STBY battery through the dc battery bus. The DCMP will operate intermittently during normal flight to maintain a No. 3 system pressure range of 2600 to 3250 psi (17926 to 22407 kPa). DCMP operation is controlled by two pressure switches installed on the accumulator isolation valve. On aircraft with ModSum 4-113673 or SB84-29-24 incorporated, the inlet tube is aligned vertically above the outlet tube.

**NOTE:** The DCMP will operate intermittently on the ground when the dc battery bus is energized by the batteries or external ground power. On aircraft with SB84-29-12 or Modsum 4-126281 incorporated: The automatic operation of the DCMP is inhibited when the parking brake is set to 'PARK'.

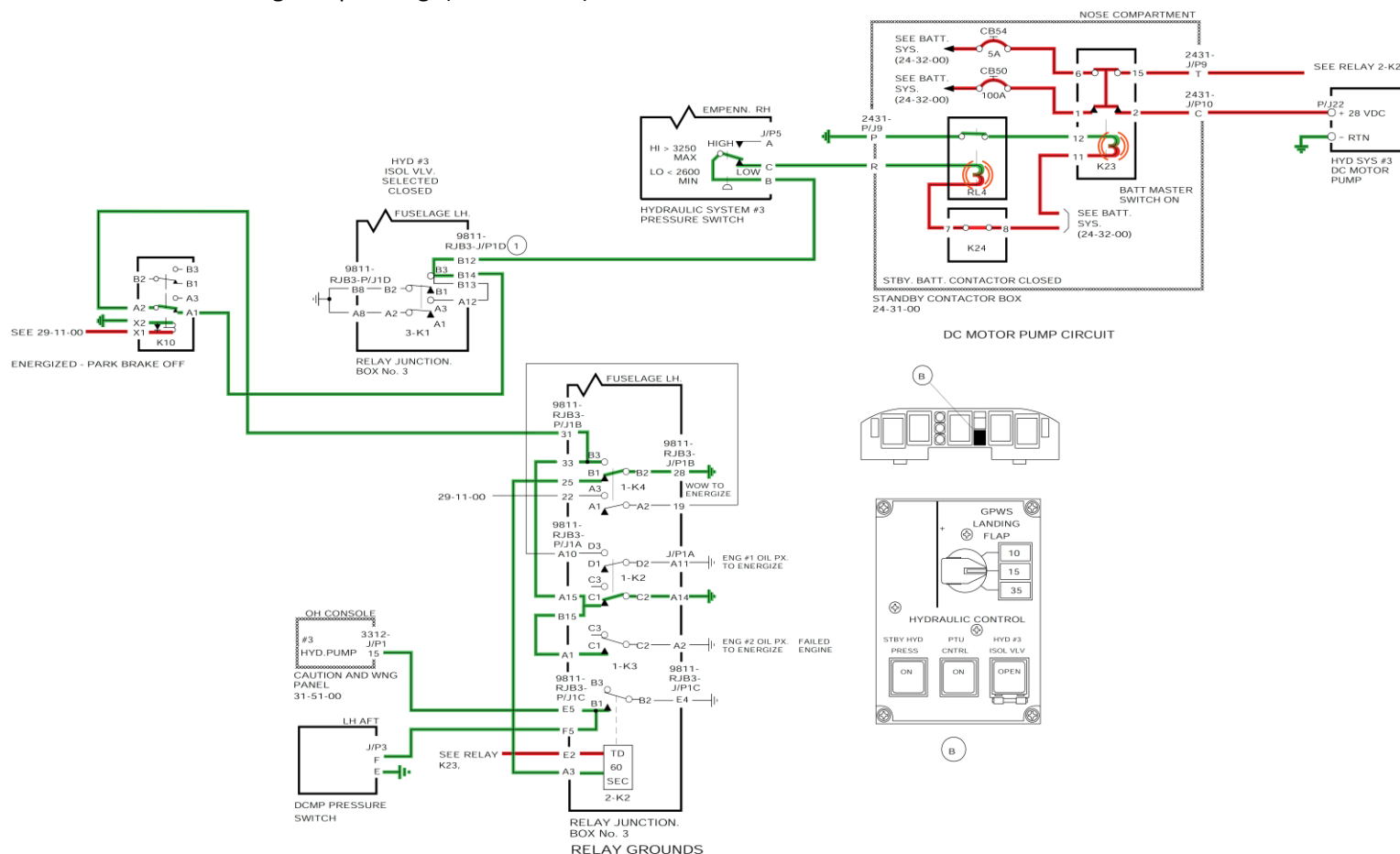


**Figure 29-42 DC Motor Pump**

## DCMP Operation (Park Brake Off - WOW - Single Engine (No.1) Operating) – Electrical Schematic

[Refer Figure 29-43](#)

Hydraulic system No.3 Pressure switch is low and relay K10 is energized by the Park Brake selected OFF and No.1 engine operating. (Oil Pressure).



**Figure 29-43 DCMP Operation (Park Brake Off - WOW - Single Engine (No.1) Operating) – Electrical Schematic**

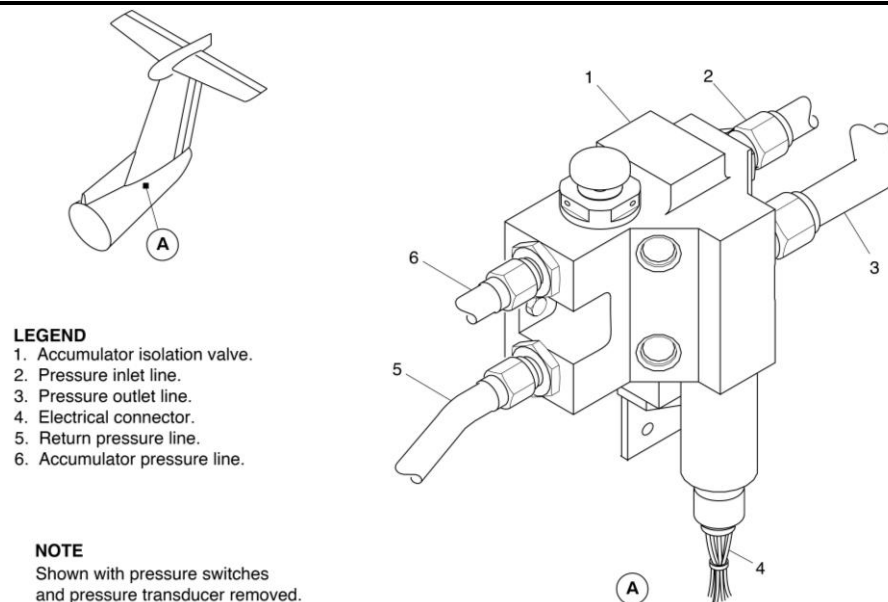
### No. 3 Hydraulic System Accumulator Isolation Valve

Refer Figure 29-44 & 29-45

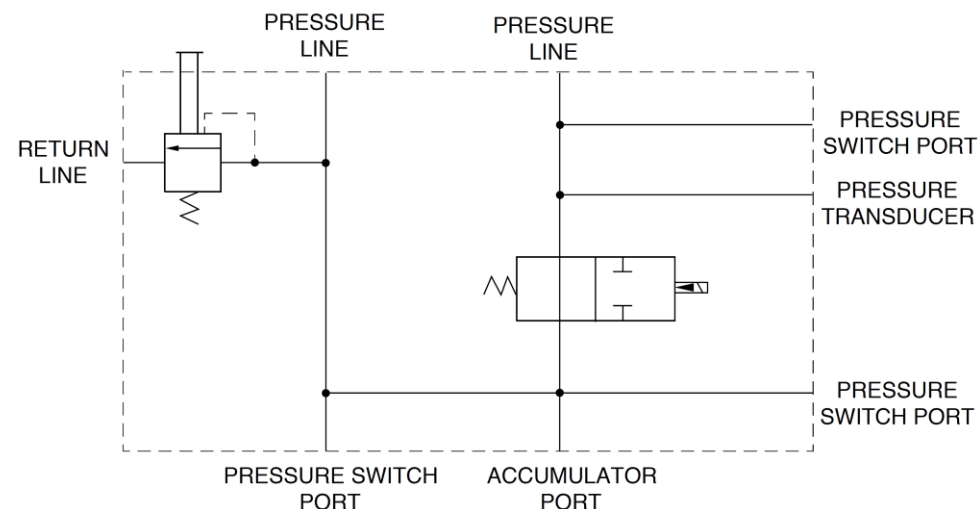
The isolation valve is used in the No. 3 hydraulic system to isolate the elevators from No. 3 hydraulic system pressure. During normal flight operation, the system is in an active standby mode with the accumulator isolation valve (energized) closed. When open, the isolation valve allows hydraulic pressure from the No. 3 hydraulic system to power the elevators. The isolation valve will open in flight if No. 1 and/or No. 2 hydraulic system pressure is lost, or, if No. 1 and No. 2 engines fail. The isolation valve can be manually opened when the HYD #3 ISOL VLV switchlight is pushed. An amber OPEN legend on the switchlight will turn on.

An additional pressure switch is installed downstream of the isolation valve. It turns on the ELEVATOR PRESS caution light if No. 1, No. 2 and No. 3 hydraulic systems are supplying power to all six elevator actuators. If this occurs, airspeed should be reduced to 200 kts (370.6km/hr) indicated air speed. If the isolation valve malfunctions open, the No. 3 hydraulic system will supply hydraulic power to the elevators, even though No. 1 and No. 2 hydraulic systems are operative. The ELEVATOR PRESS caution light will turn on. The OPEN legend in the switchlight will not turn on. The No. 3 hydraulic system supplies hydraulic pressure to both elevators when:

- No. 1 and No. 2 hydraulic systems fail
- No. 1 or No. 2 hydraulic system fails
- No. 1 and No. 2 engines fail.



**Figure 29-44 No.3 System Isolation Valve**



**Figure 29-45 No. 3 Hydraulic System Accumulator Isolation Valve/Schematic**

# No.3 Hydraulic System Isolation Valve (Open) Automatically (Due to No.1 Hydraulic System Failure In Flight) – Electrical Schematic

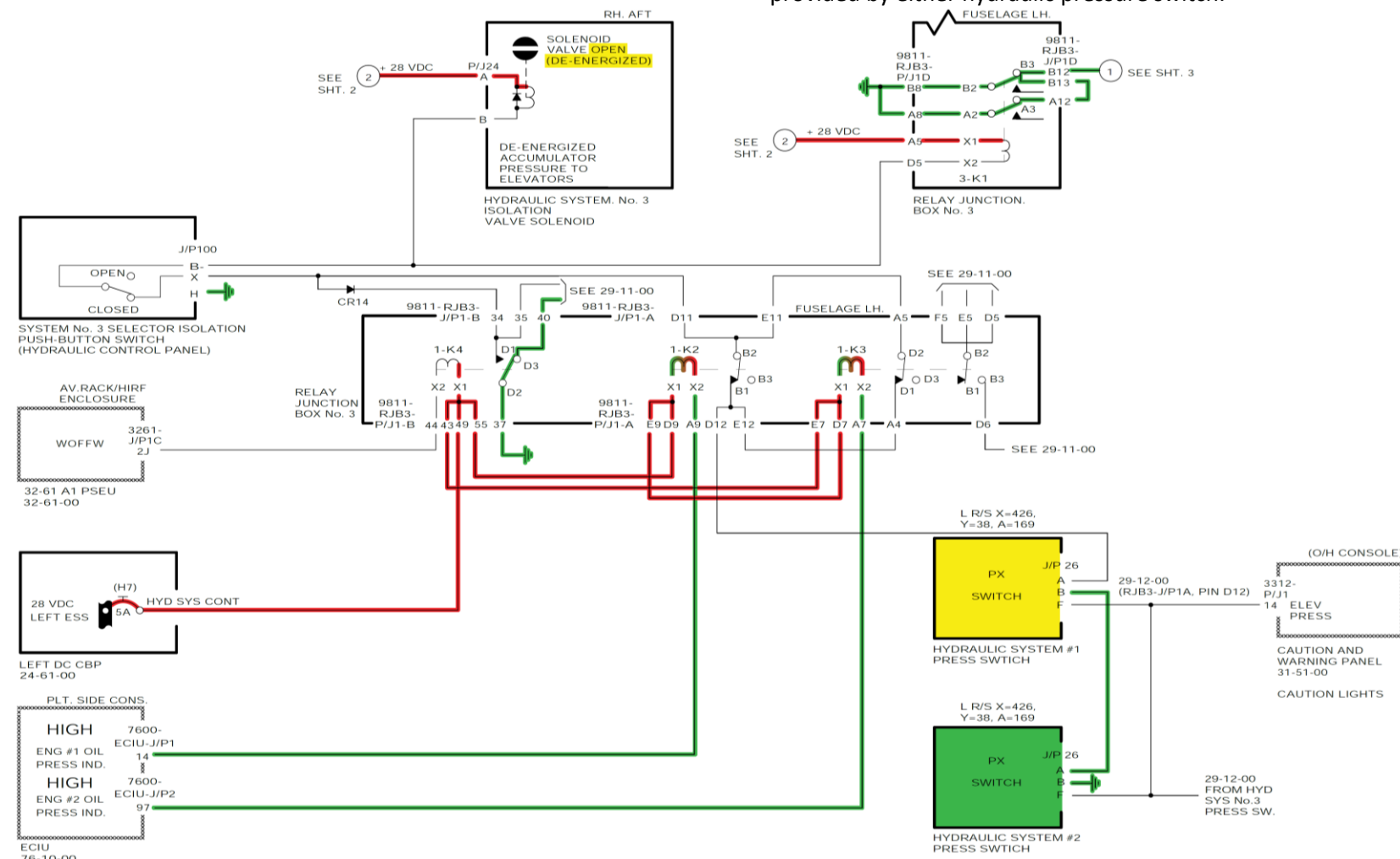
## No.3 Hydraulic System Isolation Valve (Open) Automatically (Due to No.1 Hydraulic System Failure In Flight) – Electrical Schematic

[Refer Figure 29-46](#)

## No.3 Hydraulic System Isolation Valve (Open) Automatically

**Condition:** Due to No.1 Hydraulic System Failure in Flight

The isolation valve will de-energize (Open) when the ground is no longer provided by either hydraulic pressure switch.



**Figure 29-46 No.3 Hydraulic System Isolation Valve (Open) Automatically - Due to No.1 Hydraulic System Failure In Flight**

## No.3 Hydraulic System Isolation Valve (Open) Automatically (Due to a Loss of Both Engines In Flight) – Electrical Schematic

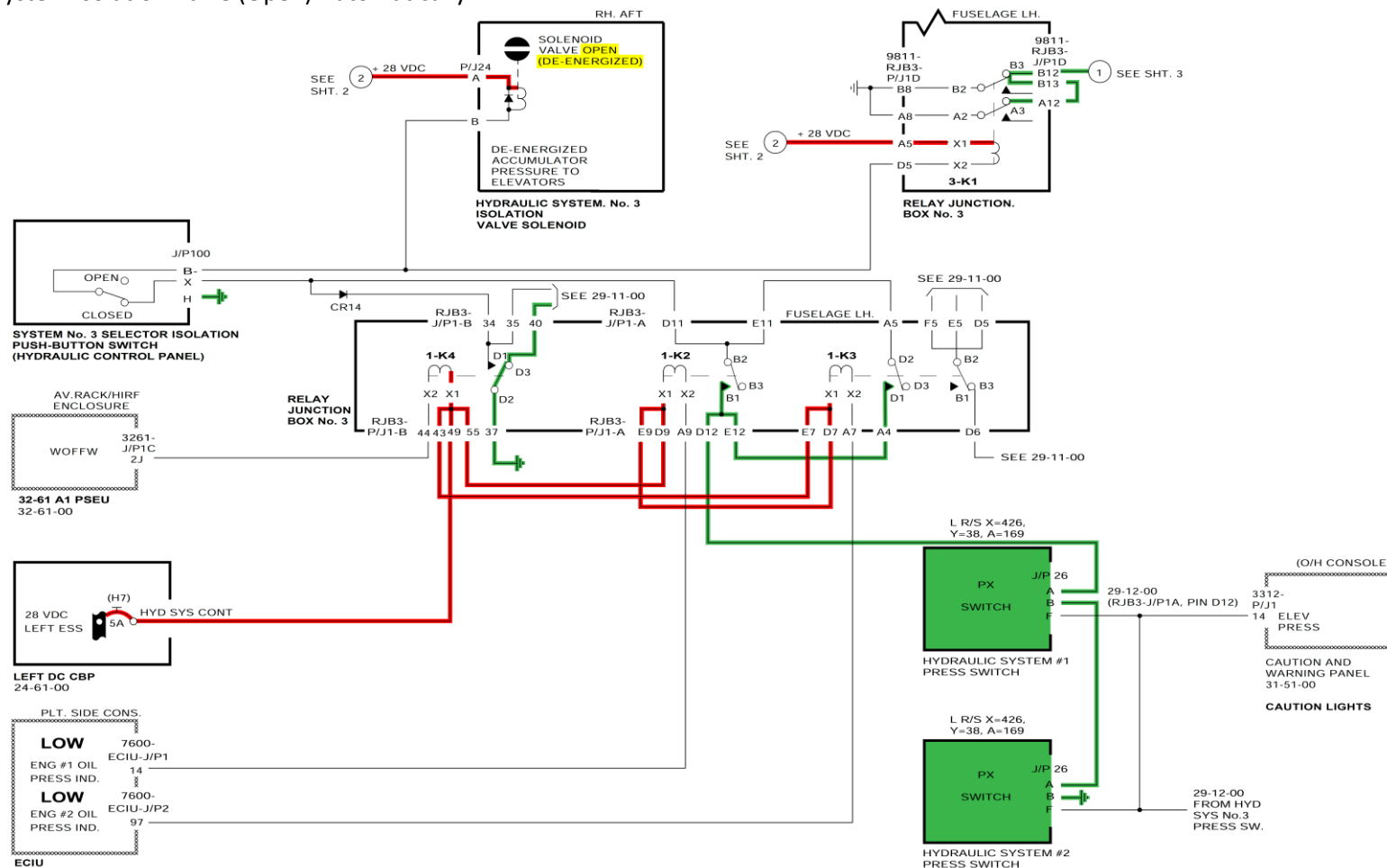
### No.3 Hydraulic System Isolation Valve (Open) Automatically (Due to a Loss of Both Engines In Flight) – Electrical Schematic

[Refer Figure 29-47](#)

No.3 Hydraulic System Isolation Valve (Open) Automatically

**Condition:** Due to a Loss of Both Engines in Flight.

The isolation valve will de-energize (Open) when the ground used to energize the 1-K2 and the 1-K3 relays is no longer provided, due to loss of oil pressure from both engines.

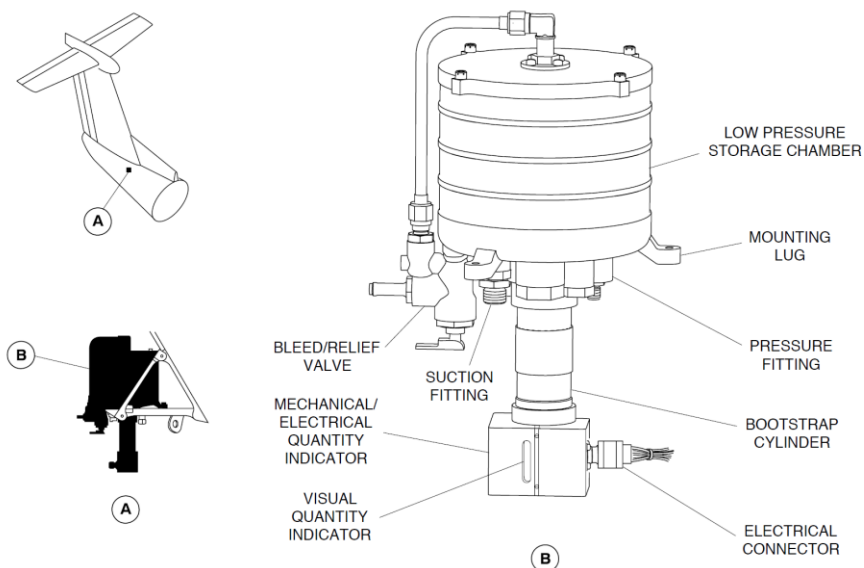


**Figure 29-47 No.3 Hydraulic System Isolation Valve (Open) Automatically (Due to a Loss of Both Engines In Flight) – Electrical Schematic**

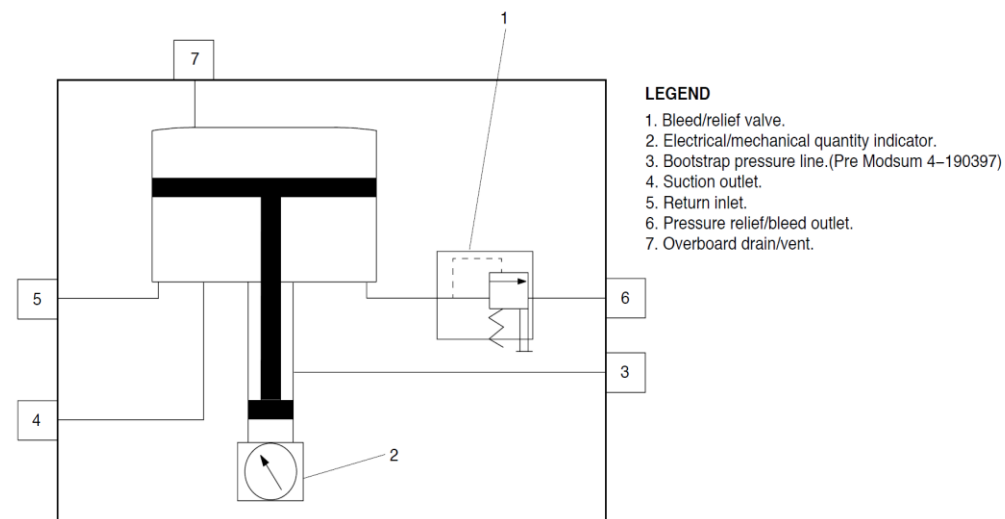
## No. 3 Hydraulic System Reservoir

Refer Figures 29-48 & 29-49

The No. 3 hydraulic system reservoir supplies the necessary fluid volume to the hydraulic system. The reservoir is a bootstrap type that controls system return pressure. A bleed/relief valve is installed in the reservoir to provide manual bleeding and pressure relief of the hydraulic system. The volume of the No. 3 hydraulic system reservoir is 150 in<sup>3</sup> (2458 cm<sup>3</sup>). The major components attached to the reservoir are bootstrap cylinder, a bleed/relief valve, and a mechanical/electrical quantity indicator.



**Figure 29-48 No.3 System Reservoir**



**Figure 29-49 No. 3 Hydraulic System Reservoir/Schematic**

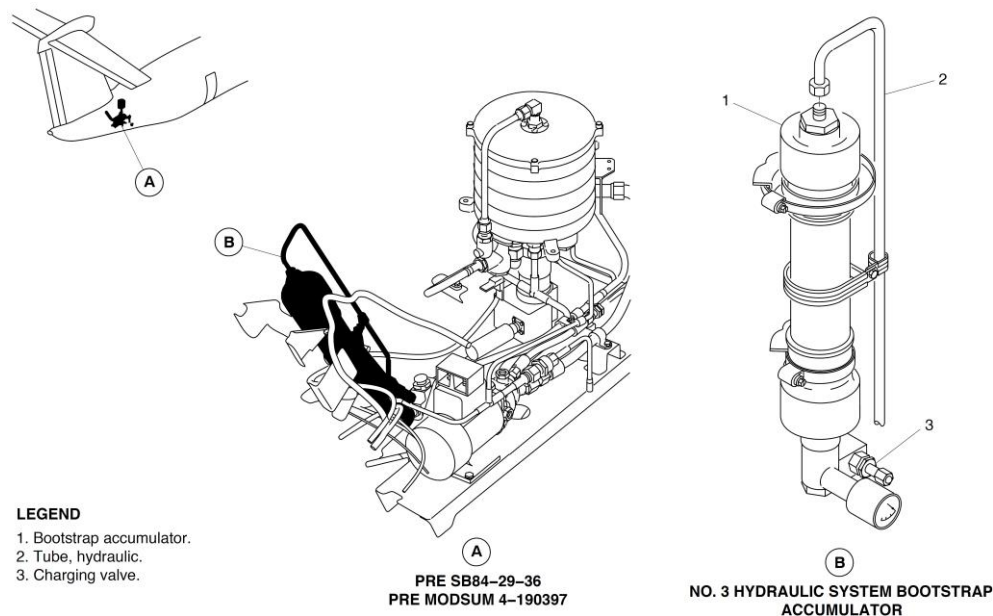


### No.3 Hydraulic Bootstrap Accumulator

[Refer Figure 29-50](#)

The bootstrap accumulator provides force to pressurize the No. 3 hydraulic system reservoir. The bootstrap contains a small-area piston exposed to the system pressure, nominally 3000 psi (20684 kPa), which provides force to drive the low-pressure piston in the reservoir and keeps it pressurized. The bootstrap accumulator in association with the relief/dump valve and check valve makes sure that the reservoir remains pressurized after overnight storage. The bootstrap piston uses two-stage head seals with the interstage cavity vented to return pressure. This arrangement keeps the sealing surface wet and limits the leakage in the event of a seal failure. The proof pressure capability of the bootstrap chamber and associated fittings/components is 4500 psi (31026 kPa) and a capacity of 25 in<sup>3</sup> (410 cm<sup>3</sup>).

On aircraft with ModSum 4-190397 or SB84-29-36 incorporated, bootstrap accumulator is not installed.

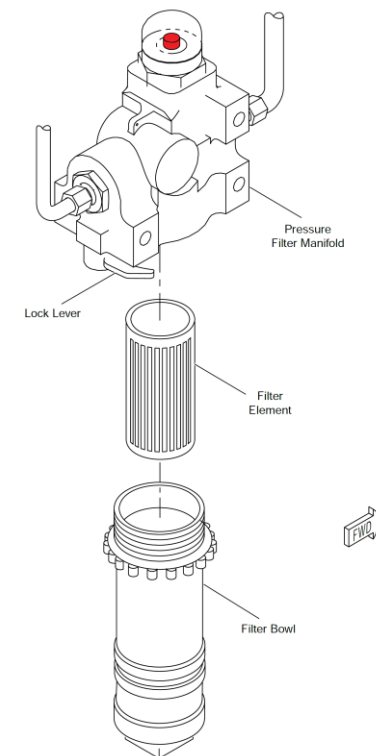


**Figure 29-50 No.3 System Bootstrap Accumulator**

### No. 3 Hydraulic System Pressure Filter

[Refer Figure 29-51](#)

The No. 3 hydraulic system pressure filter removes 100% of all hydraulic fluid contamination particles larger than 15 microns. This assembly filters the supply pressure fluid coming from the DCMP. It contains a filter element and bowl, and a differential pressure indicator. The pressure filter is a non-bypass type. A shut-off device stops fluid flow through the filter when the filter bowl is removed. A Differential Pressure Indicator (DPI) is installed on the pressure filter to give an indication of a clogged filter element. The No. 3 hydraulic system pressure filter bowl is the same as the case drain filter bowls of the No. 1 and No. 2 hydraulic system filter manifolds. These bowls are fully interchangeable.



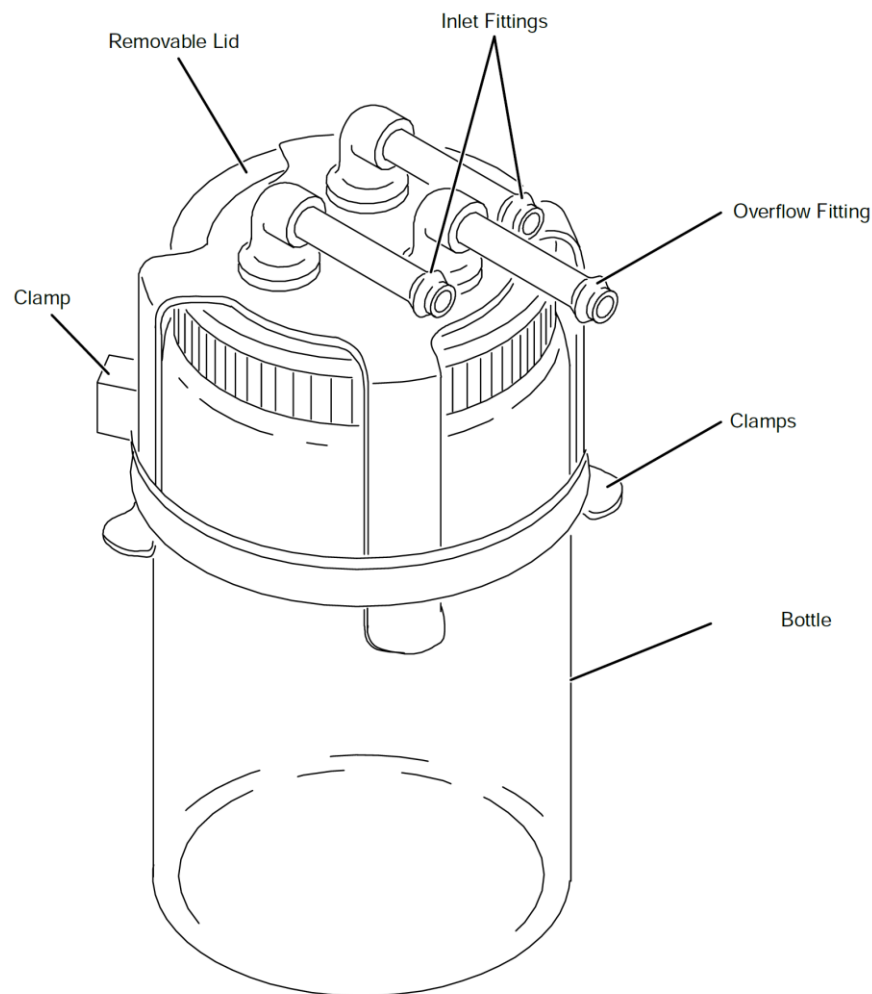
**Figure 29-51 No.3 Hydraulic System Pressure Filter**



## No. 3 Hydraulic System Ecology Bottle

[Refer Figure 29-52](#)

The No. 3 hydraulic system ecology bottle collects hydraulic fluid released from the reservoir bleed/relief valve and leakage from the DCMP shaft seal



**Figure 29-52 No.3 Hydraulic System Ecology Bottle**

## No. 3 Hydraulic System Pressure Transducer

[Refer Figure 29-53](#)

The No. 3 hydraulic system has one pressure transducer installed on the accumulator isolation valve which sends system pressure data to the MFD.

## No. 3 Hydraulic System Pressure Switches

[Refer Figure 29-53](#)

There are three pressure switches installed on the accumulator isolation valve. Two of these pressure switches control the operation of the DCMP. One of the pressure switches signals the DCMP to operate if system pressure drops to 2600 psi (17926 kPa) and any of the conditions that follow exists:

- No. 3 hydraulic system isolation valve is de-energized, or
- Aircraft is in-Air mode with the parking brake not in "PARK", or
- Either No. 1 or No. 2 engine 2 oil pressure switches indicate high pressure with the parking brake not in "PARK".

The pressure switch commands the DCMP to turn off when system pressure reaches 3250 psi (22408 kPa). The other pressure switch operates the #3 HYD PUMP caution light if the system pressure falls to 900 psi (6205 kPa), or the pump has been operating for longer than 60 seconds on the ground. An additional pressure switch is installed downstream of the isolation valve and operates the ELEVATOR PRESS caution light on the Caution and Warning panel if all three systems are supplying power to the elevator actuators.

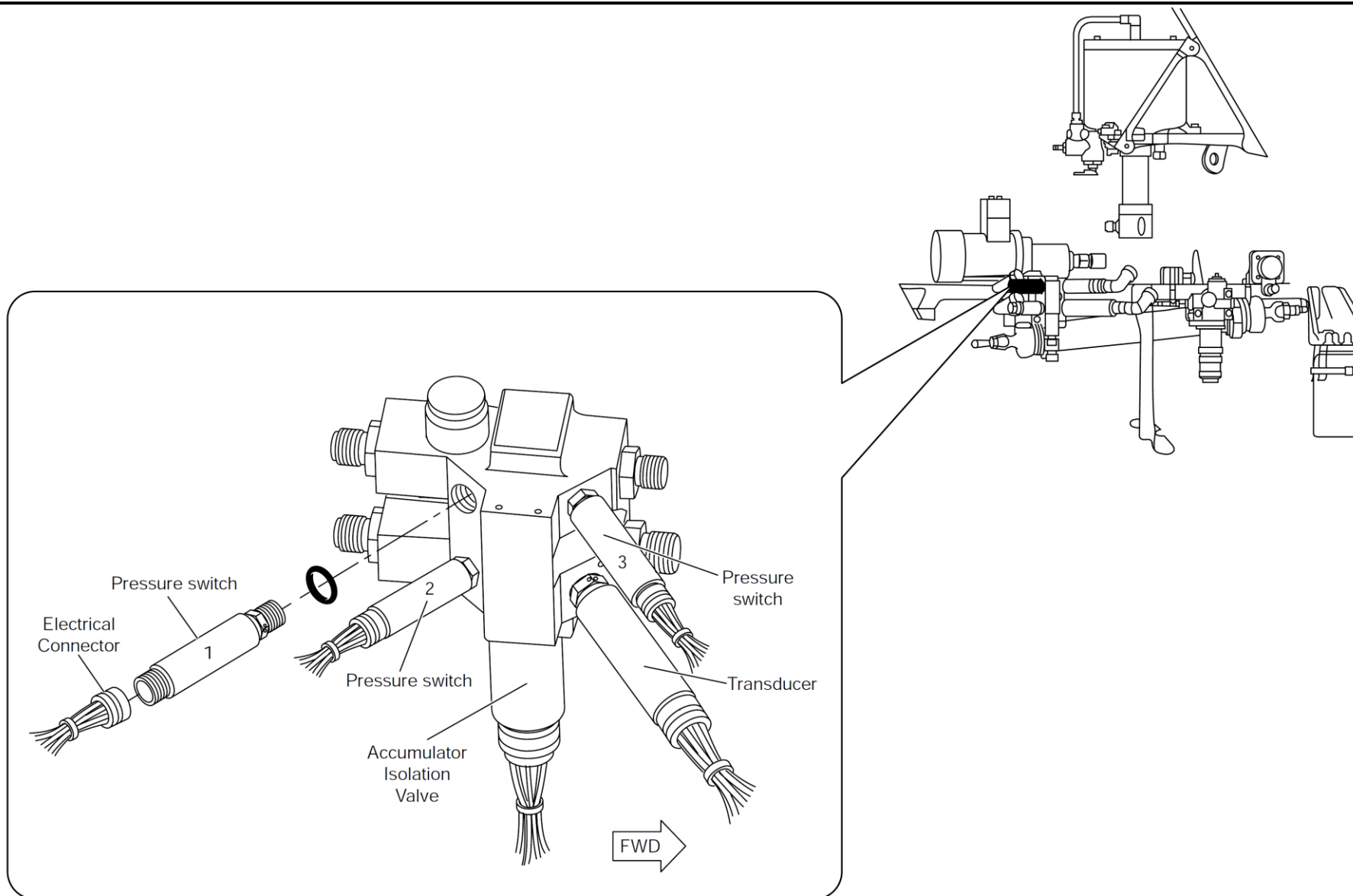


Figure 29-53 No.3 System Pressure Switches & Transducer

### HYD #3 ISOL VLV Switch light (amber)

[Refer Figure 29-54](#)

- PUSH OPEN segment amber
- Opens isolation valve
- Indicates isolation valve is open
- No. 3 system is powering the elevators
- When not selected on, the isolation valve will open automatically if No. 1 and/or No. 2 system fails

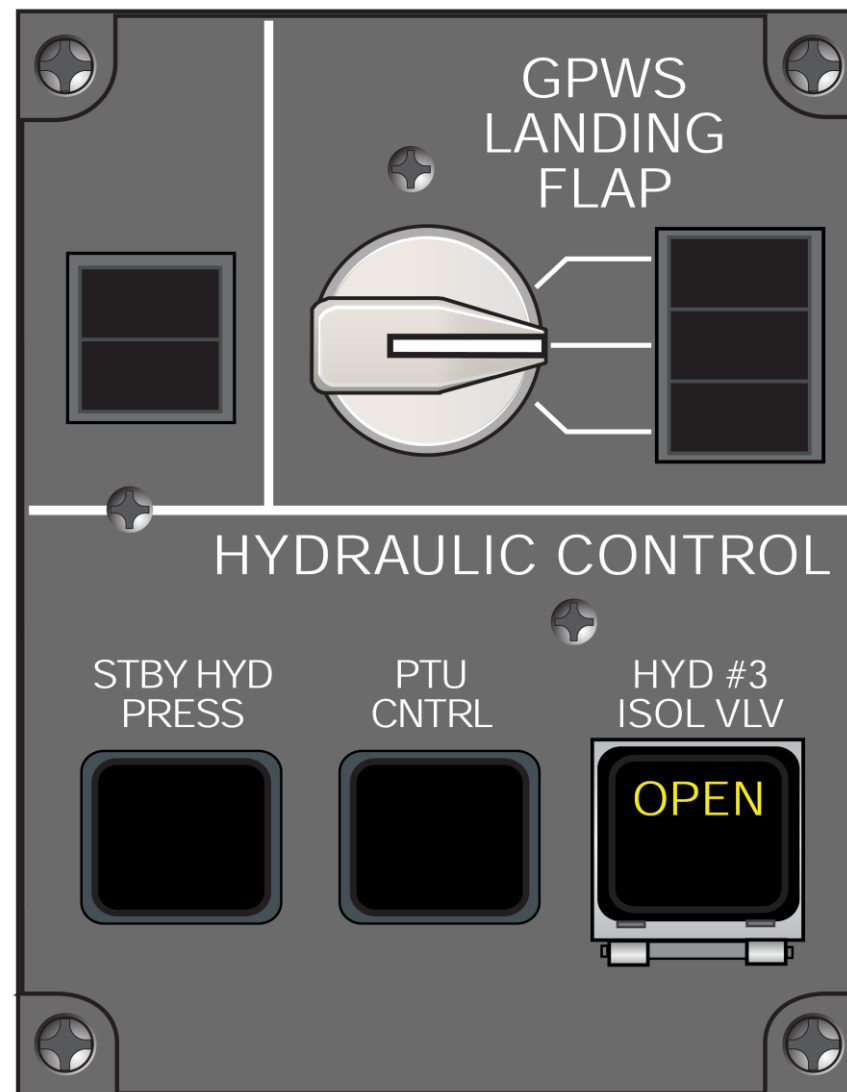


Figure 29-54 Hydraulic Control Panel

# HYDRAULIC SYSTEM 3 RESERVOIR

## FILLING INSTRUCTIONS

USE ONLY APPROVED PHOSPHATE ESTER FLUID  
FILLING SOURCE MUST BE FILTERED AND LIMITED TO  
0.5 GPM AND 70 PSI

FOR COMPLETE INSTRUCTIONS SEE MAINTENANCE MANUAL CHAPTER 12

CAUTION: DO NOT OPERATE SYSTEM IF RESERVOIR IS LESS THAN  
45 PERCENT FULL WITH SYSTEM UNPRESSURIZED

1. ATTACH THE FILLING SOURCE TO THE RESERVOIR FILL RECONNECTABLE FITTING.
- 2. NOTE PERCENT FULL AND AMBIENT TEMPERATURE.
- 3. FLUID SHOULD BE AT OR NEAR AMBIENT TEMPERATURE (AT LEAST ONE HOUR AFTER LAST OPERATION).
- 4. SERVICE SYSTEM 3 ACCUMULATOR PER LABEL NEAR ACCUMULATOR.
- 5. WITH SYSTEM DEPRESSURIZED ADD FLUID TO RESERVOIR TO FULL 0 PSI LINE.
- 6. APPLY EXTERNAL DC ELECTRIC POWER.
- 7. START DCMP. WHEN DCMP SHUTS OFF, VERIFY THAT COCKPIT INDICATION SHOWS 2600 TO 3250 PSI.
- 8. CHECK RESERVOIR QUANTITY ON INDICATOR ON RESERVOIR.
- 9. IF CHANGE IN VOLUME (Step 8 minus Step 2) EXCEEDS 25 PERCENT, FROM ZERO PRESS TO PRESSURIZED, BLEED SYSTEM PER MAINTENANCE MANUAL.
- 10. FLUID LEVEL MUST BE WITHIN FULL, 3000 PSI AND REFILL, 3000 PSI BAND LIMITS FOR FLUID TEMPERATURE AS SHOWN ON GRAPH BELOW.
- 11. FILLING IS REQUIRED IF THE QUANTITY IS ON OR BELOW THEREFILL, 3000 PSI LINE.
- 12. IF FILLING IS REQUIRED, PUMP FLUID IN UNTIL THE QUANTITY IS JUST ABOVE THE FULL, 3000 PSI LINE.
- 13. OPERATE THE BLEED VALVE ON THE RESERVOIR UNTIL THE QUANTITY IS JUST ON THE FULL, 3000 PSI LINE.
- 14. REPEAT STEPS 11 AND 12 IF FLUID FROM RESERVOIR APPEARS TO HAVE AIR ENTRAINED AS SHOWN BY FROTHY, BUBBLY APPEARANCE.
- 15. TURN OFF DCMP.
- 16. WHEN COMPLETE, DISCONNECT FILLING SOURCE AND REPLACE ALL PROTECTIVE CAPS.

Figure 29-55 Hydraulic System No.3 Filling Instruction

## AUXILIARY HYDRAULIC SYSTEMS

### Introduction

The alternate landing gear extension system supplies hydraulic power to extend the main landing gears when main hydraulic power is not available.

### General Description

The alternate landing-gear hand pump is located below an access door in the floor of the flight compartment. The hand pump must be used to extend the landing gear following isolation, or loss, of the No. 2 hydraulic system. The pump draws hydraulic fluid from an auxiliary reservoir located in the nose compartment. Hydraulic fluid pressurizes the alternate landing-gear actuators to extend the gear to the down-lock position.

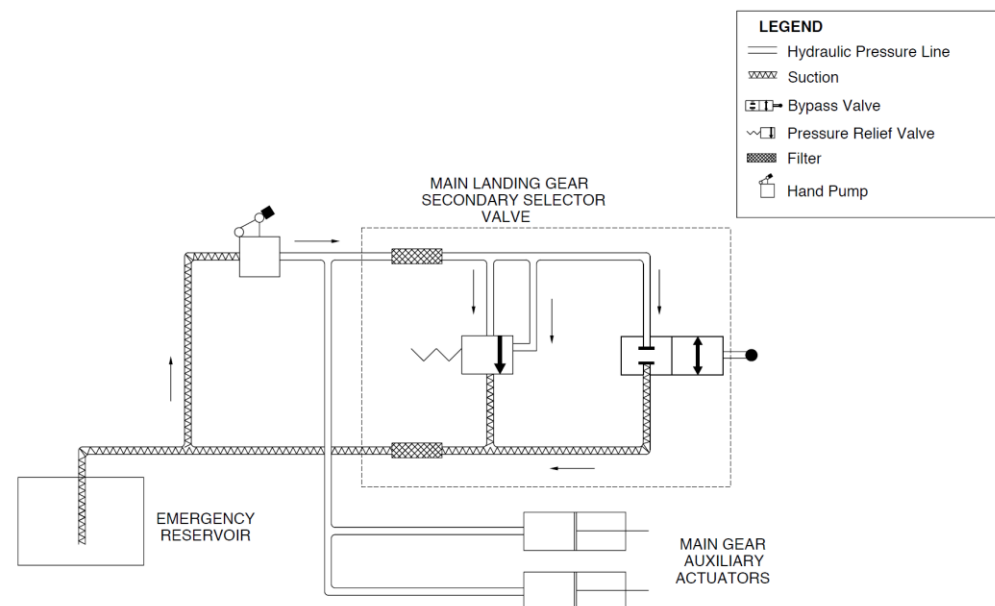
The alternate landing-gear extension system contains the components that follow:

- Hydraulic Hand Pump
- MLG Alternate Extension Selector Valve
- Alternate Landing-Gear Extension Hydraulic Reservoir

### System Operation

[Refer Figure 29-56](#)

The Main Landing Gear (MLG) alternate extension selector valve is located below the flight compartment floor and is normally in the open position. Opening the Landing Gear Alternate Extension door fully, closes the MLG selector valve and allows the hand pump lever to be inserted into the hand pump socket. Stroking the hand pump lever, provides pressure to the alternate landing-gear actuators to extend the gear to the down and locked position.



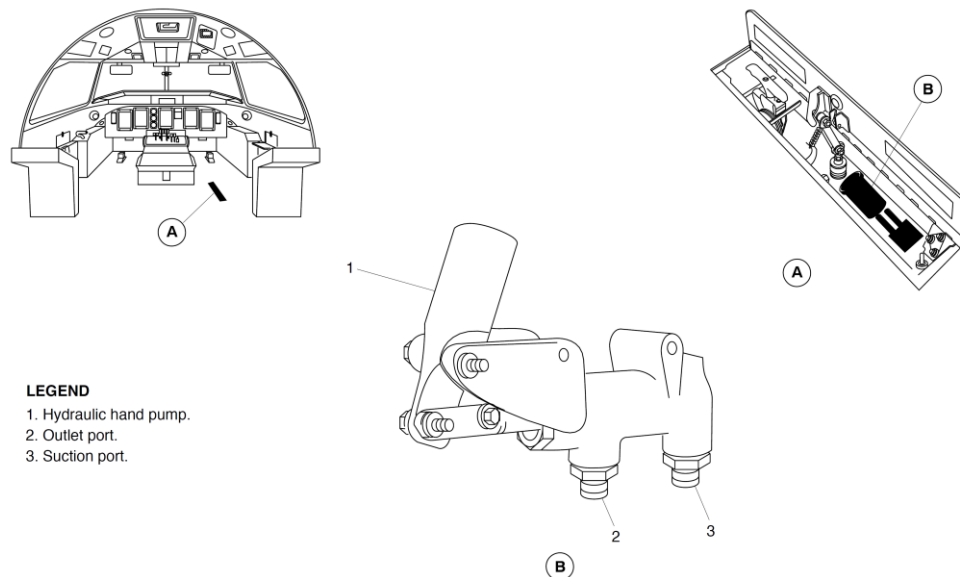
**Figure 29-56 Alternate Landing Gear Extension Hydraulic System**

## Auxiliary Hydraulic System – Component Description

### Hydraulic Hand Pump

[Refer Figures 29-57 & 29-58](#)

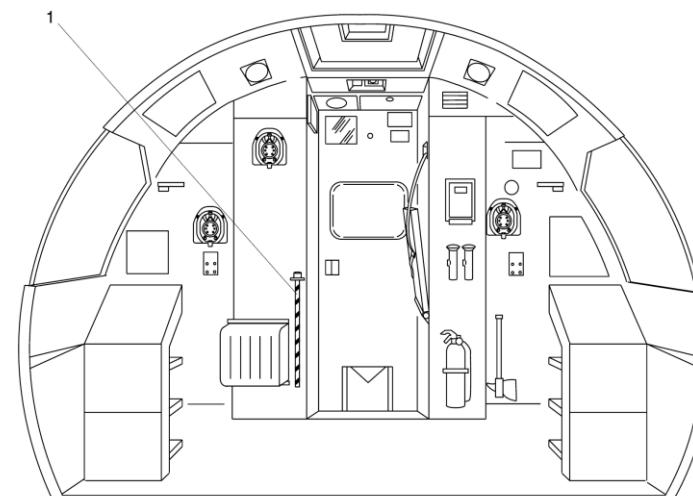
The hand pump is located below the flight compartment floor, adjacent to the copilot seat. The handpump is used to lower the landing gear following System 2 isolation. The handpump produces 0.75 in<sup>3</sup> (12.29 cm<sup>3</sup>) per stroke. A hand pump handle is installed in the flight compartment on the bulkhead behind the copilot's seat. The handle must be installed into the pump to operate it.



#### LEGEND

1. Hydraulic hand pump.
2. Outlet port.
3. Suction port.

**Figure 29-57 Alternate Hydraulic Hand Pump**



#### LEGEND

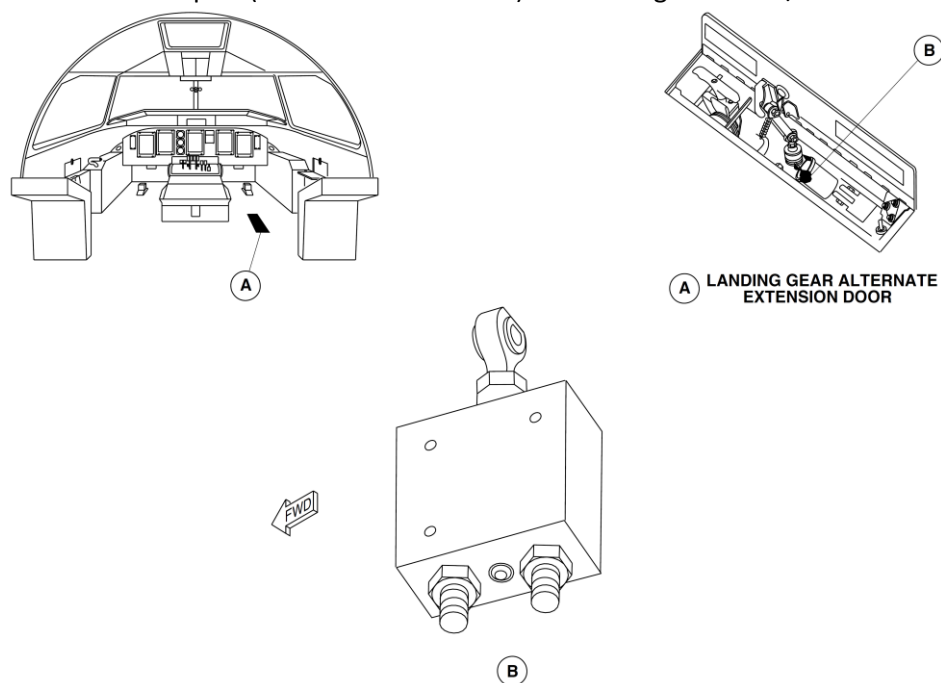
1. Alternate Landing Gear Extension Hand Pump Handle.

**Figure 29-58 Alternate Extension Hand Pump Handle**

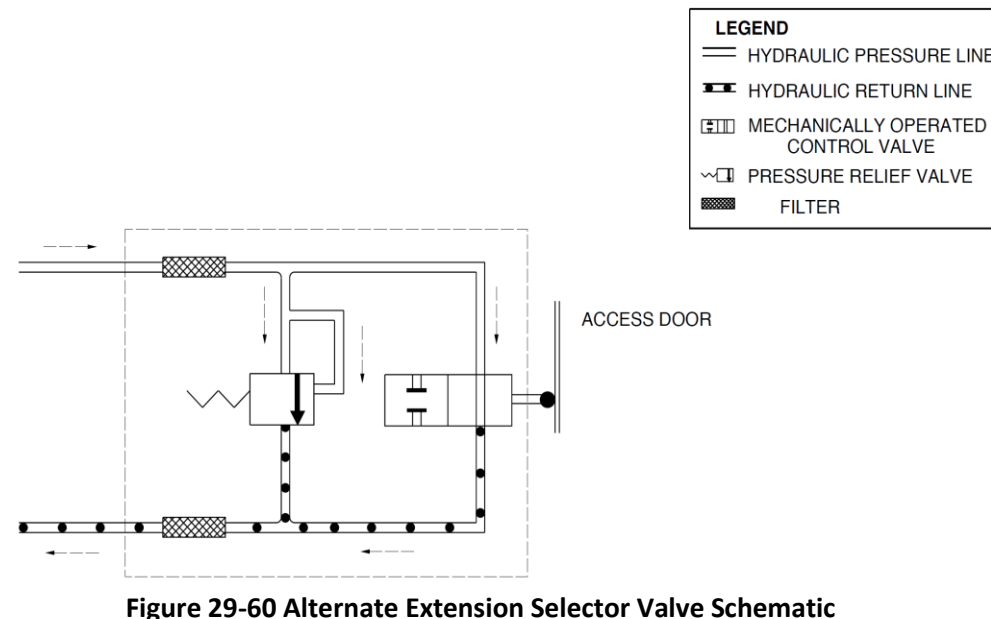
## MLG Alternate Extension Selector Valve

[Refer Figures 29-59 & 29-60](#)

The Main Landing Gear (MLG) Alternate Extension selector valve is located below the flight compartment floor adjacent to the copilot seat. The valve has a 2–way, 2–position control valve, a pressure relief valve, a filtered inlet and return port. There is a pivoting link that attaches the selector valve to the hand pump access door. The selector valve moves from the normally open position to the closed position when the access door is opened. The pressure relief valve limits the pressure that may be applied to the alternate gear down actuators by relieving handpump pressure to return. The valve cracking pressure is 3150 to 3850 psid (21718 to 26545 kPad), full flow pressure is 3500 to 4300 psid (24132 to 29647 kPad) at 0.50 gal/min (1.89 L/min) and reseal is 2385 to 3465 psid (16444 to 23890 kPad ) with leakage of 5 cm<sup>3</sup>/min.



**Figure 29-59 Alternate Extension Selector Valve**

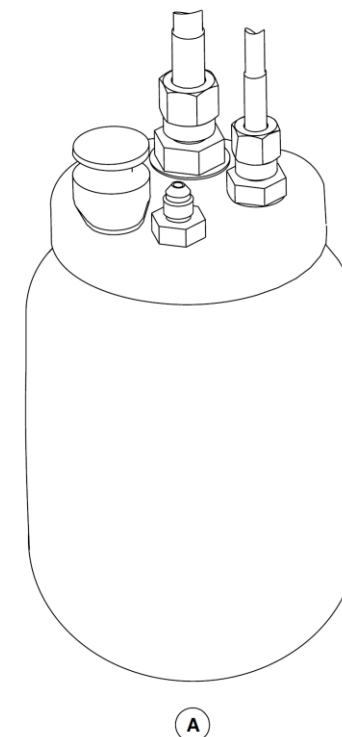
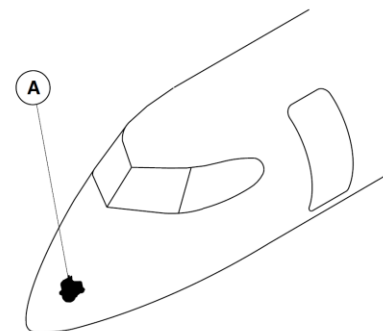


**Figure 29-60 Alternate Extension Selector Valve Schematic**

## Alternate Landing–Gear Extension Hydraulic Reservoir

[Refer Figure 29-61](#)

The alternate landing–gear extension hydraulic reservoir is located in the nose compartment of the aircraft. This reservoir supplies the hydraulic fluid source for the alternate MLG extension hand pump. The reservoir is made from aluminum and has a capacity of 1 US qt (0.95 L). The reservoir has a threaded, vented, removable lid which has three fittings installed. One fitting serves as a fill port, the second fitting is the vent and the third fitting is the outlet port to the hand pump. The lid also has a dipstick with minimum and maximum volume markings to allow filling of the reservoir to the correct fluid level.



**Figure 29-61 Alternate Landing–Gear Extension Hydraulic Reservoir**



## Introduction

Hydraulic system indications are shown on the Multi-Function Displays (MFDs), Caution and Warning panel, the HYDRAULIC CONTROL panel, and the Fire Protection Panel. The indicating system shows, in the flight compartment, the indications that follow:

- Hydraulic pressures
- Hydraulic quantities
- System faults
- System operation status.

## General Description

Hydraulic system indications are shown on the Multi-Function Displays (MFDs), Caution and Warning panel, the HYDRAULIC CONTROL panel, and the Fire Protection panel. Information is sent to the Caution and Warning panel by temperature and pressure switches. Quantity and pressure transducers give inputs to the MFD. The indicating system has the components that follow:

- Hydraulic Pressure Transducers
- Temperature Switches 225 °F (107 °C)
- Hydraulic System Pressure Switch
- Hydraulic Warning Pressure Switch

## Detailed Description

There are two Multi-Function Displays (MFDs) in the flight compartment that show hydraulic system pressure and fluid quantity. There are three hydraulic system indications:

- System No.1 and No. 2 main pressure
- No.3 hydraulic pressure
- No. 1, No. 2 and No.3 quantity indications.

The copilot's MFD shows data in analog format during normal configuration. The pilot's MFD shows data in digital format on a composite page if the copilot's MFD malfunctions. Hydraulic pressure data is sent to the MFD from

## Indicating

pressure transducers. Hydraulic fluid quantity data is sent to the MFD by electrical signals from hydraulic system reservoir quantity transducers.

### [Refer Figure 29-62](#)

The Caution and Warning panel, shows hydraulic system caution lights. Each hydraulic system includes pressure and temperature switches that turn on caution lights on this panel.

### [Refer Figure 29-63](#)

The Fire Protection Panel shows the advisory indications for the hydraulic Firewall Shut-Off Valves (FSOVs). During normal operation, an FSOV OPEN green light will be on. When the FSOV is closed, or is manually closed by pulling the PULL FUEL/HYD OFF handle, an FSOV CLOSED white light will come on.

## Pressure Transducers

There are two pressure transducers in the No.1 hydraulic system. One transducer sends system pressure data to the MFD. The other transducer sends Standby Power Unit (SPU) discharge pressure to the MFD. There is one pressure transducer installed in the No. 2 hydraulic system which sends system pressure data to the MFD. The No.3 hydraulic system has one pressure transducer installed in the accumulator isolation valve which sends system pressure data to the MFD.

## Quantity Transducers

Each hydraulic system reservoir has a quantity transducer to transmit the quantity of fluid in the reservoir to the flight compartment. The quantity transducers are installed on each of the three hydraulic system reservoirs. The quantity transducer sends an electrical signal to the data acquisition unit which shows system fluid quantity on the MFD.

## No. 1 and No. 2 Hydraulic System Pressure Switches

Two pressure switches are installed on each of the No.1 and No. 2 hydraulic system isolation valves. These pressure switches monitor hydraulic pressure

at the inlet and outlet ports of the isolation valve. When low pressure (less than 900 psi = 6205 k Pascal) is sensed at the outlet pressure switch, the #1 or #2 HYD ISO VLV caution light comes on. One EDP pressure switch is installed in the No.1 and No. 2 hydraulic system filter manifolds. These pressure switches monitor the pressure output from the EDPs. If the EDP fails, the #1 or #2 ENG HYD PUMP caution light comes on. One pressure switch is installed in the No. 2 hydraulic system, downstream from the PTU to give a pressure indication at the PTU discharge port. The PTU PBA ON light comes on if the pressure switch senses pressure.

### No. 3 Hydraulic System Pressure Switches

The No.3 hydraulic system has a pressure switch, installed downstream of the accumulator isolation valve. If the No. 1, No. 2 and No.3 system are all supplying power to the elevator actuators, the pressure switch sends a signal to the Caution and Warning panel and the ELEV PRESS caution light comes on.

### Temperature Switches

The No.1 and No. 2 hydraulic system reservoirs each have a 225 °F (107 °C) temperature switch that monitors hydraulic fluid temperature. At 225 °F (107 °C), a HYD FLUID HOT caution light on the Caution and Warning panel comes on. There is a thermal switch in the SPU that turns on the #1 STBY HYD PUMP HOT caution light if the pump windings are hot.

### Firewall Shut-off Valve (FSOV) Position Switches

[Refer Figure 29-63](#)

The FSOV position switches in the No.1 and No. 2 hydraulic systems are used to give FSOV position advisory indications on the Fire Protection Panel. The normally OPEN (green light) will go out and the CLOSED (white light) will come on when the FSOV is closed.

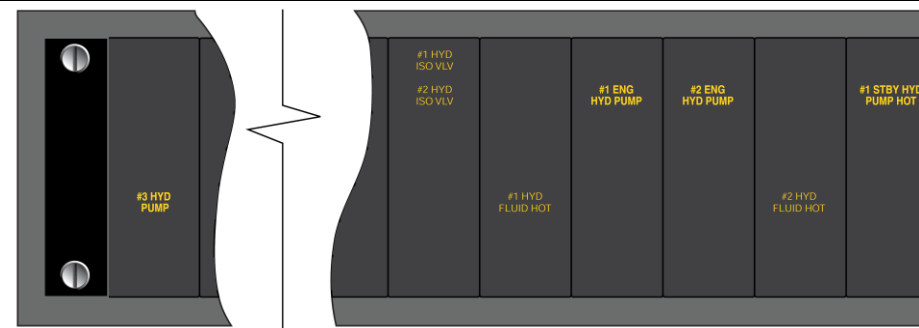


Figure 29-62 Hydraulic System Caution Indications

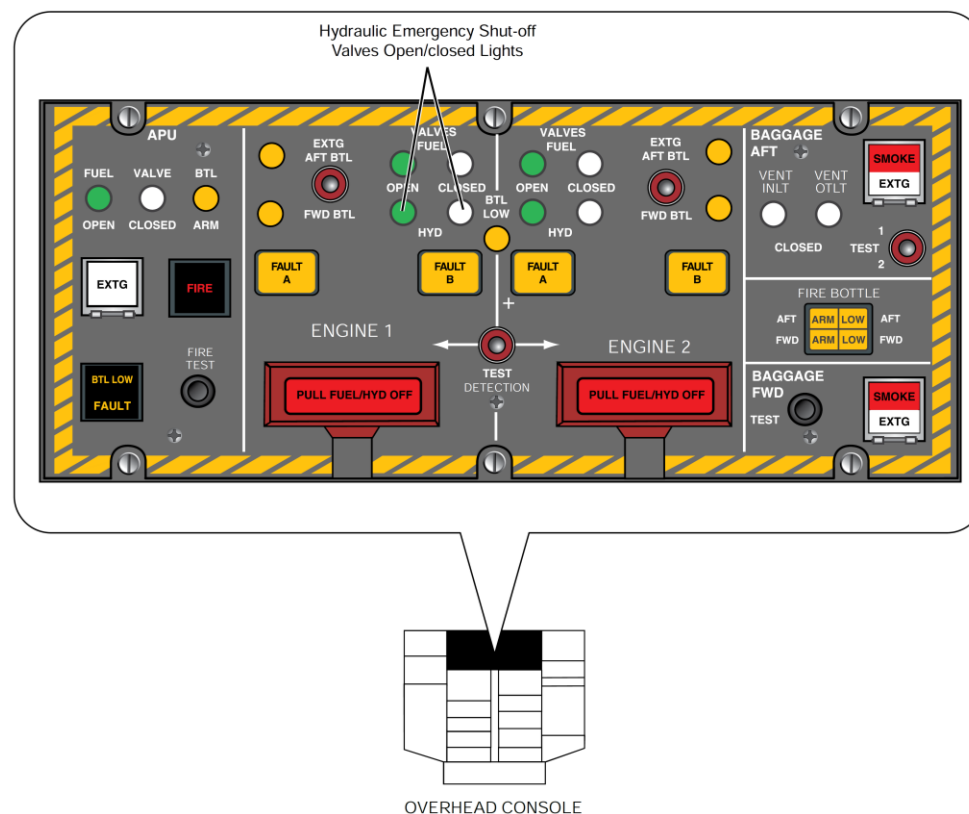


Figure 29-63 Fire Protection Panel Indication

## Hydraulic Pressures and Quantity Digital Indications

[Refer Figure 29-64](#)

### Standby Hydraulic Pressure Indicator

- Indicates pressure in No. 1 hydraulic system available from the electrically operated standby pump
- Pressures shown in psi (0 psi) (0 kPa) and indicated in white
- Digits are replaced by white dashes when parameter no longer valid

### No. 1 Main Hydraulic Pressure Indicator

- Indicates pressure in No. 1 hydraulic system regardless of pressure source
- Pressures shown in psi (3000 psi) (20684 kPa)
- Digits are replaced by white dashes when parameter no longer valid

### No. 2 Main Hydraulic Pressure Indicator

- Indicates pressure in No. 2 hydraulic system regardless of pressure source
- Pressures shown in psi (3000 psi) (20684 kPa)
- Digits are replaced by white dashes when parameter no longer valid

### No. 3 Main Hydraulic Pressure Indicator

- Indicates pressure in No. 3 hydraulic system
- Pressures shown in psi (0 psi) (0 kPa)
- Digits are replaced by white dashes when parameter no longer valid

### No. 1, 2 and 3 Hydraulic Quantity Indicators

- Indicates relative actual quantity available in the system reservoirs
- Quantities indicated in % where 100 represents the maximum quantity in each reservoir.

Reservoir quantities are as follows:

- No. 1 system 8 US qt (7.57 L)
- No. 2 system 12 US qt (11.36 L)
- No. 3 system 2.6 US qt (2.46 L)

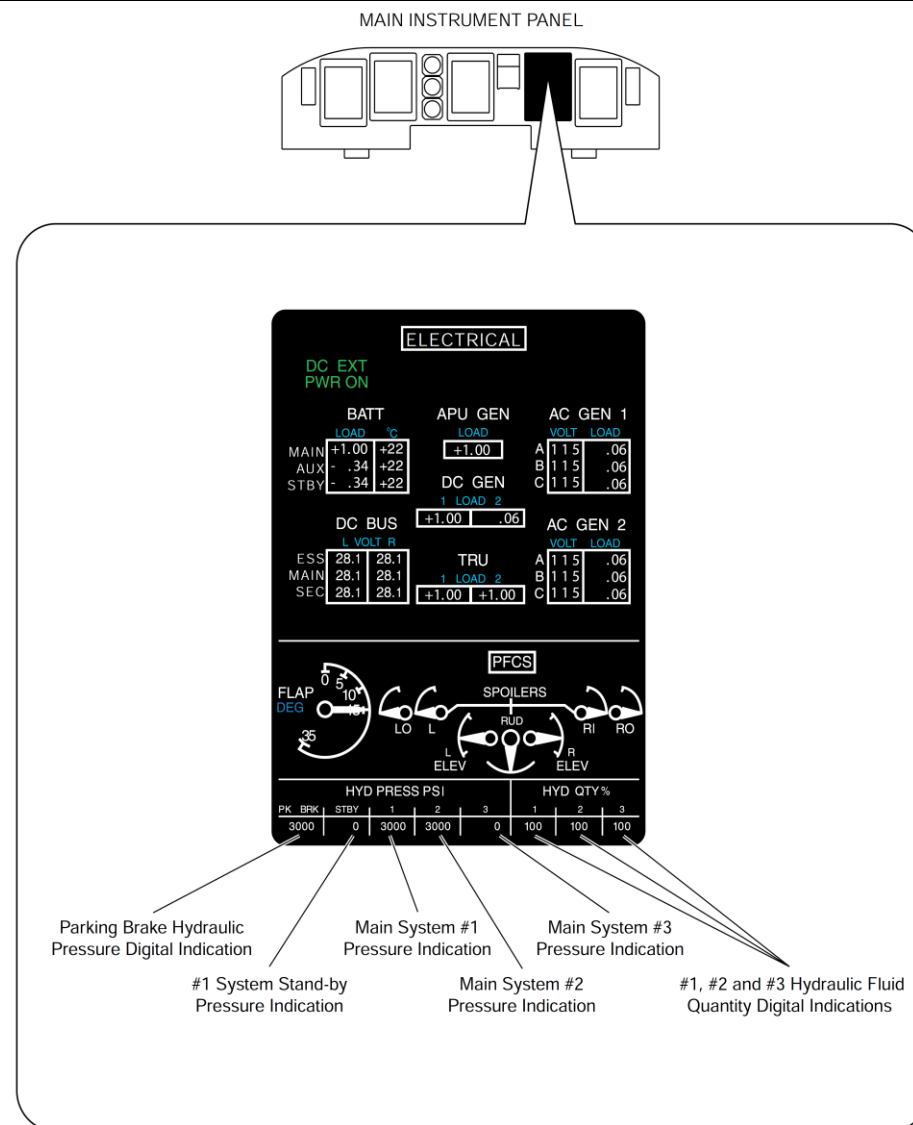


Figure 29-64 Hydraulic Systems Indications Digital

## Hydraulic Pressures and Quantity Analog Indications

[Refer Figure 29-65](#)

### Standby Hydraulic Pressure Indicator

- Indicates pressure in No. 1 hydraulic system available from the electrically operated standby pump
- Scale marked every 1000 psi (6894 kPa) from 0 to 4 and indicated in white
- Reverts to white line when parameter no longer valid.

### No. 1 Main Hydraulic Pressure Indicator

- Indicates pressure in No. 1 hydraulic system regardless of pressure source
- Scale is marked every 1000 psi (6894 Pa) from 0 to 4 and indicated in white
- Reverts to white line when parameter no longer valid.

### No. 2 Main Hydraulic Pressure Indicator

- Indicates pressure in No. 2 hydraulic system regardless of pressure source
- Scale is marked every 1000 psi (6894 kPa) from 0 to 4 and indicated in white
- Reverts to white line when parameter no longer valid.

### No. 3 Main Hydraulic Pressure Indicator

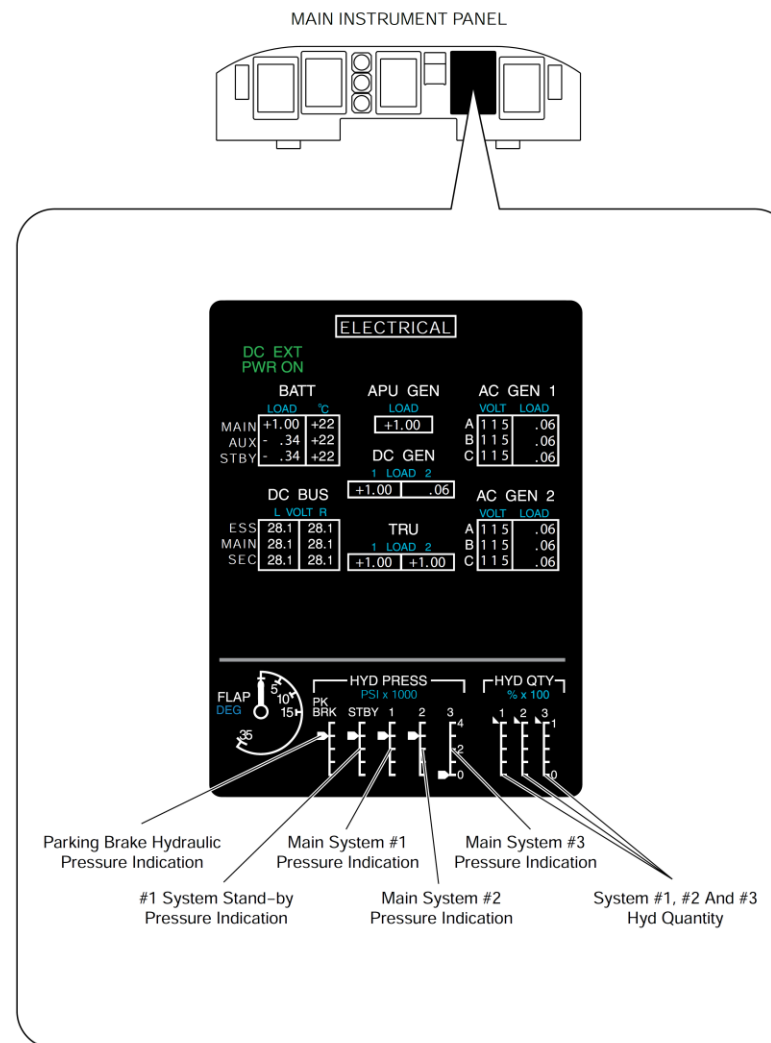
- Indicates pressure in No. 3 hydraulic system
- Scale is marked every 1000 psi (6894 kPa) from 0 to 4 and indicated in white
- Reverts to white line when parameter no longer valid.
- Digital values 0,2,4 are displayed on the right side of the No. 3 scale only.

### No. 1, 2 and 3 Hydraulic Quantity Indicators

- Indicate relative actual quantity available in the system reservoirs
- Quantities indicated in % of 100 where 100% represents the maximum quantity in each reservoir

Reservoir quantities are as follows:

- No. 1 system 8 US qt (7.57 L)
- No. 2 system 12 US qt (11.36 L)
- No. 3 system 2.6 US qt (2.46 L)



**Figure 29-65 Hydraulic Systems Indication Analog**